

Transseries for Mere Mortals

From Asymptotic Scales to Resurgence

A step-by-step tutorial for a mathematically mature undergraduate

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Scope. This book develops two closely related meanings of the word “transseries.” First, a transseries is a formal generalized series built from powers, exponentials, logarithms, and well-ordered infinite sums. Second, in asymptotic analysis and mathematical physics, a resurgent transseries is an organized family of usually divergent power-series sectors multiplied by exponentially small scales. The first viewpoint explains the algebra; the second explains how exponentially small information repairs divergent asymptotics and Stokes jumps.

Prerequisites. Single-variable calculus, elementary real analysis, linear algebra, and comfort with formal power series. A first course in complex analysis is useful only in the chapters on Borel summation and resurgence. Every specialized construction is introduced from scratch.

Sources. The exposition is original, but it was informed by the materials supplied with the request, especially Edgar’s survey [13], Costin’s construction and summability survey [8], the primer of Aniceto–Başar–Schiappa [2], and papers on composition, surreal numbers, and H -fields. Additional primary references are listed in the bibliography.

How to use this book

There are three reasonable routes through the material.

The practical route

Read Chapters 1–3, then Chapters 10–15, and finish with the worked examples in Part V. This is enough to begin calculating.

The formal-algebra route

Read Parts I–IV in order. This gives the construction of logarithmic-exponential transseries as a Hahn field, including the support conditions that make infinite algebra legal.

The analytic/resurgent route

Read Part I, the linear differential-equation example in Chapter 19, and then Part VI. This route explains factorial divergence, Borel summation, Stokes phenomena, and the role of extra exponential sectors.

The symbol x always tends to $+\infty$ unless another limit is stated. We use $f \prec g$ for “ f is asymptotically smaller than g ,” meaning $f/g \rightarrow 0$ in an analytic context and the corresponding monomial comparison in a formal context. The two meanings agree whenever a formal transseries is realized by an actual function.

Big idea

A transseries is not merely a very long power series. It is a coordinate system for asymptotic size. It can record, in one object, terms on scales such as

$$x^3, \quad x, \quad \log x, \quad 1, \quad x^{-1}, \quad e^{-x}, \quad e^{-x^2}, \quad e^{-e^x}, \dots$$

and it keeps them in a strict dominance order.

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Notation at a glance

Symbol	Meaning
$f \sim g$	$f/g \rightarrow 1$.
$f \prec g$	$f/g \rightarrow 0$, or formally: every leading monomial of f is smaller than the leading monomial of g .
$f \succ g$	$g \prec f$.
$f = O(g)$	$ f/g $ is eventually bounded.
$f = o(g)$	$f/g \rightarrow 0$.
$\mathfrak{M}$	An ordered multiplicative group of monomials.
$\text{supp } T$	The set of monomials having nonzero coefficient in the formal series T .
$\text{mag } T$	The largest monomial in $\text{supp } T$.
$\text{lc}(T)$	The coefficient of $\text{mag } T$.
$T^\dagger$	The logarithmic derivative T'/T , when $T \neq 0$.
$\mathbb{T}$	A standard field of logarithmic-exponential transseries.
$\widehat{\Phi}$	A formal, usually divergent, power series.
$\mathcal{B}, \mathcal{L}$	Borel and Laplace transforms.

Part I

Why asymptotics needs more than power series

Chapter 1

The language of asymptotic approximation

1.1 Exact formulas and asymptotic formulas answer different questions

An exact formula says what a quantity is. An asymptotic formula says which parts of it matter on a chosen limiting scale. These are different jobs.

For example,

$$\frac{x^2 + 3x + 1}{x^2 - x + 2} = 1 + \frac{4}{x} + \frac{3}{x^2} - \frac{5}{x^3} + O(x^{-4}) \quad (x \rightarrow +\infty).$$

The rational function on the left is exact. The expression on the right is more useful when x is large because it separates successive sizes. The first term is of order 1, the next of order x^{-1} , and so on.

Definition 1.1 (Asymptotic equivalence). For eventually nonzero functions f and g , we write

$$f(x) \sim g(x) \quad (x \rightarrow +\infty)$$

when $f(x)/g(x) \rightarrow 1$.

Definition 1.2 (Dominance). We write $f \prec g$ when $f/g \rightarrow 0$, and $f \succ g$ when $g \prec f$. Thus $x^{-2} \prec x^{-1} \prec 1 \prec \log x \prec x \prec e^x$.

Dominance is more informative than ordinary inequalities. Although $x^{-1} < 2x^{-1}$, the two functions have the same asymptotic size. In contrast, $x^{-2} \prec x^{-1}$ says that x^{-2} becomes negligible relative to x^{-1} .

1.2 Asymptotic scales

Definition 1.3 (Asymptotic scale). A sequence $(\phi_n)_{n \geq 0}$ is an asymptotic scale if

$$\phi_{n+1} \prec \phi_n \quad \text{for every } n.$$

The standard scale at infinity is

$$1, x^{-1}, x^{-2}, \dots$$

But many problems naturally use other scales:

$$x^\alpha, x^{\alpha-1}, \dots; \quad 1, \frac{1}{\log x}, \frac{1}{(\log x)^2}, \dots; \quad 1, e^{-x}, e^{-2x}, \dots$$

A transseries allows several such scales to be interleaved and nested.

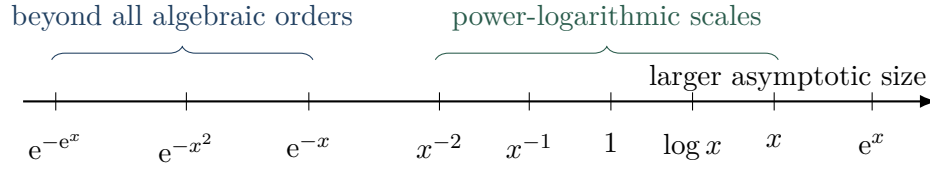


Figure 1.1: A tiny part of the dominance hierarchy at $x \rightarrow +\infty$.

1.3 Poincaré asymptotic expansions

Definition 1.4 (Asymptotic expansion). Let (ϕ_n) be an asymptotic scale. We write

$$f(x) \sim_{\text{as}} \sum_{n=0}^{\infty} a_n \phi_n(x)$$

if, for every $N \geq 1$,

$$f(x) - \sum_{n=0}^{N-1} a_n \phi_n(x) = o(\phi_{N-1}(x)).$$

Equivalently, after subtracting the first N terms, the remainder is smaller than the last retained scale.

The infinite sum in this definition need not converge. It is shorthand for an infinite list of finite-accuracy statements.

Example 1.5 (A geometric expansion). For $x > 1$,

$$\frac{1}{x+1} = \frac{1}{x} \frac{1}{1+x^{-1}} = \frac{1}{x} - \frac{1}{x^2} + \frac{1}{x^3} - \dots$$

Here the series actually converges, but convergence is stronger than required. As an asymptotic statement,

$$\frac{1}{x+1} - \sum_{n=1}^N (-1)^{n-1} x^{-n} = \frac{(-1)^N}{x^N(x+1)} = O(x^{-N-1}).$$

Example 1.6 (A divergent asymptotic expansion). The solution of

$$y' + y = \frac{1}{x}$$

that behaves like $1/x$ has the formal expansion

$$y(x) \sim_{\text{as}} \frac{1}{x} + \frac{1}{x^2} + \frac{2!}{x^3} + \frac{3!}{x^4} + \dots$$

The coefficients grow factorially, so the series diverges for every finite x . It is nevertheless an accurate asymptotic expansion when truncated near its smallest term. We will analyze it completely in Chapter 19.

1.4 Why the coefficients are unique

Proposition 1.7 (Uniqueness in a fixed scale). *If*

$$f \sim_{\text{as}} \sum_{n \geq 0} a_n \phi_n \quad \text{and} \quad f \sim_{\text{as}} \sum_{n \geq 0} b_n \phi_n$$

with respect to the same asymptotic scale, then $a_n = b_n$ for all n .

Proof. Subtract the two expansions. At order $N = 1$,

$$(a_0 - b_0)\phi_0 = o(\phi_0),$$

so $a_0 - b_0 = 0$. Suppose $a_0 = b_0, \dots, a_{N-1} = b_{N-1}$. At the next order,

$$(a_N - b_N)\phi_N = o(\phi_N),$$

which again forces $a_N = b_N$. Induction completes the proof. $\square$

The phrase “in a fixed scale” matters. A power series in x^{-1} cannot see an e^{-x} term at all. Enlarging the scale enlarges the information that can be recorded.

1.5 Truncation, optimal truncation, and error

For a convergent series, retaining more terms eventually improves the answer. For a divergent asymptotic series this is usually false. If terms first decrease and then increase, the best elementary approximation is often obtained by stopping near the smallest term.

Consider a typical Gevrey-1 term

$$\frac{n!}{x^{n+1}}.$$

The ratio of consecutive terms is $(n+1)/x$. Terms decrease while $n+1 < x$ and increase after $n+1 > x$. Thus the smallest term occurs near $n \approx x$, and its size is roughly e^{-x} by Stirling’s formula. The error left after optimal truncation is therefore exponentially small. This is the first hint that an e^{-x} sector is not decorative: it lives exactly at the scale of the best possible remainder of the power-series sector.

Chapter summary

An asymptotic expansion is an ordered sequence of approximations, not necessarily a convergent sum. Its coefficients are unique only relative to a chosen scale. Divergent series can be highly useful, and their optimal remainder often lies on a new exponentially small scale.

Chapter 2

Beyond all orders: the information power series cannot see

2.1 Flat functions

Definition 2.1 (Flat relative to powers). A function f is flat at $+\infty$ relative to the power scale if

$$f(x) = o(x^{-N}) \quad \text{for every } N \geq 0.$$

Proposition 2.2. *For every $a > 0$ and every real N , one has*

$$e^{-ax} \prec x^{-N}.$$

Proof. The claim is equivalent to $x^N e^{-ax} \rightarrow 0$. Repeated use of l'Hôpital's rule proves this for nonnegative integers N ; a larger integer bounds any real N . Alternatively, write $x^N e^{-ax} = \exp(N \log x - ax)$ and observe that $N \log x - ax \rightarrow -\infty$. $\square$

Therefore the zero power series is an asymptotic expansion of e^{-x} :

$$e^{-x} \sim_{\text{as}} 0 + 0x^{-1} + 0x^{-2} + \cdots.$$

The same is true of $17e^{-x}$, e^{-2x} , and $e^{-x}(1 + x^{-1} + x^{-2} + \cdots)$ whenever the latter expression is interpreted as an asymptotic expansion.

Common trap

The statement “two functions have the same asymptotic power series” does not mean that their difference is zero. It means only that their difference is smaller than every power retained by that scale.

2.2 An entire family hidden behind one power series

Suppose f has the power-series expansion

$$f(x) \sim_{\text{as}} \sum_{n \geq 0} a_n x^{-n}.$$

Then for every constant C ,

$$f(x) + Ce^{-x} \sim_{\text{as}} \sum_{n \geq 0} a_n x^{-n}.$$

Thus a power series can represent an entire one-parameter family. In a differential equation that parameter is often an integration constant.

This phenomenon is not rare. A differential equation of order r normally has r free constants, but a purely algebraic asymptotic expansion may appear unique. The “missing constants” often multiply exponentially small solutions of the linearized equation. Transseries restore those sectors.

2.3 A toy implicit equation that creates exponentials

Consider

$$y = x + e^{-y}. \quad (2.1)$$

If we seek only a power series in x^{-1} , the correction $y - x$ is invisible: it is smaller than every power. Set $\delta = y - x$. Then

$$\delta = e^{-x} e^{-\delta}.$$

This already suggests a series in powers of the small transmonomial e^{-x} :

$$\delta = c_1 e^{-x} + c_2 e^{-2x} + c_3 e^{-3x} + \dots.$$

In Chapter 17 we will calculate

$$y = x + e^{-x} - e^{-2x} + \frac{3}{2}e^{-3x} - \frac{8}{3}e^{-4x} + \frac{125}{24}e^{-5x} + \dots.$$

A power series says only $y \sim_{\text{as}} x$. The transseries resolves the whole exponential correction.

2.4 Nested smallness

Exponential scales themselves have scales beneath them:

$$e^{-x^2} \prec e^{-x} \prec x^{-N} \quad \text{for every fixed } N,$$

and

$$e^{-e^x} \prec e^{-x^2}.$$

Inside an e^{-x} sector we may also have an ordinary inverse-power series:

$$e^{-x} \left(1 + \frac{2}{x} + \frac{7}{x^2} + \dots \right).$$

This is the characteristic nested structure of a transseries: a series whose “coefficients” may themselves be series on a smaller scale.

2.5 Why “nonperturbative” does not mean “mysterious”

In physics, the inverse-power sector is often called perturbative and terms such as e^{-Ax} are called nonperturbative. Mathematically, both are formal pieces of one transseries. The terminology describes how the terms arise in a given approximation method, not a difference in logical status.

Intuition

A microscope with magnification x^N cannot distinguish two functions whose difference is e^{-x} . A transseries changes microscopes. After factoring out e^{-x} , it asks for the next layer of detail, perhaps a power series in x^{-1} . Then it can zoom again to e^{-2x} , and so on.

Chapter summary

Exponentially small terms are invisible to every finite algebraic order, yet they often contain integration constants and the optimal remainder of divergent series. Transseries enlarge the asymptotic scale so that this information becomes visible and computable.

Chapter 3

Your first transseries

3.1 The standard sector form

The most common transseries in applications has the form

$$\Phi(x; \sigma) = \sum_{k=0}^{\infty} \sigma^k e^{-kAx} x^{k\beta} \Phi_k(x), \quad \Phi_k(x) = \sum_{n=0}^{\infty} a_{k,n} x^{-n}, \quad (3.1)$$

where $A > 0$, $\beta \in \mathbb{R}$, and σ is a formal parameter.

Read this expression from outside inward.

1. The index k labels exponential sectors.
2. Neighboring sectors differ by the small ratio $e^{-Ax} x^\beta$.
3. Inside each sector is an inverse-power series.
4. The parameter σ usually becomes an integration constant or boundary datum after an analytic summation prescription is chosen.

The $k = 0$ sector is

$$\Phi_0(x) = a_{0,0} + a_{0,1}x^{-1} + \cdots$$

The $k = 1$ sector is exponentially smaller:

$$\sigma e^{-Ax} x^\beta \left(a_{1,0} + a_{1,1}x^{-1} + \cdots \right),$$

and the $k = 2$ sector is smaller again by another factor $\sigma e^{-Ax} x^\beta$.

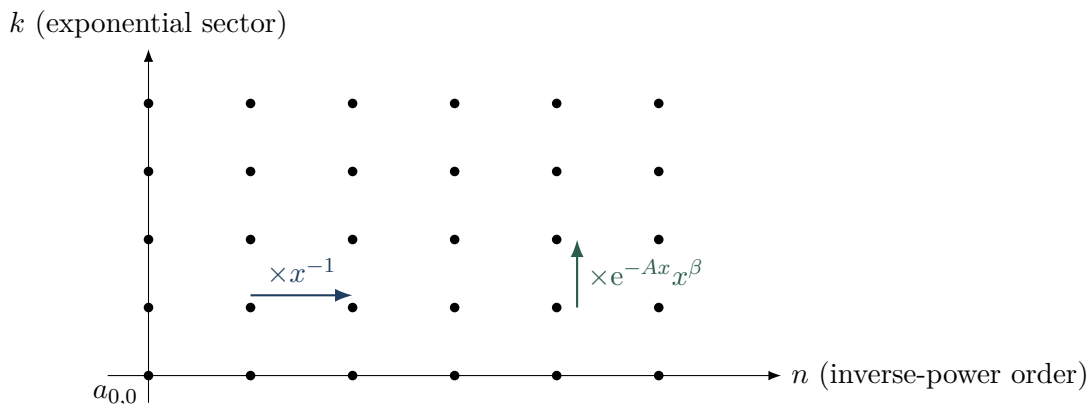
$$\sigma^0 e^{-0Ax} x^{0\beta} \Phi_0 \leftarrow \sigma e^{-1Ax} x^{1\beta} \Phi_1 \leftarrow \sigma^2 e^{-2Ax} x^{2\beta} \Phi_2 \leftarrow \sigma^3 e^{-3Ax} x^{3\beta} \Phi_3 \leftarrow \sigma^4 e^{-4Ax} x^{4\beta} \Phi_4(x)$$

power-series one-exponential two-exponential
sector sector sector

Figure 3.1: A one-parameter transseries arranged by exponentially decreasing sectors.

3.2 A two-dimensional coefficient array

It is useful to picture the coefficients $a_{k,n}$ on a lattice. Moving right increases n and multiplies the scale by x^{-1} . Moving upward increases k and multiplies it by $e^{-Ax}x^\beta$.



A finite approximation amounts to selecting a finite region of this lattice. But “rectangular truncation” is not always optimal: because the exponential step is much smaller than any fixed power step, one often computes many power terms in the first sector before including even the leading term of the next.

3.3 Three meanings of the formula

Expression (3.1) can be used in three distinct ways.

3.3.1 As a purely formal object

The symbols x^{-1} and e^{-Ax} are ordered monomials. Infinite sums are legal because their supports satisfy a well-ordering condition. No numerical value of x is substituted. This is the Hahn-field viewpoint developed in Parts II–IV.

3.3.2 As an asymptotic specification

Each sector may specify the asymptotic behavior of an actual function, with a rule for how remainders compare to omitted terms. This is the viewpoint of classical asymptotic analysis.

3.3.3 As resurgent data

The inner series Φ_k may diverge factorially. Borel–Laplace summation then turns each formal sector into an analytic function in a direction of the complex plane. Singularities of the Borel transform connect different values of k , and crossing a Stokes direction changes the effective value of σ . This is the resurgent viewpoint of Part VI.

Do not silently identify the viewpoints

A formal logarithmic-exponential transseries need not be a convergent series of real functions. Conversely, a resurgent transseries may use complex directions, branch cuts, and factorially

divergent coefficient series not captured by a naive notion of convergence. The viewpoints overlap, but analytic realization requires extra hypotheses.

3.4 Multi-parameter transseries

An r -parameter version is

$$\Phi(x; \sigma) = \sum_{k \in \mathbb{N}^r} \sigma^k e^{-(k \cdot A)x} x^{k \cdot \beta} \Phi_k(x),$$

where

$$\sigma^k = \sigma_1^{k_1} \cdots \sigma_r^{k_r}.$$

Different exponential actions A_j usually correspond to different independent solutions of a linearized equation, different saddle points of an integral, or different instanton types. When integer relations among the A_j make sectors collide, logarithms may appear; this is called resonance.

3.5 A first substitution calculation

Let

$$T = x + e^{-x} + 2x^{-1}e^{-x} + 3e^{-2x}.$$

Then, to order e^{-2x} ,

$$\begin{aligned} e^{-T} &= e^{-x} \exp\left(-e^{-x} - 2x^{-1}e^{-x} - 3e^{-2x}\right) \\ &= e^{-x} \left[1 - (1 + 2x^{-1})e^{-x} + O(e^{-2x})\right] \\ &= e^{-x} - (1 + 2x^{-1})e^{-2x} + O(e^{-3x}). \end{aligned}$$

The computation looks exactly like ordinary formal power-series algebra, except that the small expansion variable is itself a monomial, e^{-x} , and its coefficients may involve powers of x^{-1} .

3.6 What the rest of the book must justify

We have manipulated transseries informally. A rigorous theory must answer:

1. Which monomials are allowed?
2. Which infinite supports are allowed?
3. Why does the coefficient of a product involve only finitely many terms?
4. How are order, inverse, exponential, logarithm, derivative, integral, and composition defined?
5. When does a formal transseries correspond to an actual function?

Parts II–IV answer the first four questions. Part VI addresses the fifth for a large and important analytic class.

Chapter summary

The most familiar transseries is a sum of exponentially graded sectors, each carrying an inverse-power series. It may be read formally, asymptotically, or resurgently. The formal

theory must control supports; the analytic theory must control divergence and Stokes jumps.

Part II

Generalized series and ordered supports

Chapter 4

Growth rates as monomials

4.1 From numerical size to asymptotic size

Ordinary power series are built from the monomials $1, z, z^2, \dots$. At $x \rightarrow +\infty$ we want a much larger alphabet:

$$x^a, \quad e^x, \quad e^{-x^2}, \quad x^{\sqrt{2}}e^{-3x}, \quad \exp\left(\frac{x}{\log x}\right), \quad \dots$$

The basic objects are not numerical values but formal growth rates.

Definition 4.1 (Ordered monomial group). An ordered monomial group is an abelian group $\mathfrak{M}$ written multiplicatively, with identity 1, together with a total order $\succ$ compatible with multiplication:

$$\mathfrak{m} \succ \mathfrak{n} \implies \mathfrak{m}\mathfrak{g} \succ \mathfrak{n}\mathfrak{g} \quad (\mathfrak{g} \in \mathfrak{M}).$$

The intended reading is “ $\mathfrak{m}$ is asymptotically larger than $\mathfrak{n}$.” The inverse of a large monomial is small:

$$\mathfrak{m} \succ 1 \iff \mathfrak{m}^{-1} \prec 1.$$

Example 4.2 (Power-exponential monomials). For

$$\mathfrak{m} = x^a e^{bx}, \quad \mathfrak{n} = x^c e^{dx},$$

we declare $\mathfrak{m} \succ \mathfrak{n}$ if either $b > d$, or $b = d$ and $a > c$. This is just the asymptotic fact

$$\frac{x^a e^{bx}}{x^c e^{dx}} = x^{a-c} e^{(b-d)x}.$$

The exponential comparison is decided first; powers break ties.

4.2 Monomials are coordinates, not functions

A formal monomial may be inspired by a function, but the theory initially uses only its algebra and order. This separation is valuable. It lets us manipulate objects before proving convergence or analytic realization.

For example, we may declare

$$e^{-e^x} \prec e^{-x^2} \prec e^{-x} \prec x^{-N}$$

for every fixed N , and use these symbols as ordered coordinates. Later, when x is a large real number, the same order is analytically correct.

Intuition

Think of a monomial as a labeled ruler for one asymptotic scale. A transseries is a coordinate vector with respect to an enormous, ordered collection of such rulers.

4.3 Leading monomials determine order

Suppose we have a formal sum

$$T = 3e^x - 7x^4 + 2\log x + x^{-1} + e^{-x}.$$

The largest monomial is e^x , so the leading term is $3e^x$ and T is positive. The smaller terms cannot overturn the sign, just as the first nonzero digit determines the sign of a decimal written in positional notation.

Definition 4.3 (Magnitude, leading coefficient, leading term). For a nonzero formal series T , if its support has a largest monomial $\mathbf{m}$, set

$$\text{mag}(T) = \mathbf{m}, \quad \text{lc}(T) = [\mathbf{m}]T, \quad \text{dom}(T) = \text{lc}(T)\mathbf{m}.$$

Here $[\mathbf{m}]T$ denotes the coefficient of $\mathbf{m}$.

We then define

$$T > 0 \iff \text{lc}(T) > 0.$$

This order is lexicographic by asymptotic size.

Definition 4.4 (Asymptotic comparison of series). For nonzero series S, T :

$$S \prec T \iff \text{mag}(S) \prec \text{mag}(T),$$

$$S \asymp T \iff \text{mag}(S) = \text{mag}(T),$$

and

$$S \sim T \iff S - T \prec T.$$

The last relation means that S and T have the same leading term.

4.4 Large, small, and purely large parts

A series T is called

- *large* if $\text{mag}(T) \succ 1$;
- *small* if every monomial in its support is $\prec 1$;
- *purely large* if every monomial in its support is $\succ 1$.

The zero series is conventionally both small and purely large, because its support is empty.

Every transseries will have a canonical decomposition

$$T = T_{\text{pl}} + T_1 + T_{\text{sm}}, \tag{4.1}$$

where T_{pl} is purely large, $T_1 \in \mathbb{R}$ is constant, and T_{sm} is small. This decomposition is the key to defining $\exp T$.

4.5 Valuation notation

Some authors reverse the order and use an additive valuation. If monomials are written as t^γ , then smaller γ may correspond to larger asymptotic size. A valuation $v(T)$ records the exponent of the leading monomial and satisfies

$$v(ST) = v(S) + v(T), \quad v(S + T) \geq \min\{v(S), v(T)\}.$$

Cancellation can make the second inequality strict.

In this tutorial we mostly use the more visual multiplicative notation $\text{mag}(T)$. The two languages contain the same information:

$$S \prec T \iff v(S) > v(T)$$

under the usual valuation convention.

4.6 The field will be non-Archimedean

In the ordered field of transseries, x is larger than every real constant:

$$x > n \quad (n \in \mathbb{N}),$$

and x^{-1} is a positive infinitesimal:

$$0 < x^{-1} < \frac{1}{n} \quad (n \geq 1).$$

Thus the Archimedean axiom of the real numbers fails. This is not a defect; it is exactly what lets one manipulate infinity and infinitesimals algebraically.

Chapter summary

The monomial group packages asymptotic growth rates into an ordered algebra. Once a series has a largest monomial, that monomial and its coefficient determine its leading size and sign. This turns asymptotic comparison into ordinary field order.

Chapter 5

Hahn series: how infinite algebra is made legal

5.1 Why arbitrary supports do not work

Let $\mathfrak{M}$ be an ordered monomial group. A completely arbitrary formal expression

$$\sum_{\mathfrak{m} \in \mathfrak{M}} a_{\mathfrak{m}} \mathfrak{m}$$

is too large a class. It may have no leading term, and multiplication may ask us to add infinitely many unrelated coefficients.

For expansions at infinity, the series

$$1 + x^{-1} + x^{-2} + \cdots$$

behaves well: its monomials march downward from a largest one. In contrast,

$$1 + x + x^2 + x^3 + \cdots$$

has no largest monomial. It is a perfectly good power series near $x = 0$, but it is not a well-based asymptotic series at $x = +\infty$.

Definition 5.1 (Well-based subset). A subset $A \subseteq \mathfrak{M}$ is well-based if every nonempty subset of A has a largest element with respect to $\succ$.

Equivalently, A contains no infinite chain

$$\mathfrak{m}_0 \prec \mathfrak{m}_1 \prec \mathfrak{m}_2 \prec \cdots$$

of monomials growing without a largest member. It may contain infinite descending chains

$$\mathfrak{m}_0 \succ \mathfrak{m}_1 \succ \mathfrak{m}_2 \succ \cdots,$$

which is exactly how an asymptotic expansion is written.

Definition 5.2 (Hahn series). A Hahn series over a coefficient field K and monomial group $\mathfrak{M}$ is a formal sum

$$T = \sum_{\mathfrak{m} \in \mathfrak{M}} a_{\mathfrak{m}} \mathfrak{m}, \quad a_{\mathfrak{m}} \in K,$$

whose support

$$\text{supp } T = \{\mathfrak{m} \in \mathfrak{M} : a_{\mathfrak{m}} \neq 0\}$$

is well-based. The collection is denoted $K((\mathfrak{M}))$ or, in another common notation, $K[[\mathfrak{M}]]$.

5.2 Examples and nonexamples

Assume

$$e^{-x} \prec x^{-N} \prec 1 \prec x^N \prec e^x$$

for every $N > 0$.

Well-based	Not well-based
$\sum_{n \geq 0} n! x^{-n}$	$\sum_{n \geq 0} x^n$
$\sum_{n \geq 0} e^{-nx}$	$\sum_{n \geq 0} e^{nx}$
$\sum_{n \geq 0} x^{-n} e^{-n^2 x}$	$\sum_{n \geq 1} x^{-1/n}$

The last nonexample deserves attention. The monomials $x^{-1}, x^{-1/2}, x^{-1/3}, \dots$ become progressively larger and approach 1 without reaching it. The set has no largest element.

5.3 Addition

Addition is coefficientwise:

$$[\mathfrak{m}](S + T) = [\mathfrak{m}]S + [\mathfrak{m}]T.$$

The union of two well-based sets is well-based, so the support of $S + T$ is legal. Cancellation may remove the old leading term and reveal a smaller one.

Example 5.3. If

$$S = x + 3 + x^{-1}, \quad T = -x + 2 - x^{-2},$$

then

$$S + T = 5 + x^{-1} - x^{-2}.$$

The leading x terms cancel, and the magnitude drops from x to 1.

5.4 Multiplication and Neumann's lemma

Formally, multiplication should be convolution:

$$[\mathfrak{g}](ST) = \sum_{\mathfrak{m}\mathfrak{n}=\mathfrak{g}} [\mathfrak{m}]S [\mathfrak{n}]T. \quad (5.1)$$

Two facts are required:

1. only finitely many pairs $(\mathfrak{m}, \mathfrak{n})$ contribute to a fixed $\mathfrak{g}$;
2. the set of products of support monomials is again well-based.

Both follow from a standard ordered-group result usually called Neumann's lemma [16, 19, 27].

Theorem 5.4 (Neumann's lemma, two-set form). *Let $A, B \subseteq \mathfrak{M}$ be well-based. Then*

$$AB = \{ab : a \in A, b \in B\}$$

is well-based, and every $\mathfrak{g} \in \mathfrak{M}$ has only finitely many representations $\mathfrak{g} = ab$ with $a \in A$ and $b \in B$.

Proof idea. If AB had an infinite sequence moving upward with no largest element, choose corresponding products a_nb_n . By repeatedly passing to subsequences and using the well-basedness of A and B , one would force one factor sequence to move upward indefinitely, contradicting the hypothesis. A similar diagonal argument shows that infinitely many decompositions of one fixed $\mathbf{g}$ would force an upward sequence in one factor and the inverse upward sequence in the other. The full proof is an ordered-group version of the familiar fact that a coefficient of a product of ordinary power series receives only finitely many contributions. $\square$

The theorem makes (5.1) an ordinary finite sum. Hence Hahn series form a ring.

5.5 Summable families

We often need to sum a family $(T_i)_{i \in I}$ rather than a single sequence.

Definition 5.5 (Summable family). A family $(T_i)_{i \in I}$ of Hahn series is summable if

1. $\bigcup_i \text{supp } T_i$ is well-based;
2. for each monomial $\mathbf{m}$, only finitely many i have $[\mathbf{m}]T_i \neq 0$.

Then

$$\sum_{i \in I} T_i$$

is defined coefficientwise.

This is formal local finiteness. It does not ask for absolute convergence of numbers. It asks that every coordinate receive only finitely many contributions.

Proposition 5.6. *If $U \prec 1$, then the family $(U^n)_{n \geq 0}$ is summable.*

Idea. All monomials in U are small. Products of n such monomials move farther down the dominance order as n grows. Neumann's lemma implies that the union of all finite products is well-based and that a fixed monomial can occur at only finitely many total degrees. $\square$

This proposition is the formal engine behind geometric, exponential, and logarithmic power series.

5.6 Inverses

Let $T \neq 0$. Factor out its leading term:

$$T = a\mathbf{m}(1 + U), \quad a \in K^\times, \quad U \prec 1.$$

Then define

$$T^{-1} = a^{-1}\mathbf{m}^{-1} \sum_{n=0}^{\infty} (-U)^n. \quad (5.2)$$

The family is summable because U is small, and formal multiplication gives $(1 + U) \sum_{n \geq 0} (-U)^n = 1$.

Example 5.7. At infinity,

$$\begin{aligned} (x + \log x + 1)^{-1} &= x^{-1} \left(1 + \frac{\log x + 1}{x} \right)^{-1} \\ &= x^{-1} - \frac{\log x + 1}{x^2} + \frac{(\log x + 1)^2}{x^3} - \dots \end{aligned}$$

Thus, once the monomial group and support condition are fixed, Hahn series form a field.

5.7 Order and the non-Archimedean triangle law

The leading coefficient defines an order compatible with addition and multiplication. The magnitude obeys

$$\text{mag}(ST) = \text{mag}(S) \text{mag}(T),$$

and

$$\text{mag}(S + T) \preccurlyeq \max\{\text{mag}(S), \text{mag}(T)\},$$

with strict inequality only when leading terms cancel. This is the non-Archimedean analogue of a triangle inequality.

Big idea

Well-basedness replaces numerical convergence. It guarantees a leading term, legal multiplication, and legal substitution into ordinary formal power series whenever the substituted object is small.

Chapter summary

A Hahn series is a generalized power series with well-based support. Neumann's lemma ensures that products and infinite power-series substitutions are locally finite. Factoring out the leading term reduces inversion to a geometric series.

Chapter 6

Grids and ratio sets

6.1 Why finite generators are useful

The full Hahn field allows arbitrary well-based supports. For explicit calculation it is often convenient to work in a smaller class generated by a finite list of small monomials.

Definition 6.1 (Ratio set). A ratio set is a finite tuple

$$\boldsymbol{\mu} = (\mu_1, \dots, \mu_r)$$

of monomials with $\mu_j \prec 1$. For $\mathbf{k} = (k_1, \dots, k_r) \in \mathbb{Z}^r$, write

$$\boldsymbol{\mu}^{\mathbf{k}} = \mu_1^{k_1} \cdots \mu_r^{k_r}.$$

Definition 6.2 (Grid). Given a monomial $\mathbf{m}$ and a lower bound $\mathbf{k}_0 \in \mathbb{Z}^r$, the grid

$$\mathfrak{J}(\mathbf{m}, \boldsymbol{\mu}, \mathbf{k}_0) = \left\{ \mathbf{m} \boldsymbol{\mu}^{\mathbf{k}} : \mathbf{k} \geq \mathbf{k}_0 \right\}$$

uses componentwise inequality. A grid-based series has support contained in some such grid.

Usually we absorb $\mathbf{k}_0$ into $\mathbf{m}$ and write

$$\mathfrak{J} = \{ \mathbf{m} \boldsymbol{\mu}^{\mathbf{n}} : \mathbf{n} \in \mathbb{N}^r \}.$$

Example 6.3. Take

$$\mu_1 = x^{-1}, \quad \mu_2 = e^{-x}.$$

Then a grid contains monomials

$$\mathbf{m} x^{-p} e^{-qx}, \quad p, q \in \mathbb{N}.$$

This is exactly the two-dimensional sector lattice from Chapter 3.

6.2 The exponent lattice

A grid converts monomial bookkeeping into integer-vector bookkeeping. Multiplication becomes addition of exponent vectors:

$$\boldsymbol{\mu}^j \boldsymbol{\mu}^k = \boldsymbol{\mu}^{j+k}.$$

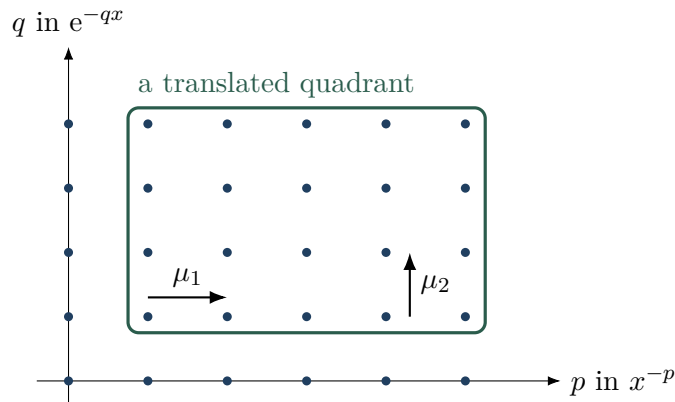


Figure 6.1: A grid generated by x^{-1} and e^{-x} . Moving right or up makes the monomial smaller.

6.3 Dickson's lemma

The key combinatorial fact is that $\mathbb{N}^r$ is well-partially-ordered under componentwise comparison.

Theorem 6.4 (Dickson's lemma). *Every subset $E \subseteq \mathbb{N}^r$ has finitely many minimal elements. Equivalently, every infinite sequence in $\mathbb{N}^r$ contains indices $i < j$ with $\mathbf{k}_i \leq \mathbf{k}_j$ componentwise.*

Proof. We prove the sequence form by induction on r . It is obvious for $r = 1$. For $r > 1$, consider the first coordinates. If some value occurs infinitely often, restrict to that subsequence and apply induction to the remaining $r - 1$ coordinates. Otherwise, pass to a subsequence with strictly increasing first coordinate, and again use induction on the remaining coordinates. The resulting pair is nondecreasing in every coordinate.

If a set had infinitely many minimal elements, listing them would produce a comparable pair, contradicting minimality. Conversely, every element lies above a minimal element; otherwise one could descend forever in the sum of the coordinates. $\square$

Dickson's lemma explains why finite ratio sets are algorithmically pleasant. It is also a basic reason for termination in monomial-ideal and Gröbner-basis arguments.

6.4 Grid-based supports are well-based

Because every generator μ_j is small, increasing an exponent vector makes the corresponding monomial smaller. Dickson's lemma prevents a subset of a grid from having infinitely many unrelated candidates for a leading monomial.

Proposition 6.5. *Every subset of a grid generated by finitely many small monomials is well-based.*

Idea. Translate the exponent set into a subset of $\mathbb{N}^r$. Its finitely many minimal vectors correspond to finitely many largest candidate monomials. Since the monomial order is total, one of these candidates is largest. The same argument applies to every nonempty subset. $\square$

6.5 Relations among generators

The map

$$\mathbf{k} \mapsto \mu^{\mathbf{k}}$$

need not be injective. For example, if $\mu_2 = \mu_1^2$, then $\mu_2 = \mu_1^2$ gives two exponent descriptions of the same monomial. Formal algorithms may either

- choose multiplicatively independent generators when possible;
- retain relations and combine coefficients during normalization;
- work with “witnesses” that certify the required dominance statements.

The mathematics does not require unique exponent vectors, but implementations must account for collisions.

6.6 Grid-based versus well-based transseries

Grid-based series form a proper subfield of the corresponding well-based Hahn field. The distinction is methodological:

	Grid-based	Well-based
Support	Contained in a finitely generated translated quadrant	Any well-based subset
Algorithms	Finite exponent-vector representation	More flexible but harder to encode
Typical applications	Explicit ODE and asymptotic calculations	Structural theorems and maximal constructions
Main caution	May need to enlarge the ratio set	Some operations require subtler support proofs

Edgar’s introductory treatment emphasizes grid-based transseries [13]; van der Hoeven and later structural treatments often use well-based variants [27, 1]. We will state most formulas in a way that applies to both, while using grids for concrete computations.

6.7 A practical truncation order

Suppose a calculation uses $\mu_1 = x^{-1}$ and $\mu_2 = e^{-x}$. A request such as “keep through x^{-4} in the perturbative sector and through $x^{-2}e^{-x}$ in the first exponential sector” describes a finite staircase in $\mathbb{N}^2$.

A good symbolic engine should not assign a single total degree $p + q$ blindly, because one factor of e^{-x} is smaller than every fixed power of x^{-1} . Instead, it should use the actual monomial order and stop at a target monomial.

Working recipe

To organize a hand calculation:

1. Choose a finite ratio set of genuinely small monomials.
2. Write every retained term as a leading monomial times a nonnegative power of the ratios.
3. Decide a target monomial below which terms will be discarded.
4. After each operation, normalize collisions and discard terms smaller than the target.

Chapter summary

A grid is a translated copy of $\mathbb{N}^r$ in the monomial group. Dickson's lemma ensures that grid supports are well-based. Grids turn transseries algebra into finite-dimensional exponent bookkeeping and are especially useful for symbolic computation.

Part III

Constructing logarithmic-exponential transseries

Chapter 7

The log-free hierarchy

7.1 Stage zero: real powers of x

We now choose a particular monomial group. At height zero, take

$$\mathfrak{M}_0 = \{x^a : a \in \mathbb{R}\},$$

with multiplication $x^a x^b = x^{a+b}$ and order

$$x^a \succ x^b \iff a > b.$$

Let

$$\mathbb{T}_0 = \mathbb{R}((\mathfrak{M}_0))$$

be the corresponding Hahn field, or its grid-based subfield if one wants finite ratio sets.

Elements of $\mathbb{T}_0$ are generalized power series such as

$$7x^{\sqrt{2}} - 3x^{1/3} + 5 + \sum_{n \geq 1} a_n x^{-n/2}.$$

The exponents need not be integers, but the support must be well-based.

Example 7.1. The series

$$\sum_{m,n \geq 1} x^{-m-n\sqrt{2}}$$

is legal and grid-based, generated by x^{-1} and $x^{-\sqrt{2}}$. Its coefficients need not have a simple closed form, but its support is contained in a finitely generated grid.

7.2 Stage one: exponentials of height-zero giants

A purely large $L \in \mathbb{T}_0$ may serve as an exponential exponent. We introduce new monomials

$$x^a e^L, \quad a \in \mathbb{R}, \quad L \in \mathbb{T}_0 \text{ purely large.}$$

They multiply by adding exponents:

$$(x^a e^L)(x^b e^K) = x^{a+b} e^{L+K}.$$

Their order is lexicographic:

$$x^a e^L \succ x^b e^K$$

if $L > K$, or if $L = K$ and $a > b$.

Why compare L and K first? Because

$$\frac{x^a e^L}{x^b e^K} = x^{a-b} e^{L-K},$$

and a nonzero purely large leading part of $L - K$ beats every power of x .

Let $\mathfrak{M}_1$ be this enlarged monomial group and

$$\mathbb{T}_1 = \mathbb{R}((\mathfrak{M}_1)).$$

Examples include

$$e^{-x^2+3x}, \quad x^{7/3} e^{2x}, \quad x^4 + e^{-x^{3/2}}, \quad \sum_{n \geq 0} x^{-n} e^{-x}.$$

7.3 Higher stages

Repeat the construction. Assuming $\mathbb{T}_n$ has been defined, set

$$\mathfrak{M}_{n+1} = \{x^a e^L : a \in \mathbb{R}, L \in \mathbb{T}_n \text{ purely large}\}, \quad (7.1)$$

with the same multiplication and lexicographic order, and let

$$\mathbb{T}_{n+1} = \mathbb{R}((\mathfrak{M}_{n+1})).$$

The inclusions

$$\mathfrak{M}_0 \subseteq \mathfrak{M}_1 \subseteq \mathfrak{M}_2 \subseteq \cdots, \quad \mathbb{T}_0 \subseteq \mathbb{T}_1 \subseteq \mathbb{T}_2 \subseteq \cdots$$

are natural because $x^a = x^a e^0$.

Definition 7.2 (Exponential height). A log-free transseries has height at most n if it lies in $\mathbb{T}_n$. Its exact height is the smallest such n .

Height counts nested exponential complexity, not the number of exponential symbols after simplification.

Expression	Typical height
$x^{\sqrt{2}} + x^{-1}$	0
$e^{-x^2} + x^3 e^x$	1
e^{-e^x}	2
$\exp(-\exp(e^x + x))$	3

The log-free field is the union

$$\mathbb{T}_{\text{lf}} = \bigcup_{n \geq 0} \mathbb{T}_n.$$

Every individual element has finite height, although the field as a whole has no maximum height.

7.4 Why the exponent is required to be purely large

Suppose an exponent A has the canonical decomposition

$$A = L + c + U,$$

with L purely large, $c \in \mathbb{R}$, and U small. Then

$$e^A = e^L e^c e^U = e^L e^c \sum_{n \geq 0} \frac{U^n}{n!}.$$

Only the purely large part creates a genuinely new monomial. The constant is a coefficient, and the small part expands as a Hahn series. Requiring L to be purely large therefore gives a normal form and prevents redundant monomials.

7.5 Height wins

A genuinely new positive exponential level outruns everything below it.

Proposition 7.3 (Height wins). *Let $L > 0$ be purely large of exact height n , and let $T \neq 0$ have height at most n . Then, for every real a ,*

$$x^a e^L \succ T, \quad x^a e^{-L} \prec T.$$

Proof sketch. Write the leading monomial of T as $x^b e^K$, where K has height at most $n - 1$ if the monomial has exact height n . The exact-height hypothesis implies that the leading term of $L - K$ is the leading term of L , hence positive. Thus $L > K$, and lexicographic ordering gives $x^a e^L \succ x^b e^K$. Replacing L by $-L$ gives the second claim. $\square$

This explains comparisons such as

$$e^{-e^x} \prec e^{-Nx} \prec x^{-M}$$

for every fixed $M, N > 0$.

7.6 Hereditary structure

A monomial $\mathfrak{m} = x^a e^L$ contains the transseries L inside its exponent, whose monomials may themselves contain exponents, and so on. A ratio set used for computation should therefore be *hereditary*: it should include enough ratios to represent all exponents occurring inside its generators.

This recursive tree is finite in depth because every standard log-free transseries has finite height. It is one reason the hierarchy is suitable for algorithms.

7.7 What is still missing

The hierarchy already contains towers of exponentials, but it does not contain $\log x$. We could try to introduce logarithms as new primitive symbols at every stage. A cleaner route is to construct all log-free objects first and then compose them with iterated logarithms.

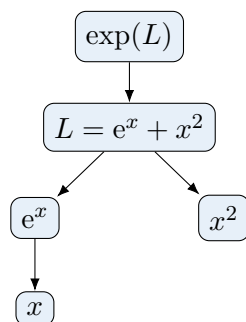


Figure 7.1: A tiny hereditary exponent tree. The construction terminates because height is finite.

Chapter summary

Starting from real powers of x , each new height adjoins exponentials of purely large transseries from the previous stage. Lexicographic ordering makes higher exponential levels dominate lower ones. Every standard transseries has finite height.

Chapter 8

Adding logarithms and defining the full transline

8.1 Iterated logarithms as changes of scale

Write

$$\log_0 x = x, \quad \log_1 x = \log x, \quad \log_2 x = \log \log x,$$

and in general

$$\log_{m+1} x = \log(\log_m x).$$

For sufficiently large real x , every fixed iterate is defined and tends to $+\infty$.

If T is log-free, substitute $\log_m x$ for x and write

$$T \circ \log_m.$$

For instance,

$$(x^a e^{-x}) \circ \log = (\log x)^a x^{-1},$$

and

$$e^{-\sqrt{x}} \circ \log_2 = \exp\left(-\sqrt{\log \log x}\right).$$

Definition 8.1 (Logarithmic depth). A transseries has depth at most m if it can be written as $T \circ \log_m$ with T log-free. Its exact depth is the least such m after identifying expressions that simplify by $\exp(\log x) = x$.

The standard logarithmic-exponential transseries field is

$$\mathbb{T} = \bigcup_{m \geq 0} \{T \circ \log_m : T \in \mathbb{T}_{\text{lf}}\}. \quad (8.1)$$

It is often called the *transline* [27, 13].

8.2 Examples of finite depth

All of the following are standard transseries:

$$\log x + \frac{1}{\log x} + \frac{1}{(\log x)^2} + \cdots,$$

$$\exp\left(\frac{x}{\log x}\right), \quad x^x = \exp(x \log x),$$

$$\exp\left(-\exp \sqrt{\log x}\right), \quad \frac{\log_3 x}{x(\log x)^2}.$$

The expression x^x is not introduced as a primitive monomial. It is normalized as $\exp(x \log x)$ and lies at finite height and depth.

8.3 Finite depth is an elementwise condition

The field contains $\log_m x$ for every fixed m , but the formal sum

$$x + \log x + \log_2 x + \log_3 x + \cdots$$

does not belong to the standard finite-depth field because one element would require unbounded logarithmic depth.

Similarly, a series whose support contains exponential towers of unbounded height is not a standard LE transseries. Larger structures—nested transseries, logarithmic hyperseries, and fields with transexponential functions—have been constructed to accommodate such behavior; see [23, 24, 21].

Common trap

“The field contains arbitrarily deep iterated logarithms” does not imply “a single series may use unbounded depth.” The first is a statement about a union of stages; the second would require a larger completion.

8.4 LE and EL conventions

The literature contains several closely related constructions.

LE-series One first builds logarithmic structure and then closes under exponentiation, or uses an equivalent staged construction.

EL-series One begins with exponential structure and then adds logarithmic changes of scale.

Grid-based versus well-based One either restricts supports to finite grids or allows arbitrary well-based supports.

Under suitable identifications, the classical constructions have large common subfields and often yield isomorphic versions of the same practical transseries field; the comparison is subtle at the level of ambient Hahn fields and chosen closures [25, 18].

In this book, “ $\mathbb{T}$ ” means a standard real LE transseries field with the usual operations. Whenever a statement depends on a particular variant, we say so.

8.5 What is not in the basic real transline

8.5.1 Persistent oscillation

Functions such as $\sin x$ and $\cos x$ do not have a total eventual dominance order: neither is eventually larger in magnitude. A real ordered field designed for nonoscillatory growth therefore does not contain them in the naive way. Complex or oscillatory transseries enlarge the monomial language, for example by using e^{ix} , but then order must be treated differently.

8.5.2 Arbitrary infinite nesting

An expression such as

$$\exp(-x + \exp(-x + \exp(-x + \cdots)))$$

requires a separate fixed-point interpretation. It is not admitted merely by writing infinitely many exponentials.

8.5.3 Uncontrolled supports

Even if every individual monomial is meaningful, their set may fail to be well-based. The sum

$$\sum_{n \geq 1} x^{-1/n}$$

remains illegal in the basic Hahn construction.

8.6 The algebraic closure properties to expect

The standard field $\mathbb{T}$ is much more than an ordered field. It supports

- real powers of positive elements;
- total exponential and logarithm on their natural ordered domains;
- differentiation and antiderivatives;
- composition $T \circ S$ when S is positive infinite;
- compositional inverses of positive infinite elements;
- solutions to broad classes of algebraic and differential equations.

Precise constructions begin in the next chapter and Part IV. Deep structural forms of these statements are developed in [27, 1].

Chapter summary

The full transline is obtained by composing finite-height log-free transseries with finitely iterated logarithms. Every element has finite height and depth. The basic real field is deliberately nonoscillatory; larger transseries and hyperseries fields relax these restrictions.

Chapter 9

The anatomy and normal forms of a transseries

9.1 Additive decomposition

For $T \in \mathbb{T}$, split its support according to comparison with 1:

$$T = T_{\text{pl}} + c + T_{\text{sm}}. \quad (9.1)$$

Here

- T_{pl} contains all monomials $\succ 1$;
- $c = [1]T \in \mathbb{R}$;
- T_{sm} contains all monomials $\prec 1$.

The decomposition is unique because the three supports are disjoint.

Example 9.1. For

$$T = e^x - 3x^2 + \log x + 7 + x^{-1} + 2e^{-x},$$

we have

$$T_{\text{pl}} = e^x - 3x^2 + \log x, \quad c = 7, \quad T_{\text{sm}} = x^{-1} + 2e^{-x}.$$

9.2 Multiplicative decomposition

Every nonzero T can be factored as

$$T = a\mathbf{m}(1 + U), \quad a = \text{lc}(T) \in \mathbb{R}^\times, \quad \mathbf{m} = \text{mag}(T), \quad U \prec 1. \quad (9.2)$$

This is the transseries analogue of scientific notation: coefficient, scale, and a relative correction.

Example 9.2.

$$T = 3x^2e^x - 5xe^x + 7 = 3x^2e^x \left(1 - \frac{5}{3x} + \frac{7}{3x^2e^x}\right).$$

The expression in parentheses is 1+small.

The additive decomposition defines exponentiation; the multiplicative decomposition defines inverse, logarithm, and real powers.

9.3 Truncation

If $\mathbf{n} \in \mathfrak{M}$ is a cutoff monomial, define the truncation

$$T|_{\succ \mathbf{n}} = \sum_{\mathbf{m} \succ \mathbf{n}} [\mathbf{m}] T \mathbf{m}.$$

Because the support is well-based, this retains an initial segment in dominance order. It may be finite or infinite, depending on the cutoff and support.

A transseries field is *truncation closed* if every such initial segment of an element is again in the field. Standard transseries fields are built to have this property. It is essential in recursive proofs and algorithms.

9.4 Asymptotic neighborhoods and formal topology

A statement

$$S - T \prec \mathbf{n}$$

means that S and T agree through every monomial at least as large as $\mathbf{n}$. This gives a non-Archimedean notion of closeness: two series are close when their first disagreement occurs very far down the support.

A sequence T_j converges coefficientwise in a fixed grid if, for every monomial, its coefficient eventually stabilizes and all supports remain in a common lower-bounded grid. For example,

$$T_N = \sum_{n=0}^N x^{-n} \longrightarrow \sum_{n=0}^{\infty} x^{-n}.$$

Numerical values need not converge for a fixed x ; this is formal convergence.

9.5 The logarithmic derivative

For nonzero T , define

$$T^\dagger = \frac{T'}{T}.$$

This operation measures relative variation. For a monomial

$$\mathbf{m} = x^a e^L,$$

we have

$$\mathbf{m}^\dagger = \frac{a}{x} + L'. \quad (9.3)$$

The logarithmic derivative converts multiplicative structure into additive structure:

$$(ST)^\dagger = S^\dagger + T^\dagger, \quad (T^b)^\dagger = bT^\dagger.$$

It will organize differentiation, integration, and comparison of rapidly varying monomials.

9.6 Dominant balance

Many equations can be attacked before any coefficient calculation by comparing leading monomials. Suppose

$$F(x, T, T', \dots) = 0.$$

One lists possible dominant monomials for each term and asks which can cancel. A term strictly larger than every other term cannot disappear, so at least two dominant contributions must have the same magnitude.

Example 9.3. For

$$y' + y = \frac{1}{x},$$

if $y \asymp ax^p$, then $y' \asymp apx^{p-1}$ is usually smaller than y . Balancing y with x^{-1} gives $p = -1$ and $a = 1$. The derivative then generates the next correction. This quickly predicts the factorial recurrence studied later.

Normal-form workflow

For any nonzero transseries T :

1. Find the leading monomial $\mathfrak{m}$ and coefficient a .
2. Factor $T = a\mathfrak{m}(1 + U)$ with $U \prec 1$.
3. Use geometric, binomial, exponential, or logarithmic power series on U .
4. Re-expand and sort monomials by dominance.
5. Truncate only after all terms capable of reaching the target order have been included.

Chapter summary

Every transseries has unique additive and multiplicative normal forms. They separate large, constant, and small parts, and isolate a leading monomial times a relative small correction. Almost every operation reduces to applying an ordinary formal power series to that correction.

Part IV

Algebra and calculus on transseries

Chapter 10

Field operations and real powers

10.1 Addition and multiplication revisited

Once all terms are normalized into monomials, addition and multiplication are exactly the Hahn-series operations from Chapter 5. The only new work is recursive normalization of expressions inside exponential and logarithmic monomials.

Example 10.1 (Multiplication by sector order). Let

$$A = 1 + a_1x^{-1} + \alpha e^{-x} + O(x^{-2}, x^{-1}e^{-x}, e^{-2x})$$

and

$$B = 1 + b_1x^{-1} + \beta e^{-x} + O(x^{-2}, x^{-1}e^{-x}, e^{-2x}).$$

Then

$$AB = 1 + (a_1 + b_1)x^{-1} + (\alpha + \beta)e^{-x} + O(x^{-2}, x^{-1}e^{-x}, e^{-2x}).$$

The omitted product $\alpha\beta e^{-2x}$ lies two exponential sectors down.

Here the notation $O(\cdots)$ is only mnemonic for a set of discarded monomials. A rigorous formal calculation instead specifies a cutoff support.

10.2 Inverse by leading-term factorization

Given

$$T = a\mathbf{m}(1 + U), \quad U \prec 1,$$

we have

$$T^{-1} = a^{-1}\mathbf{m}^{-1}(1 - U + U^2 - U^3 + \cdots).$$

Example 10.2. Let

$$T = x + \log x + e^{-x}.$$

Then

$$\begin{aligned} T^{-1} &= x^{-1} \left(1 + x^{-1} \log x + x^{-1} e^{-x} \right)^{-1} \\ &= x^{-1} - x^{-2} \log x + x^{-3} (\log x)^2 - x^{-2} e^{-x} + 2x^{-3} (\log x) e^{-x} + \cdots \end{aligned}$$

The ordering tells us that all pure inverse-power-log terms dominate the first e^{-x} term, even though we may display both for structural reasons.

10.3 Integer and rational powers

Integer powers use multiplication and inverse. For a positive transseries T and arbitrary real b , use the multiplicative normal form

$$T = a\mathfrak{m}(1 + U), \quad a > 0, U \prec 1.$$

Define

$$T^b = a^b \mathfrak{m}^b (1 + U)^b, \quad (1 + U)^b = \sum_{n=0}^{\infty} \binom{b}{n} U^n. \quad (10.1)$$

The monomial power is $\mathfrak{m}^b = \exp(b \log \mathfrak{m})$.

Example 10.3 (Square root). For

$$T = x^2 + x + e^{-x},$$

we find

$$\begin{aligned} \sqrt{T} &= x \left(1 + x^{-1} + x^{-2}e^{-x}\right)^{1/2} \\ &= x + \frac{1}{2} - \frac{1}{8x} + \frac{1}{16x^2} + \frac{1}{2}x^{-1}e^{-x} + \dots \end{aligned}$$

10.4 Formal identities are genuine identities

Because the binomial, geometric, exponential, and logarithmic families are summable on small arguments, the familiar identities hold:

$$T^a T^b = T^{a+b}, \quad (T^a)^b = T^{ab},$$

for positive T , and

$$(ST)^a = S^a T^a$$

for positive S, T . These are not numerical convergence claims; they are identities in the ordered field.

10.5 Real closedness

A field is real closed if every positive element has a square root and every odd-degree polynomial has a root, equivalently if adjoining i makes it algebraically closed. Standard fields of logarithmic-exponential transseries are real closed [27, 1].

For calculations, the practical consequence is that algebraic equations can be solved recursively by dominant balance and formal Newton expansion. The theorem is much deeper than the elementary square-root calculation above, but the local mechanism is similar: choose a leading monomial, factor it out, and solve for a small correction.

Chapter summary

Field operations reduce to Hahn-series convolution and geometric inversion. Positive transseries admit arbitrary real powers through a binomial expansion of their relative small part. Standard transseries fields are real closed.

Chapter 11

Exponential and logarithm

11.1 Exponentiating a general transseries

Take the additive decomposition

$$T = L + c + U,$$

where L is purely large, $c \in \mathbb{R}$, and U is small. Define

$$\exp T = e^L e^c \sum_{n=0}^{\infty} \frac{U^n}{n!}. \quad (11.1)$$

Each factor has a different role:

- e^L is a new transmonomial;
- e^c is an ordinary positive real coefficient;
- e^U is a legal Hahn series because $U \prec 1$.

Example 11.1. For

$$\begin{aligned} T &= x + 3 + x^{-1} + 2e^{-x}, \\ e^T &= e^3 e^x \exp(x^{-1} + 2e^{-x}) \\ &= e^3 e^x \left(1 + x^{-1} + \frac{1}{2x^2} + \cdots + 2e^{-x} + 2x^{-1}e^{-x} + \cdots \right). \end{aligned}$$

After multiplying by e^x , the term $2e^{-x}$ inside the parenthesis becomes the constant contribution $2e^3$.

11.2 Logarithm of a positive transseries

Let $T > 0$ and use the multiplicative decomposition

$$T = a \mathfrak{m}(1 + U), \quad a > 0, \quad U \prec 1.$$

Define

$$\log T = \log a + \log \mathfrak{m} + \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} U^n. \quad (11.2)$$

If $\mathfrak{m} = x^b e^L$, then

$$\log \mathfrak{m} = b \log x + L.$$

In the full transseries field this again lies in $\mathbb{T}$.

Example 11.2.

$$\begin{aligned} \log \left(x^3 e^{2x} \left(1 + \frac{\log x}{x} + e^{-x} \right) \right) &= 2x + 3 \log x \\ &+ \frac{\log x}{x} + e^{-x} - \frac{1}{2} \left(\frac{\log x}{x} + e^{-x} \right)^2 + \dots \end{aligned}$$

11.3 Exponential and logarithm are inverses

The formal power-series identities

$$\log(e^U) = U, \quad e^{\log(1+U)} = 1 + U$$

hold for small U . The large and constant factors in the normal forms are handled exactly, so

$$\log(\exp T) = T$$

for every $T \in \mathbb{T}$, and

$$\exp(\log S) = S$$

for every positive $S \in \mathbb{T}$.

Thus $\exp : \mathbb{T} \rightarrow \mathbb{T}_{>0}$ is an order-preserving group isomorphism from addition to multiplication.

11.4 Some instructive expansions

11.4.1 A small exponential

$$\exp(x^{-1}) = 1 + x^{-1} + \frac{1}{2}x^{-2} + \frac{1}{6}x^{-3} + \dots$$

11.4.2 A logarithm with a large leading term

$$\begin{aligned} \log(x + \log x) &= \log x + \log \left(1 + \frac{\log x}{x} \right) \\ &= \log x + \frac{\log x}{x} - \frac{(\log x)^2}{2x^2} + \frac{(\log x)^3}{3x^3} - \dots \end{aligned}$$

11.4.3 An exponential of mixed scales

$$\exp \left(e^{-x} + x^{-1} e^{-x} \right) = 1 + (1 + x^{-1})e^{-x} + \frac{1}{2}(1 + 2x^{-1} + x^{-2})e^{-2x} + \dots$$

11.5 Why positivity is necessary for the real logarithm

The ordered real field has no real logarithm of a negative element. A complex transseries field can define branches of $\log T$ after choosing arguments, but branch information then becomes part of the structure. In this tutorial, $\log T$ always assumes $T > 0$ unless explicitly stated otherwise.

11.6 Exponentiation changes height by at most one

If T has height n , then e^T has height at most $n + 1$. It has no greater height when the purely large part of T is absent, because then e^T is only a power series in a small transseries times a real constant.

This illustrates a recurring principle: the large part controls genuinely new asymptotic scales; the small part controls formal corrections on existing scales.

Chapter summary

To exponentiate, split a transseries additively into purely large, constant, and small parts. To take a logarithm, factor a positive transseries into coefficient, leading monomial, and $1 + \text{small}$. The ordinary power-series formulas then become exact Hahn-field identities.

Chapter 12

Differentiation

12.1 Recursive definition

Start with

$$c' = 0, \quad x' = 1, \quad (x^a)' = ax^{a-1}.$$

For a transmonomial $\mathbf{m} = x^a e^L$, define

$$\mathbf{m}' = \left(\frac{a}{x} + L' \right) \mathbf{m}. \quad (12.1)$$

Then extend termwise:

$$\left(\sum_{\mathbf{m}} a_{\mathbf{m}} \mathbf{m} \right)' = \sum_{\mathbf{m}} a_{\mathbf{m}} \mathbf{m}'. \quad (12.2)$$

A support theorem, proved recursively on height, guarantees that the family on the right is summable [13, 27].

12.2 Examples

$$\begin{aligned} (x^{\sqrt{2}} e^{-x^2})' &= \left(\frac{\sqrt{2}}{x} - 2x \right) x^{\sqrt{2}} e^{-x^2}, \\ (e^{-e^x})' &= -e^x e^{-e^x}, \\ (\log_3 x)' &= \frac{1}{x \log x \log_2 x}, \\ (x^x)' &= x^x (\log x + 1). \end{aligned}$$

The last formula follows from $x^x = \exp(x \log x)$, not from treating x as a constant exponent.

12.3 The usual differential rules

The recursive definition satisfies

$$\begin{aligned} (S + T)' &= S' + T', \\ (ST)' &= S'T + ST', \end{aligned}$$

$$(T^a)' = aT^{a-1}T' \quad (T > 0),$$

$$(e^T)' = T'e^T, \quad (\log T)' = \frac{T'}{T} \quad (T > 0).$$

These identities can be checked first on monomials and small power-series substitutions, then extended by summability.

12.4 Logarithmic derivatives

The formula

$$T^\dagger = \frac{T'}{T}$$

is especially efficient for monomials and products. For example,

$$\mathfrak{m} = x^a (\log x)^b \exp\left(cx + \frac{x}{\log x}\right)$$

has

$$\begin{aligned} \mathfrak{m}^\dagger &= \frac{a}{x} + \frac{b}{x \log x} + c + \left(\frac{x}{\log x}\right)' \\ &= c + \frac{1}{\log x} - \frac{1}{(\log x)^2} + \frac{a}{x} + \frac{b}{x \log x}. \end{aligned}$$

Therefore $\mathfrak{m}' = \mathfrak{m}\mathfrak{m}^\dagger$ without expanding the large product.

12.5 Differentiating an asymptotic series

Formal differentiation is always defined in $\mathbb{T}$, but analytic termwise differentiation of an asymptotic expansion needs hypotheses. A function can be small while its derivative oscillates wildly. The nonoscillatory Hardy-field setting avoids many such pathologies, and transseries differentiation is designed to model it.

Common trap

Do not infer an analytic derivative expansion merely from a pointwise asymptotic expansion unless the function class controls derivatives. In the formal field, differentiation is algebraically valid; analytic realization is a separate theorem.

12.6 Dominance and l'Hôpital behavior

In a Hardy-type transseries field, if S, T are nonconstant and sufficiently well behaved, comparison of S and T is reflected by comparison of their derivatives. This is the algebraic shadow of l'Hôpital's rule and is encoded in the theory of H -fields.

For hand calculations, a safer local rule is:

1. differentiate each relevant monomial using its logarithmic derivative;
2. compare the resulting products;
3. only then select the leading derivative term.

The derivative of the leading term usually dominates, but constants disappear and special cancellations can occur.

12.7 Example: the factorial recurrence appears

Assume

$$y = \sum_{n=0}^{\infty} a_n x^{-n-1}.$$

Then

$$y' = \sum_{n=0}^{\infty} -(n+1)a_n x^{-n-2}.$$

Substitution into $y' + y = x^{-1}$ gives

$$a_0 = 1, \quad a_n = na_{n-1} \quad (n \geq 1),$$

so $a_n = n!$. A perfectly routine formal derivative thus exposes the divergence mechanism.

Chapter summary

Differentiation is defined recursively on monomials and extended termwise. Logarithmic derivatives turn nested products and exponentials into manageable sums. Formal differentiation is unconditional inside $\mathbb{T}$; termwise differentiation of actual asymptotic functions requires analytic control.

Chapter 13

Integration and asymptotic antidifferentiation

13.1 Elementary antiderivatives

The familiar rules remain valid:

$$\int x^a dx = \frac{x^{a+1}}{a+1} \quad (a \neq -1), \quad \int \frac{dx}{x} = \log x,$$

$$\int \frac{dx}{x \log x} = \log_2 x, \quad \int \frac{dx}{x \log x \cdots \log_k x} = \log_{k+1} x.$$

The presence of all finite iterated logarithms removes the elementary “exceptional exponents” that would obstruct integration in a smaller power field.

13.2 Why term-by-term monomial integration is not enough

The antiderivative of e^{x^2} is not an elementary monomial. Nevertheless it has a transseries expansion. Let

$$\mathfrak{m} = e^L, \quad q = \mathfrak{m}^\dagger = L'.$$

Seek an antiderivative in the form $F = \mathfrak{m}h$. Then

$$F' = \mathfrak{m}(h' + qh).$$

Thus $F' = \mathfrak{m}$ is equivalent to

$$h' + qh = 1. \tag{13.1}$$

If differentiation of h is smaller than multiplication by q , solve recursively:

$$h_0 = q^{-1}, \quad h_{n+1} = -q^{-1}h'_n. \tag{13.2}$$

Then

$$h \sim_{\text{as}} h_0 + h_1 + h_2 + \cdots.$$

Indeed,

$$qh_0 = 1, \quad h'_n + qh_{n+1} = 0,$$

so successive residuals cancel.

13.3 Example: $\int e^{x^2} dx$

Here $q = 2x$. The recursion gives

$$h_0 = \frac{1}{2x}, \quad h_1 = \frac{1}{4x^3}, \quad h_2 = \frac{3}{8x^5}, \quad h_3 = \frac{15}{16x^7}, \dots$$

Therefore

$$\int e^{x^2} dx \sim_{\text{as}} e^{x^2} \left(\frac{1}{2x} + \frac{1}{4x^3} + \frac{3}{8x^5} + \frac{15}{16x^7} + \dots \right) + C. \quad (13.3)$$

The general coefficient is

$$\frac{(2n-1)!!}{2^{n+1}x^{2n+1}}.$$

Proof. Let

$$F_N = e^{x^2} \sum_{n=0}^{N-1} \frac{(2n-1)!!}{2^{n+1}x^{2n+1}}, \quad (-1)!! = 1.$$

Differentiating shows telescopic cancellation of all retained inverse powers, leaving

$$F'_N = e^{x^2} \left(1 - \frac{(2N-1)!!}{2^N x^{2N}} \right).$$

Hence each added term improves the relative error by x^{-2} . □

13.4 Example: a decaying Gaussian tail

For $\mathfrak{m} = e^{-x^2}$, $q = -2x$. A convenient antiderivative tending to zero is

$$\int_{\infty}^x e^{-t^2} dt \sim_{\text{as}} -\frac{e^{-x^2}}{2x} \left(1 - \frac{1}{2x^2} + \frac{3}{4x^4} - \frac{15}{8x^6} + \dots \right).$$

The alternating signs arise from the negative logarithmic derivative.

13.5 When the first recursion is not the right one

For $\mathfrak{m} = x^{-1}$, the logarithmic derivative is $q = -x^{-1}$, and the naive choice $q^{-1} = -x$ does not yield a decreasing correction scheme. This signals a change of asymptotic scale rather than failure of integration: the exact antiderivative is $\log x$.

General integration algorithms classify the leading monomial and choose a suitable asymptotic integrator. The full theorem is that the standard field $\mathbb{T}$ is closed under antiderivatives. More strongly, it is Liouville closed: every element has an antiderivative, and every nonzero element has a logarithmic antiderivative in the appropriate sense [27, 1].

13.6 Uniqueness after fixing the constant term

If $F' = T$ and $G' = T$, then $(F - G)' = 0$. The constant field of $\mathbb{T}$ is $\mathbb{R}$, so $F - G \in \mathbb{R}$. Requiring $[1]F = 0$ selects a unique formal antiderivative.

Asymptotic integration of a rapidly varying monomial

For $\mathfrak{m}$ with logarithmic derivative $q = \mathfrak{m}^\dagger$:

1. Try $F = \mathfrak{m}h$.
2. Solve $h' + qh = 1$ by $h_0 = q^{-1}$ and $h_{n+1} = -q^{-1}h'_n$.
3. Check that $h_{n+1} \prec h_n$.
4. If the sequence does not decrease, change scale; logarithms are the most common missing scale.

Chapter summary

Integration is not merely termwise division of monomials. For rapidly varying terms, factor out the monomial and solve a first-order correction equation. For critical power-log terms, new logarithmic scales appear. The full transseries field is closed under antiderivatives.

Chapter 14

Composition, Taylor expansion, and compositional inverse

14.1 Why the inner argument must be large

For ordinary functions, $T \circ S$ means “replace x in T by S .” The same idea works formally when S is positive and infinitely large:

$$S > 0, \quad S \succ 1.$$

This condition preserves the asymptotic direction $x \rightarrow +\infty$.

If

$$T = \sum_{n \geq 0} x^{-n}$$

and we substitute the small series $S = x^{-1}$, we obtain

$$\sum_{n \geq 0} x^n,$$

which is not well-based at infinity. Thus arbitrary right composition cannot be allowed.

14.2 Recursive definition

For a monomial $\mathfrak{m} = x^a e^L$, define

$$\mathfrak{m} \circ S = S^a \exp(L \circ S). \tag{14.1}$$

Then for

$$T = \sum_{\mathfrak{m}} c_{\mathfrak{m}} \mathfrak{m},$$

define

$$T \circ S = \sum_{\mathfrak{m}} c_{\mathfrak{m}} (\mathfrak{m} \circ S). \tag{14.2}$$

The difficult part is proving that the family on the right remains summable. The proof proceeds by induction on the height and depth of T ; it is one of the technically subtler foundations of the subject [14, 27, 3].

14.3 Elementary examples

$$\begin{aligned}
(x^3 + \log x) \circ e^x &= e^{3x} + x, \\
e^{-x} \circ (x + \log x) &= e^{-x} x^{-1}, \\
\log \circ (x + e^{-x}) &= \log x + \log(1 + x^{-1} e^{-x}) \\
&= \log x + x^{-1} e^{-x} - \frac{1}{2} x^{-2} e^{-2x} + \dots.
\end{aligned}$$

Composition respects the field operations:

$$\begin{aligned}
(A + B) \circ S &= A \circ S + B \circ S, \\
(AB) \circ S &= (A \circ S)(B \circ S), \\
\exp(A) \circ S &= \exp(A \circ S), \quad \log(A) \circ S = \log(A \circ S)
\end{aligned}$$

when the logarithms are defined.

14.4 The chain rule

The formal derivative satisfies

$$(T \circ S)' = (T' \circ S)S'. \quad (14.3)$$

It is proved first for monomials using (14.1), then extended to summable families.

14.5 Taylor expansion

For a familiar analytic function,

$$T(S + U) = \sum_{n \geq 0} \frac{T^{(n)}(S)}{n!} U^n.$$

In transseries the same formula holds when U is sufficiently small relative to S and to the internal scales of T :

$$T \circ (S + U) = \sum_{n=0}^{\infty} \frac{T^{(n)} \circ S}{n!} U^n. \quad (14.4)$$

The phrase “sufficiently small” is important. For a fixed grid-based T and S , one can formulate it using a finite witness set of monomial inequalities [14].

For elementary outer functions, the condition is transparent.

Example 14.1 (Logarithm). If $U \prec S$ and $S > 0$, then

$$\begin{aligned}
\log(S + U) &= \log S + \log(1 + U/S) \\
&= \log S + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} \left(\frac{U}{S}\right)^n.
\end{aligned}$$

This is exactly the Taylor series because $\log^{(n)} S = (-1)^{n-1} (n-1)! S^{-n}$.

Example 14.2 (Exponential). For any U small enough that (U^n) is summable,

$$\exp(S + U) = \exp S \sum_{n \geq 0} \frac{U^n}{n!}.$$

14.6 Positive infinite transseries form a composition group

Let

$$\mathcal{P} = \{T \in \mathbb{T} : T > 0, T \succ 1\}.$$

Then x is the identity element under composition, and every $T \in \mathcal{P}$ has a unique compositional inverse $T^{[-1]} \in \mathcal{P}$ satisfying

$$T \circ T^{[-1]} = x, \quad T^{[-1]} \circ T = x.$$

We use square brackets to distinguish compositional powers from ordinary powers:

$$T^{[2]} = T \circ T, \quad T^{[-1]} = \text{compositional inverse}.$$

14.7 How to calculate an inverse

Suppose

$$T = x + A, \quad A \prec x.$$

Seek $S = x + B$ with $B \prec x$. The equation $T \circ S = x$ becomes

$$B + A \circ (x + B) = 0. \tag{14.5}$$

This is a fixed-point equation

$$B = -A \circ (x + B).$$

Repeated substitution usually determines successively smaller terms.

Alternatively, use Newton iteration for composition:

$$S_{n+1} = S_n - \frac{T \circ S_n - x}{T' \circ S_n}. \tag{14.6}$$

When the initial leading approximation is correct, the error improves rapidly in the transseries valuation.

Example 14.3 (Inverse of $x + \log x$). Let $T = x + \log x$. Write

$$S = x - \log x + R, \quad R \prec \log x.$$

Substitution gives

$$\begin{aligned} T \circ S &= x - \log x + R + \log(x - \log x + R) \\ &= x + R - \frac{\log x - R}{x} - \frac{(\log x - R)^2}{2x^2} - \dots \end{aligned}$$

Solving recursively yields

$$T^{[-1]} = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2 - 2 \log x}{2x^2} + \dots$$

This is the same large- x pattern that appears in the Lambert W function.

14.8 Fractional iteration

Composition suggests the Abel equation

$$V \circ T = V + 1.$$

If a suitable V is found, define a fractional iterate by

$$T^{[s]} = V^{[-1]} \circ (V + s), \quad s \in \mathbb{R}.$$

Then

$$T^{[s+t]} = T^{[s]} \circ T^{[t]}.$$

The existence and uniqueness of such constructions depend on the leading form of T and on normalization choices. Transseries provide a natural language for many cases, including shallow, moderate, and deep iteration regimes studied by Edgar [15].

Chapter summary

Right composition is defined for positive infinite inner arguments and requires a nontrivial support theorem. It obeys the chain rule and suitable Taylor expansions. Positive infinite transseries form a group under composition, so formal functional inversion and fractional iteration become algebraic problems.

Chapter 15

Contractive maps and recursive equations

15.1 The non-Archimedean contraction principle

In ordinary analysis, a contraction makes numerical distances smaller. In a transseries field, a map is contractive when it pushes the leading error to a strictly smaller monomial.

Definition 15.1 (Formal contraction). On a set of transseries, a map $\mathcal{F}$ is contractive if

$$Y \neq Z \implies \mathcal{F}(Y) - \mathcal{F}(Z) \prec Y - Z.$$

Under an appropriate completeness condition on a fixed support class, a contractive map has a unique fixed point.

Theorem 15.2 (Grid-based fixed-point principle). *Let $\mathcal{A}$ be a closed affine set of grid-based transseries supported in a common grid, and suppose $\mathcal{F} : \mathcal{A} \rightarrow \mathcal{A}$ is uniformly contractive in the sense that each application lowers errors by monomials from a fixed finite ratio set. Then the iteration*

$$Y_{n+1} = \mathcal{F}(Y_n)$$

converges coefficientwise to the unique fixed point $Y = \mathcal{F}(Y)$.

Idea. Let $E_n = Y_{n+1} - Y_n$. Contractivity gives

$$E_{n+1} \prec E_n.$$

The uniform grid condition implies that, for each fixed monomial, only finitely many E_n can contribute to its coefficient. Hence

$$Y_0 + \sum_{n \geq 0} E_n$$

is a summable family and defines a limit Y . Continuity of $\mathcal{F}$ in this formal topology gives $Y = \mathcal{F}(Y)$. If Y and Z were two fixed points, then

$$Y - Z = \mathcal{F}(Y) - \mathcal{F}(Z) \prec Y - Z,$$

a contradiction. □

Costin gives a broad topological version of this principle for abstract multiseries [8]; Edgar develops a grid-based formulation for explicit computations [13].

15.2 Example: $Y = x + e^{-Y}$

Let

$$\mathcal{F}(Y) = x + e^{-Y}.$$

Start with $Y_0 = x$. Then

$$\begin{aligned} Y_1 &= x + e^{-x}, \\ Y_2 &= x + e^{-x-e^{-x}} = x + e^{-x} - e^{-2x} + \frac{1}{2}e^{-3x} + \dots \end{aligned}$$

Each iteration fixes more exponential coefficients. For two candidates $Y, Z = x + O(e^{-x})$,

$$e^{-Y} - e^{-Z} = -e^{-Z}(Y - Z) + \text{smaller terms},$$

so the error is multiplied by e^{-x} and becomes strictly smaller. The map is contractive on the appropriate affine grid.

15.3 Example: recursive logarithmic correction

Consider

$$Y = x - \log Y.$$

The first approximation is $Y_0 = x$, then

$$\begin{aligned} Y_1 &= x - \log x, \\ Y_2 &= x - \log(x - \log x) = x - \log x + \frac{\log x}{x} + \frac{(\log x)^2}{2x^2} + \dots \end{aligned}$$

The derivative of the right-hand side with respect to Y is $-1/Y \prec 1$, so corrections contract by roughly x^{-1} .

15.4 Formal implicit-function method

Suppose

$$F(x, Y) = 0$$

and a leading approximation Y_0 has been found. Write $Y = Y_0 + U$ and expand

$$0 = F(x, Y_0) + F_Y(x, Y_0)U + \frac{1}{2}F_{YY}(x, Y_0)U^2 + \dots$$

If $F_Y(x, Y_0)$ is invertible and the nonlinear remainder is smaller, solve

$$U = -F_Y(x, Y_0)^{-1}F(x, Y_0) + \text{smaller corrections}.$$

This is Newton's method in the transseries valuation.

Solving an implicit equation

1. Use dominant balance to guess the leading monomial of Y .
2. Substitute $Y = Y_0(1 + U)$ or $Y = Y_0 + U$ with U relatively small.

3. Expand all analytic operations in U .
4. Solve the largest uncanceled coefficient equation.
5. Repeat; well-basedness ensures that “next largest” is meaningful.

15.5 Differential equations and free exponential parameters

For an ODE

$$P(x, Y, Y', \dots, Y^{(r)}) = 0,$$

one first computes a base formal sector Φ_0 . Linearizing around it gives a linear differential operator $\mathcal{L}$. Exponential monomials in the kernel of $\mathcal{L}$ indicate missing free parameters.

For a first-order equation, if the linearized perturbation η satisfies

$$\eta' = Q(x)\eta,$$

then

$$\eta = C \exp \left(\int Q(x) dx \right).$$

This monomial becomes the ratio between transseries sectors. Nonlinear terms then force powers of it and couple the sector coefficients recursively.

15.6 Dominant balance and Newton polygons

When several monomials involving Y and its derivatives compete, one can plot orders and find edges whose terms can balance. This is the differential analogue of a Newton polygon for algebraic or Puiseux series. In more advanced transseries algorithms, dominant balances determine candidate leading monomials, while contractive maps construct the corrections.

Chapter summary

A formal contraction lowers the leading error rather than a numerical norm. Uniform contraction in a common grid gives existence and uniqueness by coefficientwise stabilization. This is the basic engine behind implicit solutions, compositional inverses, and many differential-equation transseries.

Chapter 16

How to compute without drowning in terms

16.1 Always choose the scale first

Before expanding, decide which monomials matter. A statement such as

“through x^{-4} in sectors $k = 0, 1$, and through the leading term in $k = 2$ ”

is much safer than “expand to order 4.” There is no universal total degree that respects exponential dominance.

16.2 A finite ratio-set representation

A practical grid-based object may be stored as

$$T = \mathfrak{m}_0 \sum_{k \in E} c_k \boldsymbol{\mu}^k,$$

where

- $\mathfrak{m}_0$ is a base monomial;
- $\boldsymbol{\mu}$ is a finite tuple of small generators;
- $E \subseteq \mathbb{N}^r$ is represented lazily or finitely up to a cutoff;
- coefficients may themselves be exact rational, algebraic, or symbolic expressions.

Recursive exponent trees encode monomials such as $\exp(-e^x + x^2)$, and a comparison routine recursively compares the purely large exponents before power factors.

16.3 Truncated arithmetic

Let τ be the smallest retained monomial. Define a projection

$$\pi_\tau(T) = T|_{\succ \tau}.$$

For addition,

$$\pi_\tau(S + T) = \pi_\tau(\pi_\tau S + \pi_\tau T).$$

For multiplication, one must retain every pair of input terms whose product can reach τ . Since supports are well-based, only finitely many such pairs occur in a fixed grid.

For inverse, exponential, or logarithm, determine how many powers of the small remainder U can contribute above τ . If $\text{mag}(U) = \mu \prec 1$, stop once

$$\mu^n \prec \frac{\tau}{\text{leading factor}}.$$

16.4 A generic algorithm

Truncated unary operation $F(T)$

1. Normalize T into the additive or multiplicative form required by F .
2. Separate the exact large/monomial factor from a small remainder U .
3. Determine a finite N such that powers U^n for $n > N$ lie below the target monomial.
4. Evaluate the ordinary formal series of F through degree N .
5. Multiply by the exact factor, combine equal monomials, sort by dominance, and discard terms below the target.
6. Verify the result by substitution or differentiation to one order beyond the requested cutoff whenever possible.

16.5 Verification by residual

The most reliable check is to substitute the truncated answer back into the original equation. If Y_N is claimed correct through τ , compute the residual

$$R_N = F(x, Y_N, Y'_N, \dots).$$

The leading monomial of R_N should be strictly below the accuracy implied by τ . This catches sign errors, missing cross-sector products, and incorrect logarithmic powers.

16.6 Common bookkeeping errors

Discarding too early A term small inside an exponent may become important after multiplication by a large outside monomial.

Using ordinary degree x^{-100} is still much larger than e^{-x} . A degree that counts both as one step destroys the dominance order.

Forgetting generator closure Differentiation introduces logarithmic derivatives; composition introduces substituted generators; inversion may require a new small ratio. The ratio set must be enlarged.

Ignoring resonance Two distinct multi-indices may yield the same monomial. Their coefficients must be combined, and logarithmic sectors may be forced in differential equations.

Confusing formal and numerical convergence A formally summable family need not converge when a finite real x is substituted. Conversely, an analytic function may have a divergent but useful formal transseries.

16.7 Computer algebra

Van der Hoeven developed algorithms for generalized series and transseries that perform comparison, arithmetic, composition, differential operations, and asymptotic solving [27, 28]. A persistent challenge is zero testing: after recursive operations, deciding whether a symbolic transseries expression is identically zero may be nontrivial. Recent work gives a zero test for an important class of differentially algebraic transseries [5].

Chapter summary

Effective transseries computation is target-driven. Choose a monomial cutoff, use finite ratio sets, expand only powers capable of reaching that cutoff, and verify by residual. The actual dominance order, not a guessed total degree, must guide truncation.

Part V

Worked transseries calculations

Chapter 17

An implicit equation with an exponentially small correction

17.1 The problem

Solve, for large positive x ,

$$y = x + e^{-y}. \quad (17.1)$$

A power-series ansatz in x^{-1} finds only $y \sim_{\text{as}} x$, because the correction is beyond all powers. We therefore set

$$y = x + \delta, \quad \delta \prec 1.$$

Then

$$\delta = e^{-x} e^{-\delta}. \quad (17.2)$$

The natural small ratio is $z = e^{-x}$.

17.2 Direct coefficient recursion

Assume

$$\delta = c_1 z + c_2 z^2 + c_3 z^3 + \cdots.$$

Expand the right side:

$$ze^{-\delta} = z \left(1 - \delta + \frac{\delta^2}{2} - \frac{\delta^3}{6} + \cdots \right).$$

Matching powers of z gives

$$c_1 = 1, \quad c_2 = -c_1 = -1,$$

$$c_3 = -c_2 + \frac{1}{2}c_1^2 = \frac{3}{2},$$

$$c_4 = -c_3 + c_1 c_2 - \frac{1}{6}c_1^3 = -\frac{8}{3}.$$

Continuing,

$$y = x + e^{-x} - e^{-2x} + \frac{3}{2}e^{-3x} - \frac{8}{3}e^{-4x} + \frac{125}{24}e^{-5x} - \frac{54}{5}e^{-6x} + \cdots. \quad (17.3)$$

17.3 Closed coefficient formula from Lambert W

Equation (17.2) is equivalent to

$$\delta e^\delta = e^{-x}.$$

By definition of the Lambert W function,

$$\delta = W(e^{-x}).$$

The Lagrange inversion formula gives

$$W(z) = \sum_{n=1}^{\infty} \frac{(-n)^{n-1}}{n!} z^n, \quad |z| < e^{-1}. \quad (17.4)$$

Substituting $z = e^{-x}$ reproduces (17.3). For $x > 1$, this particular transseries actually converges.

Derivation by Lagrange inversion. Write $\delta = z\phi(\delta)$ with $\phi(u) = e^{-u}$. Lagrange inversion says

$$[z^n]\delta(z) = \frac{1}{n}[u^{n-1}]\phi(u)^n = \frac{1}{n}[u^{n-1}]e^{-nu}.$$

Since

$$[u^{n-1}]e^{-nu} = \frac{(-n)^{n-1}}{(n-1)!},$$

the coefficient is $(-n)^{n-1}/n!$. □

17.4 Why a contraction proves uniqueness

On the affine set $x + e^{-x}\mathbb{R}[[e^{-x}]]$, define

$$\mathcal{F}(Y) = x + e^{-Y}.$$

If $Y - Z$ begins at e^{-kx} , then

$$\mathcal{F}(Y) - \mathcal{F}(Z) = -e^{-Z}(Y - Z) + \text{smaller terms}$$

begins at $e^{-(k+1)x}$. Thus the map pushes every error one full sector down. The fixed point is unique in this support class.

17.5 Residual check

Let

$$y_N = x + \sum_{n=1}^N c_n e^{-nx}.$$

If the coefficients are correct through N , then

$$y_N - x - e^{-y_N} = O(e^{-(N+1)x}).$$

This check is easy to automate and is more reliable than inspecting individual coefficient formulas.

Big idea

A correction can be invisible to every inverse power and still possess a perfectly ordinary convergent series in a different small monomial. The right question is not “does a power series exist?” but “which asymptotic ratio generates the correction?”

Chapter 18

Lambert W at infinity: logarithms inside powers

18.1 The equation

Consider

$$y + \log y = x, \quad y > 0. \quad (18.1)$$

Exponentiating gives

$$ye^y = e^x,$$

so

$$y = W(e^x).$$

The dominant balance is $y \sim x$, but the logarithm contributes a correction of size $\log x$.

18.2 Successive approximation

Start with $y_0 = x$. Then

$$y = x - \log y$$

suggests

$$y_1 = x - \log x.$$

Substitute again:

$$\begin{aligned} y_2 &= x - \log(x - \log x) \\ &= x - \log x - \log\left(1 - \frac{\log x}{x}\right) \\ &= x - \log x + \frac{\log x}{x} + \frac{(\log x)^2}{2x^2} + \cdots. \end{aligned}$$

This is not yet the final coefficient at x^{-2} because the x^{-1} correction feeds back into the logarithm. Systematic recursion resolves it.

18.3 Polynomial recursion

Let $L = \log x$ and seek

$$y = x - L + \sum_{n=1}^{\infty} \frac{P_n(L)}{x^n}, \quad (18.2)$$

where P_n is a polynomial. Put $t = x^{-1}$ and

$$U(t) = -Lt + \sum_{n \geq 1} P_n(L)t^{n+1}.$$

Since $y = x(1 + U)$, equation (18.1) becomes

$$\sum_{n \geq 1} P_n(L)t^n + \log(1 + U(t)) = 0.$$

Therefore

$$P_n(L) = -[t^n] \log \left(1 - Lt + \sum_{j=1}^{n-1} P_j(L)t^{j+1} \right). \quad (18.3)$$

The coefficient P_n depends only on earlier polynomials.

The first six are

$$\begin{aligned} P_1(L) &= L, \\ P_2(L) &= \frac{L^2}{2} - L, \\ P_3(L) &= \frac{L^3}{3} - \frac{3L^2}{2} + L, \\ P_4(L) &= \frac{L^4}{4} - \frac{11L^3}{6} + 3L^2 - L, \\ P_5(L) &= \frac{L^5}{5} - \frac{25L^4}{12} + \frac{35L^3}{6} - 5L^2 + L, \\ P_6(L) &= \frac{L^6}{6} - \frac{137L^5}{60} + \frac{75L^4}{8} - \frac{85L^3}{6} + \frac{15L^2}{2} - L. \end{aligned}$$

Hence

$$\begin{aligned} W(e^x) &= x - L + \frac{L}{x} + \frac{L(L-2)}{2x^2} + \frac{L(2L^2-9L+6)}{6x^3} \\ &\quad + \frac{L(3L^3-22L^2+36L-12)}{12x^4} + O\left(\frac{L^5}{x^5}\right). \end{aligned} \quad (18.4)$$

18.4 Closed formula with Stirling numbers

Let $\left[\begin{smallmatrix} n \\ k \end{smallmatrix} \right]$ denote the unsigned Stirling number of the first kind. The classical large-argument expansion can be written

$$W(z) = L_1 - L_2 + \sum_{k=0}^{\infty} \sum_{m=1}^{\infty} \frac{(-1)^k}{m!} \left[\begin{smallmatrix} k+m \\ k+1 \end{smallmatrix} \right] \frac{L_2^m}{L_1^{k+m}}, \quad (18.5)$$

where $L_1 = \log z$ and $L_2 = \log \log z$. Setting $z = e^x$ gives $L_1 = x$ and $L_2 = \log x$.

Formula (18.5) is an excellent example of a log-power transseries: the coefficient at each power x^{-n} is a polynomial in $\log x$.

18.5 Compositional interpretation

The map

$$T(y) = y + \log y$$

is positive infinite and has a compositional inverse in $\mathbb{T}$. The expansion above is precisely $T^{[-1]}(x)$. Thus the Lambert W asymptotic is not an isolated trick; it is an instance of the general composition-group theorem.

18.6 A useful accuracy estimate

Since $L/x \rightarrow 0$, each successive displayed term is smaller than the preceding one in the transseries order:

$$x \succ L \succ \frac{L}{x} \succ \frac{L^2}{x^2} \succ \dots$$

For fixed truncation order N , substitution into $y + \log y - x$ leaves a residual of order

$$O\left(\frac{(\log x)^{N+1}}{x^{N+1}}\right).$$

Chapter summary

The large Lambert W expansion illustrates how logarithms become polynomial coefficients of inverse powers. A coefficient-extraction recursion determines each polynomial, while Stirling numbers give a closed double sum. Formally, the calculation is compositional inversion in $\mathbb{T}$.

Chapter 19

The Euler differential equation and a hidden constant

19.1 Formal solution

Consider

$$y' + y = \frac{1}{x}. \quad (19.1)$$

Seek

$$\hat{\Phi}_0(x) = \sum_{n=0}^{\infty} a_n x^{-n-1}.$$

Substitution gives

$$a_0 = 1, \quad a_n = na_{n-1},$$

so

$$\hat{\Phi}_0(x) = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}}. \quad (19.2)$$

The series diverges for every finite x because the ratio of consecutive terms is $(n+1)/x$.

19.2 The missing homogeneous solution

The homogeneous equation

$$y' + y = 0$$

has solution Ce^{-x} . Therefore the complete formal transseries solution is

$$\hat{y}(x; C) = \sum_{n=0}^{\infty} \frac{n!}{x^{n+1}} + Ce^{-x}. \quad (19.3)$$

Every value of C has the same inverse-power asymptotic expansion. The transseries reveals the integration constant that the power sector hides.

19.3 Exact analytic solution

Multiplying (19.1) by e^x gives

$$(e^x y)' = \frac{e^x}{x}.$$

Hence, on a chosen branch,

$$y(x) = e^{-x}(\text{Ei}(x) + C), \quad (19.4)$$

where $\text{Ei}'(x) = e^x/x$. Repeated integration by parts yields

$$\text{Ei}(x) \sim_{\text{as}} \frac{e^x}{x} \left(1 + \frac{1}{x} + \frac{2!}{x^2} + \frac{3!}{x^3} + \cdots \right),$$

which reproduces (19.2).

19.4 Optimal truncation

The n th term has size $n!/x^{n+1}$. It is smallest near $n \approx x$. Stirling's formula gives the minimum scale

$$\left. \frac{n!}{x^{n+1}} \right|_{n \approx x} \asymp \sqrt{\frac{2\pi}{x}} e^{-x}.$$

Thus the smallest achievable error from the power sector alone is on the same exponential scale as the homogeneous solution.

Big idea

The “missing” Ce^{-x} term lives at precisely the scale where the divergent power expansion stops improving. Optimal truncation points toward the next transseries sector.

19.5 Borel preview

Under the order-one Borel transform

$$\mathcal{B}(x^{-n-1}) = \frac{\xi^n}{n!},$$

series (19.2) becomes

$$\mathcal{B}\hat{\Phi}_0(\xi) = \sum_{n \geq 0} \xi^n = \frac{1}{1 - \xi}. \quad (19.5)$$

The factorial divergence has become a simple pole at $\xi = 1$. Laplace integration above or below this pole gives two lateral sums whose difference is proportional to e^{-x} . Part VI develops this in detail.

Chapter summary

The simplest linear ODE already displays the whole transseries story: a factorially divergent perturbative sector, an exponentially small homogeneous sector carrying a free constant, optimal truncation at the exponential scale, and a Borel singularity encoding the same exponential.

Chapter 20

Nonlinearity creates a ladder of sectors

20.1 The logistic equation

Consider

$$y' = y - y^2. \quad (20.1)$$

The equilibrium $y = 1$ has linearized perturbation $\eta' = -\eta$, so the natural small monomial near $x = +\infty$ is e^{-x} .

The exact one-parameter solution tending to 1 is

$$y(x) = \frac{1}{1 + Ae^{-x}}. \quad (20.2)$$

Expanding geometrically,

$$y = 1 - Ae^{-x} + A^2e^{-2x} - A^3e^{-3x} + \dots. \quad (20.3)$$

20.2 Sector recursion without using the exact solution

Write

$$y = 1 + \sum_{k \geq 1} c_k e^{-kx}.$$

Then

$$y' = - \sum_{k \geq 1} k c_k e^{-kx},$$

and

$$y - y^2 = - \sum_{k \geq 1} c_k e^{-kx} - \left(\sum_{k \geq 1} c_k e^{-kx} \right)^2.$$

At sector $k \geq 2$,

$$(k-1)c_k = \sum_{j=1}^{k-1} c_j c_{k-j}. \quad (20.4)$$

The first coefficient c_1 is free. The recurrence gives

$$c_k = c_1^k.$$

Taking $c_1 = -A$ recovers (20.3).

This is the basic nonlinear mechanism: a first exponential perturbation is free, but its products force all higher sectors.

20.3 A more realistic Riccati example

Now consider

$$y' + y = \frac{1}{x} + y^2. \quad (20.5)$$

The base power sector

$$\Phi_0(x) = \sum_{n \geq 0} a_n x^{-n-1}$$

satisfies

$$a_0 = 1, \quad a_n = na_{n-1} + \sum_{j=0}^{n-1} a_j a_{n-1-j}. \quad (20.6)$$

Thus

$$\Phi_0 = \frac{1}{x} + \frac{2}{x^2} + \frac{8}{x^3} + \frac{44}{x^4} + \frac{296}{x^5} + \frac{2312}{x^6} + \cdots.$$

The coefficients again grow rapidly.

Linearize by writing $y = \Phi_0 + \eta$. To first order,

$$\eta' + \eta = 2\Phi_0\eta,$$

so

$$\eta' = (-1 + 2\Phi_0)\eta.$$

Therefore

$$\begin{aligned} \eta &= C \exp\left(-x + 2 \int \Phi_0 dx\right) \\ &= C e^{-x} x^2 \left(1 - \frac{4}{x} - \frac{8}{x^3} - \frac{52}{x^4} + \cdots\right). \end{aligned}$$

The x^2 factor comes from integrating the leading $2/x$ term.

20.4 Full sector equation

Use the ansatz

$$y = \sum_{k=0}^{\infty} \sigma^k e^{-kx} \Phi_k(x).$$

Substitution into (20.5) gives

$$\Phi'_k + (1 - k)\Phi_k = \sum_{j=0}^k \Phi_j \Phi_{k-j}. \quad (20.7)$$

For $k \geq 1$,

$$\Phi'_k + (1 - k - 2\Phi_0)\Phi_k = \sum_{j=1}^{k-1} \Phi_j \Phi_{k-j}. \quad (20.8)$$

The $k = 1$ equation is homogeneous and carries the free parameter. Every $k \geq 2$ equation is forced by lower sectors. Dominant balance suggests

$$\Phi_k(x) \asymp x^{2k} \times \text{an inverse-power series.}$$

20.5 Resonance warning

In multi-parameter problems, different combinations of actions may produce the same exponential monomial. Then the right side of a sector equation can resonate with the homogeneous solution, forcing powers of $\log x$ or polynomial factors in the transseries parameters. The simple one-action Riccati example avoids this complication.

Chapter summary

Linearization identifies the first exponential scale and its free coefficient. Nonlinear products then generate a ladder of higher sectors. Sector equations are typically linear in the unknown sector and forced by lower ones, making the transseries recursively computable.

Chapter 21

A singular perturbation and a boundary-layer exponential

21.1 A small parameter changes the asymptotic variable

Transseries need not use $x \rightarrow \infty$. Consider $\varepsilon \rightarrow 0^+$ in

$$\varepsilon \frac{dy}{dx} + y = \frac{1}{1+x}, \quad x > 0. \quad (21.1)$$

Formally invert the operator $1 + \varepsilon D$:

$$y_{\text{out}} = (1 + \varepsilon D)^{-1} \frac{1}{1+x} = \sum_{n=0}^{\infty} (-\varepsilon D)^n \frac{1}{1+x}.$$

Since

$$D^n \frac{1}{1+x} = \frac{(-1)^n n!}{(1+x)^{n+1}},$$

we obtain

$$\hat{y}_{\text{out}} = \sum_{n=0}^{\infty} \frac{n! \varepsilon^n}{(1+x)^{n+1}}. \quad (21.2)$$

For fixed x , this series diverges for every $\varepsilon \neq 0$.

21.2 The homogeneous exponential

The homogeneous equation

$$\varepsilon y' + y = 0$$

has solution

$$C e^{-x/\varepsilon}.$$

Thus the complete formal transseries is

$$\hat{y}(x, \varepsilon; C) = \sum_{n=0}^{\infty} \frac{n! \varepsilon^n}{(1+x)^{n+1}} + C e^{-x/\varepsilon}. \quad (21.3)$$

The exponential is smaller than every power of ε for fixed $x > 0$.

21.3 Exact solution

An integrating factor gives

$$y(x) = e^{-x/\varepsilon} \left[C + \frac{1}{\varepsilon} \int^x \frac{e^{t/\varepsilon}}{1+t} dt \right]. \quad (21.4)$$

Repeated integration by parts produces the outer series (21.2). The lower limit of the integral and the boundary condition determine C .

21.4 Boundary layers

Suppose a boundary condition at $x = 0$ disagrees with the value predicted by the outer expansion. The discrepancy is repaired by $Ce^{-x/\varepsilon}$, which changes rapidly on the stretched scale

$$X = \frac{x}{\varepsilon}.$$

In the boundary layer, the exponential is order one; away from the boundary, it is beyond all algebraic orders in ε .

This is a concrete reason transseries appear in singular perturbation theory: no finite power series in ε can satisfy both the differential equation and a mismatched boundary condition.

21.5 The same model in different clothes

Set

$$X = \frac{1+x}{\varepsilon}.$$

After a simple rescaling, the factorial series in (21.2) reduces to the Euler series of Chapter 19. Its Borel singularity therefore lies at the action corresponding to $e^{-x/\varepsilon}$.

Chapter summary

Singular perturbations naturally produce power series in a small parameter plus exponentials such as $e^{-x/\varepsilon}$. The power series may diverge factorially, while the exponential carries boundary data invisible to all algebraic orders.

Part VI

Divergence, Borel summation, and resurgence

Chapter 22

Factorial divergence is information, not failure

22.1 A formal expansion need not converge

Return to a formal inverse-power series

$$\widehat{\Phi}(z) = \sum_{n=0}^{\infty} \frac{a_n}{z^{n+1}}, \quad z \rightarrow \infty. \quad (22.1)$$

The hat reminds us that the series is formal: at first we make no claim that the sum converges. This is not an embarrassment. In asymptotic analysis, the useful question is whether finite truncations approximate a function in a controlled way.

Definition 22.1 (Poincaré asymptotic expansion). Let S be an unbounded region in the complex plane. We write

$$f(z) \sim_{\text{as}} \sum_{n=0}^{\infty} \frac{a_n}{z^{n+1}} \quad (z \rightarrow \infty \text{ in } S)$$

if, for every fixed $N \geq 0$,

$$f(z) - \sum_{n=0}^{N-1} \frac{a_n}{z^{n+1}} = O(z^{-N-1}).$$

Thus, with the normalization in (22.1), the remainder after terms $0, \dots, N-1$ is $O(z^{-N-1})$.

The definition asks us to fix N first and then let z grow. It does *not* ask us to send N to infinity for a fixed z . Those two limiting processes need not agree.

Example 22.2 (The Euler series again). The equation

$$y'(z) + y(z) = \frac{1}{z}$$

led in Chapter 19 to

$$\widehat{E}(z) = \sum_{n=0}^{\infty} \frac{n!}{z^{n+1}}. \quad (22.2)$$

For every fixed nonzero z , the ratio of consecutive terms is

$$\frac{(n+1)!/z^{n+2}}{n!/z^{n+1}} = \frac{n+1}{z},$$

whose magnitude eventually exceeds 1. The series therefore diverges for every finite z . Nevertheless, its first several terms can approximate an exact solution extremely well when z is large.

22.2 Why the terms first decrease and then increase

Suppose, schematically,

$$a_n \sim K \frac{\Gamma(n + \beta)}{A^{n+\beta}} \quad (n \rightarrow \infty), \quad (22.3)$$

where $A \neq 0$. Then the n th term of (22.1) has size approximately

$$|K| \frac{\Gamma(n + \beta)}{|A|^{n+\Re\beta} |z|^{n+1}}.$$

The ratio of successive magnitudes is approximately

$$\frac{n + \Re\beta}{|Az|}.$$

Thus the terms decrease while $n \lesssim |Az|$ and increase afterward. The smallest term occurs near

$$N_*(z) \approx |Az|. \quad (22.4)$$

Using Stirling's formula, its size is roughly an algebraic factor times

$$e^{-|Az|}. \quad (22.5)$$

Big idea

Factorial divergence and exponentially small scales are two views of the same phenomenon. If coefficients grow like $n!/A^n$, the optimally truncated error is usually of size e^{-Az} . The divergent tail is pointing toward the next transseries sector.

22.3 Optimal truncation

For a convergent series, adding more terms eventually improves the answer. For a divergent asymptotic series, the sensible rule is different.

Working recipe

To use a factorially divergent expansion numerically:

1. Estimate the magnitude of successive terms.
2. Stop near the least term, usually where their ratio crosses 1.
3. Treat the least-term size as the natural scale of the unresolved remainder.
4. If greater accuracy is needed, describe that remainder by another exponential sector rather than by blindly adding more terms.

For the Euler series at real $z > 0$, the n th term is $n!/z^{n+1}$, so the least term lies near $n = z$. Stirling's formula gives

$$\left. \frac{n!}{z^{n+1}} \right|_{n \approx z} \asymp \sqrt{\frac{2\pi}{z}} e^{-z}.$$

The homogeneous solution of the differential equation is exactly Ce^{-z} . The agreement of exponential scales is not an accident.

22.4 Gevrey order

Factorial growth can be measured systematically.

Definition 22.3 (Gevrey order). A formal series $\sum_{n \geq 0} a_n z^{-n-1}$ is *Gevrey of order s* in a coefficientwise sense if there are constants $C, A > 0$ such that

$$|a_n| \leq C A^n \Gamma(1 + sn) \quad (n \geq 0). \quad (22.6)$$

The most common case in ordinary differential equations is $s = 1$, where the bound is essentially $C A^n n!$.

A convergent inverse-power series is Gevrey-0 in this informal terminology. Gevrey-1 series are the natural domain of the ordinary Borel transform. Higher Gevrey orders call for Borel–Leroy transforms or repeated changes of scale, which we discuss in Chapter 28.

22.5 Flat functions explain nonuniqueness

A function $r(z)$ is *flat* relative to the inverse-power scale in a sector if

$$r(z) = O(z^{-N})$$

for every N . For example, e^{-z} is flat as $z \rightarrow +\infty$. Consequently, if f has a power-series asymptotic expansion, then so does $f + Ce^{-z}$, with exactly the same coefficients.

This proves that a Poincaré expansion by itself generally does not determine a function. One must also provide exponentially small data, boundary conditions, analyticity in a sufficiently wide sector, or a summation prescription.

There is an important refinement. Strong Gevrey bounds in sectors wider than a critical angle can force uniqueness; this is the content of Watson-type uniqueness theorems. At the elementary level, the lesson is simply that angular analytic information can rule out flat additions that are invisible on one ray.

22.6 Where factorials come from

Several common mechanisms create factorial coefficients.

1. **Repeated differentiation.** Since

$$\frac{d^n}{dz^n} \frac{1}{z} = (-1)^n \frac{n!}{z^{n+1}},$$

iterating a differential operator naturally produces $n!$.

2. **Repeated integration by parts.** Laplace-type integrals generate successive derivatives of an amplitude, again producing factorial growth.
3. **Nearby singularities.** Taylor coefficients of a function are controlled by its nearest complex singularities. After the Borel transform, this elementary principle becomes a direct explanation of large-order growth.
4. **Combinatorial proliferation.** Nonlinear recursion may create a number of diagrams, trees, or partitions growing like $n!$.

22.7 Large order as a diagnostic tool

Suppose numerical coefficients satisfy (22.3). Then

$$\frac{a_n}{a_{n-1}} \sim \frac{n + \beta - 1}{A}. \quad (22.7)$$

A plot of a_n/a_{n-1} against n is therefore approximately a straight line. Its slope estimates $1/A$, while the intercept estimates $(\beta - 1)/A$. After removing these leading factors, subleading $1/n$ corrections reveal coefficients of a neighboring transseries sector.

Common trap

The nearest action need not be real or unique. A conjugate pair of actions can make the coefficients oscillate, and several actions of equal modulus can interfere. Ratio tests then require refinement, such as even/odd subsequences, Richardson extrapolation, or Borel–Padé analysis.

Chapter summary

A divergent asymptotic series can be highly predictive. Factorial-over-power coefficient growth determines a least-term index, an exponentially small error scale, and often the location of a Borel singularity. Divergence is the first visible trace of the larger transseries.

Chapter 23

The Borel–Laplace microscope

23.1 The basic trick

The ordinary Borel transform divides out the factorial growth that causes a Gevrey-1 series to diverge. For

$$\widehat{\Phi}(z) = \sum_{n=0}^{\infty} \frac{a_n}{z^{n+1}},$$

define

$$(\mathcal{B}\widehat{\Phi})(\xi) = \widehat{\varphi}(\xi) := \sum_{n=0}^{\infty} \frac{a_n}{n!} \xi^n. \quad (23.1)$$

If a_n grows no faster than $CA^n n!$, the new series has a positive radius of convergence, at least $1/A$.

The directional Laplace transform reverses the operation:

$$(\mathcal{L}_\theta \widehat{\varphi})(z) := \int_0^{e^{i\theta}\infty} e^{-z\xi} \widehat{\varphi}(\xi) d\xi, \quad (23.2)$$

provided the ray avoids singularities, the continued Borel transform grows slowly enough, and $\Re(ze^{i\theta}) > 0$.

Indeed,

$$\int_0^{e^{i\theta}\infty} e^{-z\xi} \xi^n d\xi = \frac{n!}{z^{n+1}}, \quad (23.3)$$

with compatible choices of direction. Formally applying $\mathcal{L}_\theta \mathcal{B}$ therefore restores the original series term by term.

Definition 23.1 (Directional Borel sum). If the Borel transform converges near 0, can be analytically continued along the ray of angle θ , and has at most exponential growth there, then

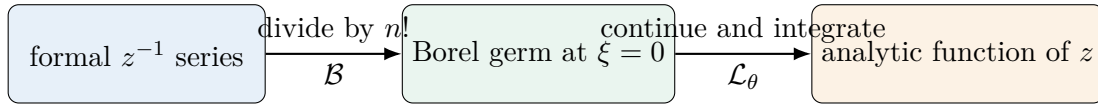
$$\mathcal{S}_\theta \widehat{\Phi} := \mathcal{L}_\theta \mathcal{B}\widehat{\Phi} \quad (23.4)$$

is the directional Borel sum of $\widehat{\Phi}$.

A constant term c is simply carried along: $\mathcal{S}_\theta(c + \widehat{\Phi}) = c + \mathcal{S}_\theta \widehat{\Phi}$.

23.2 A three-stage journey

It helps to keep the domains separate.



The first arrow is algebraic. The difficult information lies in the middle: how can the germ be analytically continued, where are its singularities, and how does it grow? The second arrow converts this Borel-plane geometry into exponentially small behavior in the original variable.

23.3 The Euler series becomes a rational function

For (22.2),

$$\mathcal{B}\widehat{E}(\xi) = \sum_{n=0}^{\infty} \xi^n = \frac{1}{1-\xi}. \quad (23.5)$$

A violently divergent factorial series has become the simplest possible meromorphic function. Its first singularity is a pole at $\xi = 1$.

If the integration ray avoids the positive real axis, then

$$\mathcal{S}_\theta \widehat{E}(z) = \int_0^{e^{i\theta}\infty} \frac{e^{-z\xi}}{1-\xi} d\xi. \quad (23.6)$$

Differentiating under the integral sign verifies that this is an actual solution of

$$y' + y = \frac{1}{z}.$$

The Borel transform has therefore converted a formal solution into analytic solutions.

23.4 Why Borel singularities create exponentials

Suppose the Borel transform has a singularity at $\xi = A$. In a Laplace integral, translate $\xi = A + u$ near that point:

$$e^{-z\xi} = e^{-Az} e^{-zu}.$$

Every contribution localized near A carries the factor e^{-Az} . Thus

$$\boxed{\text{Borel singularity at } A \quad \longleftrightarrow \quad \text{exponential scale } e^{-Az}.} \quad (23.7)$$

This is the basic Borel-plane dictionary behind resurgence.

23.5 Watson's lemma

The Laplace transform automatically has the original formal expansion.

Theorem 23.2 (Watson’s lemma, elementary form). *Suppose φ is analytic near 0 with*

$$\varphi(\xi) = \sum_{n=0}^{N-1} b_n \xi^n + O(\xi^N)$$

and has sufficient growth control on a Laplace ray. Then

$$\int_0^{e^{i\theta}\infty} e^{-z\xi} \varphi(\xi) d\xi = \sum_{n=0}^{N-1} \frac{b_n n!}{z^{n+1}} + O(z^{-N-1}) \quad (23.8)$$

inside every closed subsector where the integral converges uniformly.

Proof idea. Split the ray at a small fixed distance ρ . On $0 \leq |\xi| \leq \rho$, insert the Taylor polynomial and integrate each monomial using (23.3). The Taylor remainder contributes $O(z^{-N-1})$ after the substitution $u = z\xi$. On the remaining part of the ray, the exponential kernel supplies a bound of order $e^{-c|z|}$, which is smaller than every inverse power. The details are estimates, not new ideas. $\square$

23.6 Algebra in the Borel plane

Multiplication of formal series becomes convolution. If

$$\widehat{\Phi}(z) = \sum_{n \geq 0} a_n z^{-n-1}, \quad \widehat{\psi}(z) = \sum_{n \geq 0} b_n z^{-n-1},$$

then

$$\mathcal{B}(\widehat{\Phi}\widehat{\psi}) = (\mathcal{B}\widehat{\Phi}) * (\mathcal{B}\widehat{\psi}), \quad (f * g)(\xi) = \int_0^\xi f(\eta)g(\xi - \eta) d\eta. \quad (23.9)$$

The beta integral

$$\int_0^\xi \eta^m (\xi - \eta)^n d\eta = \frac{m!n!}{(m+n+1)!} \xi^{m+n+1}$$

proves the identity coefficient by coefficient.

Convolution explains why nonlinear equations produce new singularities at sums of old singularities. If one factor is singular near A and another near B , their convolution can be singular near $A + B$. In the original variable, this is just multiplication of exponentials:

$$e^{-Az} e^{-Bz} = e^{-(A+B)z}.$$

23.7 What can go wrong

Borel summation is powerful but not automatic.

1. The Borel series may have zero radius of convergence because growth is faster than $n!$.
2. A singularity may lie on the desired integration ray.
3. Analytic continuation may exist only along restricted paths.
4. The continued Borel transform may grow too fast for an ordinary Laplace integral.
5. Different lateral paths may give different answers.

The next chapters turn the fourth and fifth items from obstacles into useful structure.

23.8 Borel–Padé summation in practice

Often only finitely many coefficients $a_0, \dots, a_N$ are known. One can still probe the Borel plane.

Working recipe

A basic Borel–Padé calculation proceeds as follows.

1. Form the truncated Borel polynomial $\sum_{n=0}^N a_n \xi^n / n!$.
2. Replace it by a near-diagonal Padé approximant, a rational function whose Taylor series matches as many coefficients as possible.
3. Inspect stable strings or clusters of poles as approximations to branch cuts and isolated singularities.
4. Integrate the rational approximation along a safe ray.
5. Vary the Padé order and ray slightly to test numerical stability.

Spurious pole-zero pairs, called Froissart doublets, can appear. A numerical plot is evidence, not a proof of analytic continuation. Nevertheless, Borel–Padé analysis is often an excellent microscope for discovering the actions and Stokes directions of an unknown problem.

Chapter summary

The Borel transform removes a factorial, turning a divergent Gevrey-1 series into a locally convergent germ. Analytic continuation exposes singularities; the Laplace transform returns to the original variable. A singularity at A produces the scale e^{-Az} , and multiplication becomes convolution.

Chapter 24

Stokes phenomena: when an invisible term switches on

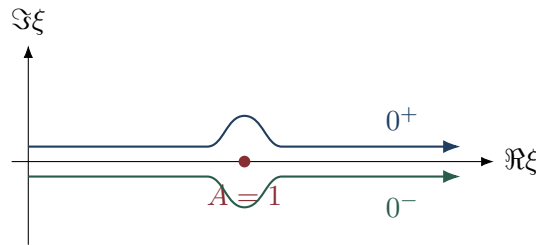
24.1 A singular direction

For the Euler Borel transform $1/(1 - \xi)$, the positive real Laplace ray runs through the pole at $\xi = 1$. There are two natural detours:

$$\mathcal{S}_{0+}\hat{E}(z) = \int_0^{\infty e^{+i0}} \frac{e^{-z\xi}}{1 - \xi} d\xi, \quad (24.1)$$

$$\mathcal{S}_{0-}\hat{E}(z) = \int_0^{\infty e^{-i0}} \frac{e^{-z\xi}}{1 - \xi} d\xi. \quad (24.2)$$

The notation means that the contour passes just above or just below the positive axis.



24.2 The Euler jump by one residue

The difference of the two contours encloses the pole. With the orientation in (24.2), the residue theorem gives

$$\mathcal{S}_{0+}\hat{E}(z) - \mathcal{S}_{0-}\hat{E}(z) = 2\pi i e^{-z}. \quad (24.3)$$

Indeed, the residue of $e^{-z\xi}/(1 - \xi)$ at $\xi = 1$ is $-e^{-z}$, and the small contour resulting from “upper minus lower” is clockwise, supplying the second minus sign.

The two sums have the same inverse-power expansion. Their difference is flat, exactly as the discussion in Chapter 22 predicted.

24.3 The transseries parameter absorbs the jump

The complete formal solution of the Euler equation is

$$\hat{y}(z; C) = \hat{E}(z) + Ce^{-z}. \quad (24.4)$$

To represent the same analytic solution above and below the Stokes ray, require

$$\mathcal{S}_{0+}\hat{E} + C_+e^{-z} = \mathcal{S}_{0-}\hat{E} + C_-e^{-z}.$$

Using (24.3), we obtain

$$C_+ = C_- - 2\pi i. \quad (24.5)$$

The function itself need not jump. What jumps is the asymptotic coordinate C used to describe it.

Big idea

A Stokes phenomenon is a change of asymptotic coordinates. A contribution that was exponentially subdominant can acquire a new coefficient when a singular direction is crossed. The full transseries remains consistent because its parameters transform at the same time.

24.4 Stokes and anti-Stokes terminology

For an exponential e^{-Az} , two geometrically important sets of rays occur:

$$\begin{aligned} \Im(Az) = 0, & \quad \text{the exponential has constant phase,} \\ \Re(Az) = 0, & \quad \text{the exponential changes between decay and growth.} \end{aligned}$$

Different communities interchange the names *Stokes line* and *anti-Stokes line* for these sets. To avoid ambiguity, we will say *Borel singular direction* for a Laplace direction hitting a singularity, and *equal-magnitude line* when two saddle exponentials have equal magnitude.

24.5 Dominance can change without discontinuity

Consider

$$F(z) = 1 + e^{-z}.$$

For $\Re z > 0$, the exponential is small; for $\Re z < 0$, it is large. The function is entire and does not jump at $\Re z = 0$. Only the useful asymptotic organization changes. A genuine Stokes jump is subtler: it concerns how analytic continuation and asymptotic representation interact when an exponentially small sector is switched into the expansion.

24.6 The Stokes automorphism

Let $\mathcal{S}_{\theta-}$ and $\mathcal{S}_{\theta+}$ denote lateral resummations of a whole formal transseries just below and above a singular direction. We define the formal Stokes automorphism $\mathfrak{S}_\theta$ by the convention

$$\mathcal{S}_{\theta+} = \mathcal{S}_{\theta-} \circ \mathfrak{S}_\theta. \quad (24.6)$$

It is an automorphism because it must respect addition, multiplication, and the formal equations satisfied by the transseries. In the Euler example it acts by a translation of the transseries parameter.

The exact sign of a quoted “Stokes constant” depends on conventions: direction of the Laplace kernel, orientation of lateral contours, normalization of the exponential sector, and whether the automorphism maps lower data to upper data or conversely. A reliable calculation always states these choices.

24.7 Smooth switching at finite size

The discontinuous Stokes jump belongs to an ideal asymptotic description as $|z| \rightarrow \infty$. At large but finite $|z|$, optimal truncation reveals a rapid yet smooth transition, typically on an angular scale of order $|z|^{-1/2}$. In many classical problems the multiplier is approximated by an error function. This is called Berry smoothing.

There is no contradiction. A step function is the limiting profile of a smooth transition whose width shrinks as the asymptotic parameter grows.

Chapter summary

When a Laplace ray crosses a Borel singularity, its two lateral sums can differ by an exponentially small term. The corresponding transseries parameter changes so that one analytic solution has compatible descriptions on both sides. Stokes jumps are changes of asymptotic coordinates, not mysterious discontinuities of the underlying function.

Chapter 25

Airy functions: two saddles and a Stokes geometry

25.1 Why Airy's equation is the standard laboratory

Airy's equation

$$y''(x) - xy(x) = 0 \quad (25.1)$$

is linear, exactly solvable, and already contains the geometry of competing exponentials. It describes the universal local behavior near a simple turning point in many differential equations.

Seek a WKB-type solution

$$y(x) = \exp(S(x)).$$

Then

$$S'' + (S')^2 = x.$$

At leading order, $(S')^2 \sim x$, so

$$S(x) \sim \pm \frac{2}{3}x^{3/2}. \quad (25.2)$$

The two fundamental exponential scales are therefore

$$e^{-\zeta}, \quad e^{+\zeta}, \quad \zeta = \frac{2}{3}x^{3/2}.$$

25.2 The algebraic prefactor

Write

$$y(x) = e^{\lambda\zeta} x^\alpha u(x), \quad \lambda = \pm 1, \quad u(x) \sim 1 + \dots.$$

Substitution into (25.1) shows that the next balance requires

$$\alpha = -\frac{1}{4}.$$

Thus the two formal WKB sectors begin as

$$y_{\pm}(x) \sim x^{-1/4} e^{\pm \frac{2}{3}x^{3/2}} \left(1 + \sum_{n \geq 1} \frac{c_n^{(\pm)}}{x^{3n/2}} \right). \quad (25.3)$$

The coefficient series are divergent of Gevrey order 1 in the variable ζ^{-1} .

For real $x > 0$, conventional normalizations give

$$\text{Ai}(x) \sim_{\text{as}} \frac{e^{-\zeta}}{2\sqrt{\pi} x^{1/4}} \left(1 - \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} - \cdots \right), \quad (25.4)$$

$$\text{Bi}(x) \sim_{\text{as}} \frac{e^{+\zeta}}{\sqrt{\pi} x^{1/4}} \left(1 + \frac{5}{48x^{3/2}} + \frac{385}{4608x^3} + \cdots \right). \quad (25.5)$$

25.3 The contour integral and its saddles

One representation is

$$\text{Ai}(x) = \frac{1}{2\pi i} \int_{\mathcal{C}} \exp\left(\frac{t^3}{3} - xt\right) dt, \quad (25.6)$$

where $\mathcal{C}$ runs between sectors in which the cubic exponential decays. The phase

$$\Phi(t) = \frac{t^3}{3} - xt$$

has critical points

$$\Phi'(t) = t^2 - x = 0, \quad t_{\pm} = \pm\sqrt{x}.$$

Their phase values are

$$\Phi(t_+) = -\frac{2}{3}x^{3/2} = -\zeta, \quad \Phi(t_-) = +\frac{2}{3}x^{3/2} = +\zeta.$$

These are precisely the two WKB exponentials.

A steepest-descent contour can be decomposed into elementary contours, called Lefschetz thimbles, attached to the saddles. As $\arg x$ changes, the valid thimble decomposition changes by integer combinations. This geometric jump is the integral version of a Stokes automorphism.

25.4 Action differences matter

The difference between the saddle actions is

$$2\zeta = \frac{4}{3}x^{3/2}. \quad (25.7)$$

Consequently, the late coefficients of the expansion around one saddle know about the other saddle through an exponential scale $e^{-2\zeta}$. Relative to the dominant factor already extracted from a sector, this is the size of the neighboring saddle contribution.

25.5 Rotating the argument

Let $x = re^{i\phi}$. Then

$$\zeta = \frac{2}{3}r^{3/2}e^{3i\phi/2}.$$

The two exponentials have equal magnitude when $\Re\zeta = 0$, that is, on rays

$$\phi = \frac{\pi}{3}, \pi, -\frac{\pi}{3} \pmod{2\pi}.$$

Between these rays, one saddle is exponentially dominant and the other subdominant. Analytic continuation of Ai is smooth, but its most economical asymptotic representation changes from sector to sector.

Intuition

Imagine describing a landscape at night with two lamps. In one region the first lamp overwhelms the second, so the second is invisible to a power-series camera. After rotating the landscape, the dim lamp becomes comparable or dominant. The lamp did not suddenly appear; only the hierarchy of brightness changed.

25.6 What the Airy example teaches

Airy's equation packages several universal ideas:

1. exponential actions arise by integrating a leading momentum, $\int \sqrt{x} dx$;
2. algebraic prefactors arise at the next asymptotic order;
3. each saddle carries a divergent fluctuation series;
4. differences of saddle actions govern large-order behavior;
5. Stokes jumps are changes in contour or saddle decomposition.

The same pattern survives in far more complicated ordinary differential equations, integrals, quantum-mechanical problems, and path integrals.

Chapter summary

Airy's equation has two formal sectors, proportional to $x^{-1/4}e^{\pm 2x^{3/2}/3}$ times divergent series. Its contour integral has two matching saddles. Rotating x changes which saddle is dominant and how the integration contour decomposes, giving a concrete geometric model of Stokes phenomena.

Chapter 26

Resurgence: sectors talking to one another

26.1 From isolated summation to a network

Borel summation treats one divergent series. Resurgence adds a stronger claim: the singularities encountered while continuing its Borel transform are not arbitrary obstacles. Their local expansions reproduce the Borel transforms of other sectors in the same problem. Information seems to “re-emerge” at those singular points, which motivated Écaille’s term *resurgence*.

A typical one-parameter transseries has the form

$$\widehat{\Phi}(z, \sigma) = \sum_{k=0}^{\infty} \sigma^k e^{-kAz} z^{k\beta} \widehat{\Phi}_k(z), \quad \widehat{\Phi}_k(z) = \sum_{n=0}^{\infty} \frac{a_n^{(k)}}{z^{n+1}}. \quad (26.1)$$

The sector $k = 0$ is often called perturbative. The sectors $k \geq 1$ are nonperturbative or multi-instanton sectors. The terminology comes from physics, but the algebra applies to ordinary differential equations and integrals just as well.

26.2 Endless continuability, informally

A convergent germ $\widehat{\varphi}(\xi)$ at the Borel origin is called resurgent when it can be analytically continued along every finite-length path after one avoids a locally finite collection of singular points, with suitable control on the resulting singularities. Precise definitions differ according to the class of singularities and the amount of uniformity required [12, 22, 10].

The adjective *endless* means that continuation can be repeated: reaching one singularity does not terminate the story. One can go around it, examine the new germ, and continue farther.

26.3 The simplest large-order calculation

Suppose the Borel transform of $\widehat{\Phi}_0$ has its nearest singularity at $\xi = A$ and, near that point, behaves like a simple pole

$$\widehat{\varphi}_0(\xi) \sim \frac{r}{\xi - A}. \quad (26.2)$$

Expanding the pole about the origin gives

$$\frac{r}{\xi - A} = -\frac{r}{A} \frac{1}{1 - \xi/A} = -r \sum_{n=0}^{\infty} \frac{\xi^n}{A^{n+1}}.$$

Since the coefficient of ξ^n in the Borel transform is $a_n^{(0)}/n!$, we expect

$$a_n^{(0)} \sim -r \frac{n!}{A^{n+1}}. \quad (26.3)$$

For the Euler series, $A = 1$ and $r = -1$, so this gives $a_n = n!$ exactly.

This elementary argument is Darboux's principle in Borel clothing: the nearest singularity controls the late Taylor coefficients.

26.4 A neighboring sector inside a singularity

A more characteristic resurgent singularity is not merely a pole. Schematically, one may find

$$\widehat{\varphi}_0(A + \xi) \sim \frac{S_{01}}{2\pi i} \widehat{\varphi}_1(\xi) \log \xi + \text{less singular terms}, \quad (26.4)$$

where S_{01} is a Stokes constant and $\widehat{\varphi}_1 = \mathcal{B}\widehat{\Phi}_1$. Expanding the right-hand side about the origin and applying coefficient asymptotics yields a full large-order relation of the form

$$a_n^{(0)} \sim \frac{S_{01}}{2\pi i} \sum_{m \geq 0} a_m^{(1)} \frac{\Gamma(n - m + \gamma)}{A^{n-m+\gamma}} + \text{contributions from other singularities}. \quad (26.5)$$

The shift γ depends on the powers of z used to normalize the two sectors. Formula (26.5) is schematic but its message is exact: *late coefficients of one sector are determined by early coefficients of another.*

26.5 Reading the neighboring sector from data

Suppose A , γ , and the leading Stokes constant have been estimated. Then one can peel off the leading factorial growth:

$$\frac{2\pi i A^{n+\gamma} a_n^{(0)}}{S_{01} \Gamma(n + \gamma)} \sim a_0^{(1)} + \frac{A a_1^{(1)}}{n + \gamma - 1} + \frac{A^2 a_2^{(1)}}{(n + \gamma - 1)(n + \gamma - 2)} + \text{eedspace1em} \dots$$

Successive $1/n$ corrections recover $a_1^{(1)}, a_2^{(1)}, \dots$. In good examples, dozens of low-order perturbative coefficients can numerically predict the first few nonperturbative coefficients with striking accuracy.

26.6 Why nonlinear equations generate a lattice of actions

Suppose an equation contains products of sectors with actions A and B . Their product has action $A + B$. Iterating nonlinearities creates an additive semigroup

$$\Omega = \{n_1 A_1 + \dots + n_r A_r : n_j \in \mathbb{N}\}.$$

Correspondingly, Borel singularities tend to occur at points of Ω and its signed or analytically continued variants. This is the analytic counterpart of the transmonomial lattice generated by $e^{-A_1 z}, \dots, e^{-A_r z}$.

Common trap

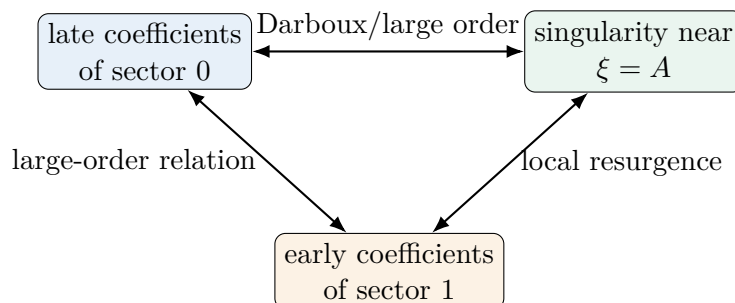
The action lattice can have coincidences. If two different integer combinations give the same action, the problem is resonant. Logarithmic sectors and polynomial factors in z may then be forced. A naive ansatz containing only e^{-kAz} times ordinary power series may be incomplete.

26.7 Resurgence does not mean convergence

A resurgent series is usually still divergent. Resurgence says that its Borel transform has a controlled and highly organized continuation, not that the original series converges. Nor is every divergent series resurgent. Some Borel transforms have natural boundaries or singular sets too dense for endless continuation.

26.8 The conceptual triangle

Three structures determine one another:



This triangle is the practical heart of resurgence.

Chapter summary

A resurgent transseries is a network of sectors. Borel singularities of one sector encode other sectors, and the resulting local singular expansions imply large-order relations among their coefficients. Nonlinearity adds actions and therefore creates whole lattices of exponential sectors and Borel singularities.

Chapter 27

Alien derivatives and bridge equations

27.1 Why invent another derivative?

An ordinary derivative measures local variation with respect to z . To study resurgence, we need an operator that measures the singular behavior reached by analytic continuation in the Borel plane. Écaille's *alien derivative* Δ_ω extracts the new resurgent information located at the Borel singularity ω .

The word *alien* emphasizes that the operator is nonlocal in the original variable. It can detect an exponentially small sector that no finite number of ordinary derivatives of a power-series asymptotic expansion would reveal.

A full construction requires the algebra of singular germs and careful choices of paths. Here we focus on the rules that make the operator useful.

27.2 A schematic extraction rule

If the continued Borel transform of $\widehat{\Phi}_0$ near ω contains

$$\frac{S_{01}}{2\pi i} \widehat{\varphi}_1(\xi - \omega) \log(\xi - \omega),$$

then, under a common normalization,

$$\Delta_\omega \widehat{\Phi}_0 = S_{01} \widehat{\Phi}_1. \quad (27.1)$$

More complicated singularities produce combinations of sectors. The numerical factor assigned to S_{01} varies among conventions, but the invariant content is the coupling between sectors.

Alien derivatives satisfy a Leibniz rule in their standard logarithmic normalization:

$$\Delta_\omega(FG) = (\Delta_\omega F)G + F(\Delta_\omega G). \quad (27.2)$$

This makes them derivations of the resurgent algebra.

27.3 Pointed alien derivatives

It is often convenient to attach the exponential weight to the operator:

$$\dot{\Delta}_\omega := e^{-\omega z} \Delta_\omega. \quad (27.3)$$

With a standard choice of alien normalization, the pointed derivative is chosen to commute with ordinary differentiation,

$$[\partial_z, \dot{\Delta}_\omega] = 0. \quad (27.4)$$

Other references place the exponential weight or sign elsewhere. The safe practice is to check the stated commutator before importing a formula.

27.4 The Stokes automorphism as an exponential

Along a singular direction θ , let Γ_θ be the set of Borel singularities on the ray. Formally, the Stokes automorphism can be written

$$\mathfrak{S}_\theta = \exp\left(\sum_{\omega \in \Gamma_\theta} e^{-\omega z} \Delta_\omega\right) = \exp\left(\sum_{\omega \in \Gamma_\theta} \dot{\Delta}_\omega\right), \quad (27.5)$$

with path-ordering or refined “lateral” alien operators inserted when the singularity geometry demands it. This resembles the familiar identity “finite translation equals the exponential of a derivative.” The alien derivatives are infinitesimal generators of a finite Stokes jump.

27.5 The bridge equation

Now suppose $\hat{\Phi}(z, \sigma)$ is a formal solution of a differential equation. Apply $\dot{\Delta}_\omega$ to that equation. Because the pointed alien derivative commutes with ∂_z and obeys a Leibniz rule, the result solves the linearized equation around $\hat{\Phi}$. But differentiating the original solution with respect to its free parameter σ also solves that linearized equation. Under suitable one-dimensionality assumptions, the two must be proportional:

$$\dot{\Delta}_\omega \hat{\Phi}(z, \sigma) = S_\omega(\sigma) \partial_\sigma \hat{\Phi}(z, \sigma). \quad (27.6)$$

This is a *bridge equation*: it bridges analytic continuation in the Borel plane and ordinary differentiation in transseries-parameter space.

For the simplest one-action problem, the first relation may reduce to

$$\dot{\Delta}_A \hat{\Phi} = S_1 \partial_\sigma \hat{\Phi}, \quad (27.7)$$

where S_1 is a Stokes constant.

27.6 Parameter translation

Insert (27.7) into (27.5). If no other alien derivatives contribute along the ray, then

$$\mathfrak{S}_0 = \exp(S_1 \partial_\sigma).$$

But

$$\exp(S_1 \partial_\sigma) F(\sigma) = F(\sigma + S_1),$$

so

$$\mathfrak{S}_0 \hat{\Phi}(z, \sigma) = \hat{\Phi}(z, \sigma + S_1). \quad (27.8)$$

The Euler parameter jump in (24.5) is the elementary prototype of this general statement, after conventions and normalizations are matched.

27.7 Median summation

Assume real coefficients and a Stokes ray along the positive real direction. The two lateral Borel sums are often complex conjugates, even though the desired answer is real. With the convention (24.6), define the median resummation by

$$\mathcal{S}_{\text{med}} = \mathcal{S}_{0-} \circ \mathfrak{S}_0^{1/2} = \mathcal{S}_{0+} \circ \mathfrak{S}_0^{-1/2}. \quad (27.9)$$

It divides the Stokes jump symmetrically between the two sides. Under appropriate reality conditions, the result is real and ambiguity-free.

For a pure parameter translation $\sigma \mapsto \sigma + S_1$, the half-power translates by $S_1/2$. Thus median summation can be understood as choosing a half-shifted transseries parameter before taking a lateral sum.

27.8 Several parameters and resonance

With actions $A_1, \dots, A_r$, introduce parameters $\sigma = (\sigma_1, \dots, \sigma_r)$ and sectors

$$\sigma^n e^{-(n \cdot A)z} \widehat{\Phi}_n(z).$$

Bridge equations become vector fields on parameter space:

$$\dot{\Delta}_\omega \widehat{\Phi} = \sum_{j=1}^r S_{\omega,j}(\sigma) \frac{\partial \widehat{\Phi}}{\partial \sigma_j}. \quad (27.10)$$

The Stokes automorphism is then the time-one flow of a generally nonlinear formal vector field.

When action combinations coincide, logarithms and additional parameters may be necessary. Alien calculus still works, but the parameter-space vector field can mix sectors in a triangular or resonant fashion.

Common trap

Alien calculus is convention-sensitive notation for invariant analytic facts. Never compare bare Stokes constants from two sources until the sector normalizations, Borel transform, Laplace kernel, lateral orientation, and bridge equation convention have all been aligned.

Chapter summary

Alien derivatives extract singular information at specified Borel locations. Their exponential generates Stokes automorphisms. Bridge equations identify alien differentiation with vector fields in transseries-parameter space, often turning a Stokes jump into an ordinary parameter translation. Median summation uses half of this transformation to cancel lateral ambiguity symmetrically.

Chapter 28

Beyond one Borel sum

28.1 Borel–Leroy transforms

If coefficients grow like $\Gamma(n + \beta)$ rather than exactly $n!$, it is convenient to divide by that gamma factor. One version of the Borel–Leroy transform is

$$\mathcal{B}_\beta \left(\sum_{n \geq 0} a_n z^{-n-\beta} \right) = \sum_{n \geq 0} \frac{a_n}{\Gamma(n + \beta)} \xi^{n+\beta-1}. \quad (28.1)$$

The inverse is still a Laplace integral because

$$\int_0^\infty e^{-z\xi} \xi^{n+\beta-1} d\xi = \frac{\Gamma(n + \beta)}{z^{n+\beta}}.$$

This formulation keeps algebraic prefactors attached to an exponential sector without awkward index shifts.

28.2 k -summability and higher Gevrey order

For growth comparable to $\Gamma(1 + n/k)$ or, after a reciprocal convention, $\Gamma(1 + kn)$, one uses a Borel transform of order k together with a Laplace kernel such as $\exp[-(z\xi)^k]$. The exact formulas vary with normalization. The principle is stable:

1. divide out the characteristic gamma growth;
2. analytically continue in the transformed plane;
3. use a matching integral kernel to restore the original monomials.

Different levels of an equation can possess different Gevrey orders. A single ordinary Borel transform then cannot regularize every level at once.

28.3 Multisummability

A formal series is *multisummable* when it can be decomposed into finitely many pieces, each summable at its own Gevrey level, in a compatible sequence of directions. Roughly,

$$\widehat{\Phi} = \widehat{\Phi}^{(k_1)} + \cdots + \widehat{\Phi}^{(k_r)}, \quad k_1 < \cdots < k_r,$$

and each component is k_j -summable after the faster levels have been handled. Multisummability is a fundamental theorem for broad classes of formal solutions of analytic differential equations.

28.4 Acceleration

Sometimes the analytically continued Borel transform grows faster than any ordinary exponential, so the Laplace integral still diverges. Écaille's *acceleration* changes the Borel variable and kernel to slow this growth, then continues the summation process. One can think of it as replacing an inadequate microscope by one adapted to a more rapidly separated asymptotic scale.

Acceleration is one reason the full theory of transseries goes beyond the simple slogan “Borel transform and Laplace transform.” Yet the slogan remains the correct first model.

28.5 Hyperasymptotics

Optimal truncation leaves an exponentially small remainder. Hyperasymptotics expands that remainder using contributions from adjacent saddles or Borel singularities, then optimally truncates those expansions in turn.

For a Laplace-type integral with saddles $s_0, s_1, \dots$, the first expansion around s_0 has a remainder controlled by the nearest action difference A_{01} . One rewrites the remainder as an integral associated with s_1 , expands it, and obtains an error of a still smaller exponential order. Repeating this process creates a hierarchy:

$$\text{power accuracy} \longrightarrow e^{-A_{01}z} \text{ accuracy} \longrightarrow e^{-(A_{01}+A_{12})z} \text{ accuracy} \longrightarrow \dots$$

The resulting special functions that smooth truncated saddle contributions are often called *terminants*.

28.6 Exponential asymptotics

Exponential asymptotics is a practical methodology, especially common in singular perturbation theory:

1. compute the late-term ansatz of a divergent regular expansion;
2. infer the singulant $A(x)$ and prefactor from its recurrence;
3. truncate near the least term;
4. derive a differential equation for the exponentially small remainder;
5. resolve its rapid change across a Stokes curve, often obtaining an error function profile.

This can reveal beyond-all-orders waves, capillary ripples, boundary-layer leakage, and other effects without constructing the complete alien calculus.

28.7 Transasymptotic rearrangement

A sector expansion

$$\sum_{k \geq 0} \sigma^k e^{-kAz} \widehat{\Phi}_k(z)$$

is ordered by exponentials when σ is fixed and $\Re(Az) \gg 1$. Near a region where

$$\tau = \sigma e^{-Az} z^\beta$$

is no longer tiny, it may be better to hold τ fixed and sum over k first:

$$\hat{\Phi}(z, \sigma) = \sum_{n \geq 0} \frac{F_n(\tau)}{z^n}. \quad (28.2)$$

The coefficient functions F_n resum infinitely many exponential sectors. This is called a transasymptotic rearrangement. It can extend an approximation through regions where the original sector-by-sector ordering breaks down.

28.8 A practical hierarchy of methods

Method	What it adds	Typical use
Poincaré asymptotics	finitely many algebraic terms	coarse large-parameter approximation
Optimal truncation	best use of a divergent series	exponential estimate of the remainder
Borel summation	analytic value for one Gevrey series	reconstructing a canonical sector
Resurgent transseries	coupled exponential sectors	Stokes jumps and ambiguity cancellation
Hyperasymptotics	re-expansion of remainders	exponentially improved numerical accuracy
Multisummability	several Gevrey levels	singular differential equations with multiple scales
Transasymptotics	sectors resummed into coefficient functions	transition regions and global matching

Chapter summary

Ordinary Borel summation handles one Gevrey-1 level. Borel–Leroy transforms, k -summability, multisummability, and acceleration handle other growth scales. Hyperasymptotics repeatedly expands optimally truncated remainders, while transasymptotic rearrangement sums sectors in a new order when the exponential parameter becomes order one.

Chapter 29

A reusable analytic workflow

29.1 Step 1: find the formal sectors

Start from the equation, recurrence, or integral and identify all leading balances. For an ODE, substitute

$$y \sim e^{-A(z)} z^\beta$$

into the linearized equation. For an integral, locate all relevant saddles and endpoint singularities. Every independent action is a candidate exponential scale.

29.2 Step 2: derive coefficient recurrences

For each sector, write

$$\Phi_k(z) = e^{-A_k z} z^{\beta_k} \sum_{n \geq 0} a_n^{(k)} z^{-n}$$

or the appropriate nonconstant-action version. Substitute and solve in dominance order. Keep the normalization $a_0^{(k)}$ explicit; changing it inversely changes the corresponding Stokes constant.

29.3 Step 3: inspect late terms

Compute enough coefficients to test

$$a_n^{(0)} \sim K \frac{\Gamma(n + \gamma)}{A^{n+\gamma}}.$$

Use ratios, Richardson extrapolation, and known symmetries. Complex-conjugate or equidistant singularities may require separating subsequences or fitting a sum of oscillatory contributions.

29.4 Step 4: enter the Borel plane

Divide by the appropriate gamma factor. Determine the radius of convergence, then seek analytic continuation. With exact equations, the Borel transform may satisfy an integral or differential equation. With finite data, use conformal maps and Padé approximants.

29.5 Step 5: locate singular directions

Each Borel singularity A determines directions for which the Laplace ray hits it. Each action also determines curves where $\Re(Az) = 0$, across which exponential dominance changes. Draw both geometries; they answer related but not identical questions.

29.6 Step 6: compute discontinuities

For a pole, use residues. For logarithmic or algebraic branch points, compute the discontinuity across a cut. Match that discontinuity to the Borel transform of a neighboring sector. This determines a Stokes constant once sector normalizations are fixed.

29.7 Step 7: choose the physical or analytic parameter

Boundary conditions, contour choices, regularity, reality, or quantization conditions determine transseries parameters. Across a Stokes direction, transform them according to the Stokes automorphism rather than keeping their numerical coordinates artificially fixed.

29.8 Step 8: validate at three levels

A strong analysis performs three independent checks:

1. **Formal residual:** substituting a truncation leaves an error at the predicted next transmonomial.
2. **Large-order check:** coefficients of one sector reproduce the predicted action and early data of another.
3. **Numerical check:** lateral or median resummation agrees with direct integration or an exact solution in overlapping domains.

Agreement at all three levels makes normalization or sign errors much less likely.

Chapter summary

The analytic workflow runs from dominant balance to sectors, from sectors to late terms, from late terms to Borel singularities, and from singularities to Stokes transformations. Formal, large-order, and numerical residual checks form a useful triangle of verification.

Part VII

Where transseries live and where they are used

Chapter 30

Hardy fields, H -fields, and asymptotic differential algebra

30.1 Functions modulo eventual agreement

Two real functions defined for all sufficiently large x determine the same *germ at $+\infty$* if they eventually agree. A germ forgets behavior on every bounded interval and retains only the tail. This is exactly the viewpoint needed for asymptotics.

A *Hardy field* is a field of such germs that is closed under differentiation. Rational functions form a Hardy field. So do many fields built from algebraic functions, exponentials, and logarithms. Every nonzero element of a Hardy field has eventually constant sign; otherwise its infinitely many zeros would conflict with field and derivative closure through Rolle-type arguments. Consequently, Hardy fields carry a natural eventual order.

Transseries are formal rather than initially analytic, but their order and derivative behave as though they were germs of eventually well-behaved functions. H -fields axiomatize this behavior.

30.2 The axioms of an H -field

Let K be an ordered differential field with constant field

$$C = \{f \in K : f' = 0\}.$$

Write $f > C$ when f is larger than every constant. Let

$$\mathcal{O} = \{f \in K : |f| \leq c \text{ for some } c \in C_{>0}\}$$

be the convex hull of the constants, and let

$$\mathfrak{o} = \{f \in K : |f| < c \text{ for every } c \in C_{>0}\}$$

be its infinitesimal ideal.

Definition 30.1 (H -field). An ordered differential field K is an H -field if:

1. whenever $f > C$, one has $f' > 0$;
2. every bounded element has a unique constant part up to an infinitesimal:

$$\mathcal{O} = C + \mathfrak{o}. \tag{30.1}$$

The first axiom says that a function infinitely larger than constants is increasing. The second says that every bounded element looks like “constant plus something tending to zero.” Both are familiar facts for tame function germs.

Suitable fields of logarithmic-exponential transseries are H -fields. Their constant field is $\mathbb{R}$, their infinitesimals are precisely the series dominated by 1, and the leading monomial decides eventual sign.

30.3 The natural valuation

The dominance relation defines a valuation. Declare

$$f \succ g \iff f \preccurlyeq g \text{ and } g \preccurlyeq f.$$

Two nonzero elements have the same value when they have the same asymptotic magnitude up to constant factors. We may write

$$v(f) > v(g) \iff f \prec g,$$

so larger valuation means smaller magnitude.

For a transseries, the value is determined by the leading monomial. The familiar non-Archimedean law

$$v(f + g) \geq \min\{v(f), v(g)\}$$

is merely the statement that adding two quantities cannot produce something larger than the larger summand, though cancellation can make it smaller.

30.4 The asymptotic couple

Differentiation interacts with valuation through the logarithmic derivative. If $\gamma = v(f)$, define schematically

$$\psi(\gamma) = v\left(\frac{f'}{f}\right). \tag{30.2}$$

Under the usual hypotheses, this depends only on γ , not on the chosen f . The ordered abelian value group Γ together with $\psi : \Gamma^{\neq 0} \rightarrow \Gamma$ is an *asymptotic couple*.

For simple monomials:

$$\begin{aligned} (x^a)^\dagger &= \frac{a}{x}, \\ (e^{x^2})^\dagger &= 2x, \\ ((\log x)^b)^\dagger &= \frac{b}{x \log x}. \end{aligned}$$

The map ψ records these derivative scales abstractly. It is a compact combinatorial skeleton of asymptotic differentiation.

30.5 Liouville closure

A differential field is *Liouville closed* if it contains antiderivatives and exponential integrals in the differential-algebraic sense: for every f , there are $g \neq 0$ and h with

$$\frac{g'}{g} = f, \quad h' = f,$$

and the underlying ordered field is real closed. A sufficiently rich standard field of transseries has these closure properties. This formalizes our earlier claims that one can integrate transseries and solve first-order linear differential equations inside the field.

The word “Liouville” here is differential-algebraic and should not be confused with Liouville’s theorem in complex analysis.

30.6 Newtonianity

For a polynomial equation, Newton polygons compare powers and valuations to find dominant balances. Differential polynomials require a differential analogue. An H -field is called *newtonian*, roughly, when every differential polynomial of Newton degree one has a zero in the valuation ring. This is an asymptotic version of algebraic henselianity.

The technical definition is beyond our scope, but the idea is concrete: formal Newton iteration should not fail merely because the unknown occurs with derivatives. Transseries possess enough monomials and closure operations to solve broad classes of asymptotic differential equations recursively.

30.7 Model theory

Aschenbrenner, van den Dries, and van der Hoeven developed a model theory of H -fields centered on transseries [1]. In an appropriate language, properties such as Liouville closure, newtonianity, and the absence of certain asymptotic gaps give an axiomatic description of the first-order behavior of the standard transseries field. This connects asymptotic expansion to questions normally asked about algebraically closed fields or real closed fields: Which statements are consequences of the axioms? Which extensions are existentially closed? Which decision procedures are possible?

The surprise is that an object built from expressions such as $e^{x+e^{-x}}$ has a coherent global algebraic theory, not merely a collection of ad hoc expansion rules.

30.8 From formal series to actual germs

A formal transseries may be realized by an analytic germ through summation. For special classes, this realization respects order, algebra, composition, and differentiation. One seeks an injective map

$$\mathfrak{s} : \mathbb{T}_{\text{summable}} \longrightarrow \mathcal{H}$$

into a Hardy field or field of analytic germs such that

$$\begin{aligned} \mathfrak{s}(T + U) &= \mathfrak{s}(T) + \mathfrak{s}(U), \\ \mathfrak{s}(TU) &= \mathfrak{s}(T)\mathfrak{s}(U), \\ \mathfrak{s}(T') &= (\mathfrak{s}T)', \end{aligned}$$

$$\mathfrak{s}(T \circ U) = (\mathfrak{s}T) \circ (\mathfrak{s}U)$$

whenever the operations are defined. Results of this kind connect formal logarithmic-exponential series with analyzable germs [25, 26, 6].

Not every formal transseries has a unique analytic realization without a chosen summation structure. Stokes data is precisely part of what must be supplied.

Chapter summary

Hardy fields describe germs of tame functions at infinity; H -fields abstract their ordered differential behavior. Transseries form a central H -field whose valuation comes from dominance and whose derivative is encoded by an asymptotic couple. Differential-algebraic closure and model theory explain why transseries solve asymptotic equations so systematically.

Chapter 31

Surreal numbers and transseries

31.1 A family resemblance

Conway's surreal numbers form a proper class **No** containing the real numbers, all ordinal numbers, infinitesimals, and infinitely large quantities. Every surreal has a normal form

$$a = \sum_{i < \alpha} r_i \omega^{a_i}, \quad (31.1)$$

where α is an ordinal, $r_i \in \mathbb{R} \setminus \{0\}$, and the exponents a_i strictly decrease. This looks remarkably like a Hahn series whose monomial group is generated recursively by powers of the simplest infinite surreal ω .

Compare it with a transseries normal form

$$T = \sum_{\mathbf{m} \in S} c_{\mathbf{m}} \mathbf{m}.$$

Both use ordered supports, leading terms, and transfinite sums. The main difference is emphasis: surreal numbers begin as a universal ordered field, while transseries begin as a universal language for asymptotic functions and therefore carry composition and differentiation as essential operations.

31.2 Exponentiation on the surreals

Gonshor defined an exponential map

$$\exp : \mathbf{No} \rightarrow \mathbf{No}_{>0}$$

extending the real exponential and compatible with the ordered-field structure. Its inverse is a logarithm. This allows surreal counterparts of iterated exponentials, iterated logarithms, and logarithmic-exponential monomials.

One may therefore send the transseries variable x to an infinite surreal such as ω and interpret many transseries as surreal numbers. The leading transmonomial becomes the leading Conway monomial.

31.3 Embedding transseries

There are structure-preserving embeddings of major transseries fields into the surreals, compatible with order and exponential structure. In suitable frameworks, differential structure can also be

made compatible [4, 1]. Symbolically,

$$\mathbb{T} \hookrightarrow \mathbf{No}, \quad x \mapsto \omega.$$

For example,

$$x + \log x + e^{-x} \mapsto \omega + \log \omega + e^{-\omega}.$$

The infinitesimal $e^{-\omega}$ is smaller than every power ω^{-n} , exactly mirroring the transseries dominance relation.

31.4 Derivations on the surreals

A derivation on $\mathbf{No}$ should satisfy the ordinary algebraic rules, annihilate real constants, respect exponentiation,

$$D(e^a) = e^a D(a),$$

and behave well with summable families. Constructing such a derivation on a proper class with enormous supports is nontrivial. Berarducci and Mantova built surreal derivations with strong closure properties, making the analogy with transseries precise [4].

There is no requirement that an abstract infinite number possess a derivative. The derivative is additional structure selected to make ω play the role of a variable tending to infinity.

31.5 Why the surreals are larger

The ordinary logarithmic-exponential transseries field permits only finite exponential height and logarithmic depth for each individual element. The surreals contain vastly longer transfinite patterns. They also contain many objects with no immediate interpretation as ordinary asymptotic germs.

Conversely, function composition $T \circ U$ has no automatic counterpart in an arbitrary ordered exponential field: it depends on distinguishing a variable from a value. Thus neither slogan “transseries are just surreal numbers” nor “surreal numbers are just transseries” is accurate. There is a deep embedding and shared Hahn-series geometry, but the intended structures differ.

31.6 Germs, transseries, and surreals

A useful three-way picture is

$$\boxed{\text{analytic germs}} \longleftrightarrow \boxed{\text{formal transseries}} \longrightarrow \boxed{\text{surreal values at } \omega}.$$

The first arrow requires summation or asymptotic expansion and is generally not one-to-one without analytic restrictions. The second is algebraic and can be made into a rich embedding. Research relating germs, transseries, and surreal numbers asks how much of this triangle can simultaneously preserve order, exponential, composition, and differentiation [3, 20].

Chapter summary

Surreal normal forms and transseries are both Hahn-like sums of ordered monomials. Exponentiation and suitable derivations permit embeddings of major transseries fields into the surreals, often by sending x to ω . The surreals are much larger, while transseries carry a more explicit function-like structure through composition and asymptotic differentiation.

Chapter 32

Dynamical systems, normal forms, and iteration

32.1 Normal forms remove terms scale by scale

Near a fixed point, a differential equation or iteration can often be simplified by a change of coordinates. Ordinary power series remove nonresonant polynomial terms. Exponentially small or logarithmic effects remain after every algebraic order and require transseries.

Suppose that, after a coordinate change near infinity,

$$x_{n+1} = F(x_n), \quad F(x) = x + 1 + \frac{a}{x} + O(x^{-2}).$$

An Abel coordinate V satisfies

$$V(F(x)) = V(x) + 1. \tag{32.1}$$

One expects

$$V(x) = x - a \log x + \frac{b_1}{x} + \cdots,$$

together with exponentially small corrections that encode global analytic data. Once V is known, fractional iterates are

$$F^{\circ t}(x) = V^{-1}(V(x) + t).$$

This is the construction introduced formally in Chapter 14; resurgence describes how different analytic normalizations of V are related.

32.2 Dulac series

Near a singular point of a planar analytic vector field, transition maps can involve powers and logarithms such as

$$\sum_{\alpha,k} c_{\alpha,k} x^\alpha (\log x)^k. \tag{32.2}$$

These are called Dulac series or generalized Dulac expansions. Iterated transition maps can require exponential refinements as well. Such expansions play an important role in the study of limit cycles and local return maps.

The support conditions are again essential. Without a well-ordered dominance structure, composition of two return-map expansions could assign infinitely many contributions to one coefficient.

32.3 Quasianalyticity

A class of functions is quasianalytic if a function is determined by its full asymptotic expansion: no nonzero flat function belongs to the class. Ordinary smooth functions are not quasianalytic because e^{-1/x^2} is flat at 0. But carefully restricted classes of Dulac or analyzable functions can be quasianalytic.

Quasianalyticity matters in dynamics because a return map with zero formal expansion should then be the identity, ruling out hidden cycles or degeneracies. Transseries plus a summation theorem can provide exactly the additional structure needed to turn a formal expansion into a uniqueness statement.

32.4 Separatrix splitting

In nearly integrable systems, stable and unstable manifolds may coincide to all orders in a perturbation parameter yet differ by

$$\Delta(\varepsilon) \sim C\varepsilon^\beta e^{-A/\varepsilon}.$$

Every coefficient of the ordinary power series for Δ is zero. The actual splitting is beyond all algebraic orders and can create chaotic intersections. Transseries are therefore not a decorative refinement: the leading dynamical effect may live entirely in an exponential sector.

32.5 Difference equations

Difference equations naturally combine composition and asymptotics. A simple example is

$$y(x+1) - y(x) = \frac{1}{x}. \quad (32.3)$$

A formal solution begins

$$y(x) \sim \log x - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} - \cdots,$$

a version of the asymptotic expansion of the digamma function. Bernoulli numbers make the series divergent, and exponentially small periodic information is not captured by the inverse-power series alone.

Difference equations also lead beyond finite logarithmic depth. Solving

$$F(x+1) = \exp(F(x))$$

or constructing regular iteration of the exponential motivates transexponential and hyperseries extensions [23, 21, 24].

32.6 Parabolic germs and Écaille–Voronin invariants

A holomorphic germ tangent to the identity,

$$f(z) = z + az^{k+1} + \cdots,$$

has formal normal forms that do not completely determine its analytic conjugacy class. Sectorial Fatou coordinates differ on overlaps by exponentially small maps. These differences, organized into Écaille–Voronin invariants, are a canonical example of analytic data invisible to ordinary formal power series.

The moral is the same as for the Euler equation: formal normalization leaves flat ambiguities, and Stokes-type transition data classifies the analytic objects.

Chapter summary

Transseries enter dynamics through normal forms, fractional iteration, Dulac maps, separatrix splitting, and difference equations. They record effects that vanish to every power of a perturbation parameter and organize sectorial coordinate changes whose mismatches are analytic invariants.

Chapter 33

Saddles, instantons, and mathematical physics

33.1 The universal saddle expansion

Consider an integral

$$I(g) = \int_{\mathcal{C}} e^{-S(x)/g} a(x) dx, \quad g \rightarrow 0. \quad (33.1)$$

Critical points x_j of S satisfy $S'(x_j) = 0$. Expanding near one saddle gives

$$I_j(g) \sim e^{-S(x_j)/g} g^{\beta_j} \sum_{n=0}^{\infty} a_{j,n} g^n.$$

The full integral is a linear combination

$$I(g) \sim \sum_j \sigma_j e^{-S_j/g} g^{\beta_j} \widehat{\Phi}_j(g). \quad (33.2)$$

This is a transseries with actions S_j and parameters determined by how the contour decomposes into steepest-descent cycles.

The Airy integral of Chapter 25 is the simplest nontrivial example. In more dimensions, saddles can be isolated critical points, critical manifolds, complex solutions, or singular endpoints.

33.2 Instantons

In quantum mechanics or field theory, a Euclidean classical solution connecting distinct vacua is called an instanton. Its action A produces a factor

$$e^{-A/g} \quad \text{or} \quad e^{-A/\hbar}.$$

Fluctuations around the instanton produce a divergent power series. Configurations with k instantons produce sectors $e^{-kA/g}$, often with powers and logarithms of g arising from zero modes and interactions.

The term *instanton sector* is therefore a physical realization of the formal sector index k in (26.1).

33.3 Ambiguity cancellation

In many problems, the perturbative Borel transform has a singularity on the physical positive axis. Its two lateral sums differ by an imaginary quantity of order $e^{-A/g}$. A nonperturbative saddle sector has its own ambiguous continuation, with the opposite imaginary jump. The sum is real and unambiguous once all required sectors are included.

Symbolically,

$$\mathrm{Im} \mathcal{S}_{0\pm} \hat{\Phi}_0 + \mathrm{Im} \mathcal{S}_{0\pm} \left(e^{-A/g} \hat{\Phi}_1 \right) = 0.$$

This cancellation is sometimes called the Bogomolny–Zinn-Justin mechanism in quantum-mechanical examples. Resurgence explains it as one Stokes automorphism acting consistently on the whole transseries.

33.4 Perturbation theory predicts nonperturbative physics

Large-order behavior of perturbative coefficients can reveal:

1. the smallest instanton or complex-saddle action;
2. the power-law prefactor from fluctuation zero modes;
3. Stokes constants;
4. low-order coefficients around a nonperturbative saddle.

Conversely, a semiclassical saddle calculation predicts the divergence pattern of high perturbative orders. This two-way relation is one of the most useful consequences of resurgence.

33.5 Renormalons and other singularities

Not every Borel singularity in quantum field theory is associated with a finite-action classical solution. Renormalization-group effects can produce *renormalon* singularities. Their status and interpretation depend on the theory, observable, and regularization. A complete transseries may therefore need sectors motivated by Borel-plane analysis rather than by obvious real instantons.

This is a useful warning against identifying “nonperturbative” exclusively with “real classical solution.” Complex saddles, singular configurations, and collective effects can all contribute.

33.6 Exact WKB

For a Schrödinger-type equation

$$-\hbar^2 \psi''(x) + Q(x) \psi(x) = 0, \tag{33.3}$$

set

$$\psi(x) = \exp\left(\frac{1}{\hbar} \int^x S(t, \hbar) dt\right).$$

The Riccati equation

$$S^2 + \hbar S' = Q$$

produces a divergent WKB series. Its Borel sums change across Stokes curves emanating from turning points. Period integrals

$$A_\gamma = \oint_\gamma \sqrt{Q(x)} dx$$

act as instanton actions. Exact quantization conditions combine perturbative periods and exponentially small tunneling sectors into resurgent relations.

33.7 Many modern arenas

Resurgent transseries appear in matrix models, topological strings, gauge theories, hydrodynamic gradient expansions, nonlinear wave equations, and semiclassical spectral problems. The details differ, but the reusable pattern is stable:

$$\text{saddles/actions} \longleftrightarrow \text{Borel singularities} \longleftrightarrow \text{large-order growth} \longleftrightarrow \text{Stokes data}.$$

The primer by Aniceto, Başar, and Schiappa develops this viewpoint across many physical examples [2]; Dorigoni gives an accessible entry to alien calculus [10].

Common trap

In an infinite-dimensional path integral, the “integral” may itself be formal, and existence of all required saddles or resummations can be difficult. The finite-dimensional saddle picture is a guide, not an automatic theorem for every quantum field theory.

Chapter summary

Small-coupling integrals and semiclassical problems decompose into saddle sectors $e^{-S_j/g}$ times divergent fluctuation series. Instantons are particular saddles; large perturbative orders encode their data. Stokes ambiguities cancel only after all coupled sectors are transformed and summed consistently.

Chapter 34

Computer algebra and numerical exploration

34.1 Why representation matters

A computer cannot store an infinite support. It stores a finite truncation plus a description of the admissible monomial grid. A practical transseries object might contain:

1. a ratio set $\boldsymbol{\mu} = (\mu_1, \dots, \mu_r)$;
2. an exponent lower bound $\mathbf{k}_0 \in \mathbb{Z}^r$;
3. a finite map from exponent vectors to coefficients;
4. a truncation cut expressed in dominance order;
5. recursive expression nodes for exponentials, logarithms, and composition.

The mathematical support theorem guarantees that omitted terms cannot contribute above the cut in infinitely many hidden ways.

34.2 Canonicalization

Different generator expressions can denote the same monomial:

$$(e^{-x})^2 = e^{-2x}, \quad x^{-1}e^{-\log x} = x^{-2}.$$

A computer algebra system must normalize logarithmic exponents, combine exponentials, and detect relations among ratio generators. Otherwise equal terms will fail to collect and leading-term comparisons will be wrong.

One robust approach represents a transmonomial recursively by a normalized exponent L in e^L , with L itself a purely large transseries, together with explicit powers from lower logarithmic levels. Comparison then descends through the recursive structure.

34.3 Truncated multiplication

Given finite maps for T and U , naive multiplication forms every pair of stored terms. Better algorithms exploit order:

1. sort terms by decreasing dominance;

2. use a priority queue of pairwise products;
3. emit the largest unseen product;
4. merge duplicates;
5. stop as soon as the next possible product lies below the truncation cut.

This is analogous to multiplying sparse multivariate series with an admissible monomial order.

34.4 Recursive expansion engines

Operations such as inverse, logarithm, exponential, and composition reduce to universal templates after leading-term factorization:

$$\begin{aligned}
 (1 + U)^{-1} &= \sum_{n \geq 0} (-U)^n, \\
 \log(1 + U) &= \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n, \\
 e^U &= \sum_{n \geq 0} \frac{U^n}{n!}, \\
 T(x + U) &= \sum_{n \geq 0} \frac{T^{(n)}(x)}{n!} U^n.
 \end{aligned}$$

The engine needs a proof that $U \prec 1$ in the relevant asymptotic scale and a termination test showing that U^n eventually lies below the requested cut.

34.5 Equality is harder than expansion

For a normalized finite truncation, equality is easy. Exact zero testing for rich transseries expressions is subtler because cancellation may occur after deep functional identities or normalization across exponential levels. Differential algebra and model theory supply decision procedures for important classes, while recent work develops explicit algorithms for larger logarithmic-exponential transseries fields [5].

No symbolic system should silently infer exact zero merely because every term computed so far vanished. That proves only that the expression lies below the current cut.

34.6 Numerical Borel analysis

A practical numerical pipeline is:

1. generate N coefficients at high precision;
2. divide by the expected gamma growth;
3. build Padé or conformal-Borel approximants;
4. locate stable singularity patterns;
5. integrate laterally with adaptive quadrature;
6. compare the jump with the predicted neighboring sector;
7. fit transseries parameters to boundary or initial data.

Conformal mapping can greatly improve Padé resolution. If the nearest Borel singularity is known to be at $A > 0$ with a cut $[A, \infty)$, a map from the cut plane to the unit disk places the entire first sheet inside a convergent Taylor region. The precise map should reflect the known singularity geometry rather than being chosen mechanically.

34.7 Richardson acceleration

Suppose a sequence has

$$q_n = q + \frac{c_1}{n} + \frac{c_2}{n^2} + \cdots.$$

Richardson transforms eliminate successive inverse powers of n , accelerating convergence to q . Large-order tests routinely use this after dividing out the leading $\Gamma(n + \beta)A^{-n}$ growth.

A simple first transform is

$$R_n = (n + 1)q_{n+1} - nq_n = q + O(n^{-2}).$$

Repeated or generalized versions can expose Stokes constants hidden behind slow $1/n$ corrections.

34.8 Verification by exact arithmetic when possible

Many coefficient recurrences use rational numbers and symbolic parameters. Compute them exactly before converting to floating point. Exact arithmetic can reveal a missed cancellation, a hidden factorization, or an integer sequence. Use high-precision numerical arithmetic only for analytic continuation and resummation steps that genuinely require it.

34.9 A minimal data contract

A reliable transseries library should distinguish:

Formal equality

proved after complete normalization;

Equality modulo a cut

agreement of all terms above a specified monomial;

Asymptotic equivalence

agreement of leading terms;

Numerical agreement

closeness after a specified summation and parameter choice.

Conflating these notions is a common source of plausible but incorrect symbolic output.

Chapter summary

Computer algebra stores finite truncations on certified monomial grids. Canonicalization, dominance-aware multiplication, and cut-aware recursive expansion are essential. Exact equality is deeper than matching a truncation. For divergent sectors, high-precision Borel–Padé and large-order analysis connect symbolic coefficient generation to numerical resummation.

Chapter 35

Extensions, boundaries, and several meanings of “transseries”

35.1 Two traditions

The word *transseries* is used in two overlapping traditions.

1. In ordered algebra and model theory, a transseries is a Hahn-like formal sum of logarithmic-exponential monomials, equipped with order, derivative, exponential, logarithm, and composition.
2. In asymptotic analysis and physics, a transseries is often written as a sum of exponential sectors, each carrying a divergent power series and one or more free parameters.

The first tradition emphasizes legal infinite algebra and universal closure. The second emphasizes Borel summation, Stokes data, and analytic reconstruction. They meet when the sector prefactors and fluctuations are embedded into a common ordered monomial field, but neither vocabulary should be forced into the other’s narrowest notation.

35.2 Finite height and depth

An ordinary logarithmic-exponential transseries has finite exponential height and finite logarithmic depth for each element. It may contain

$$\exp(\exp(x) + x^2)$$

but not automatically an object requiring infinitely many nested exponentials in one element. Likewise, each transseries uses only finitely many iterated logarithms as basic scales.

This elementwise finiteness still allows a field containing objects of every finite height and depth. It is analogous to the set of all finite polynomials: there is no uniform degree bound, but each polynomial has finite degree.

35.3 Transexponentials and hyperseries

Functional equations involving iteration can demand new scales. A transexponential grows faster than every finite iterate of the ordinary exponential while satisfying an iteration relation adapted

to exponentiation. Logarithmic hyperseries and related constructions systematically adjoin such transfinite levels [23, 24, 21].

These extensions preserve the central design principles: ordered monomials, well-based supports, recursive closure, and a derivative compatible with asymptotic growth. What changes is the ordinal length of the scale hierarchy.

35.4 Oscillation

Real transseries ordered by eventual magnitude do not directly encode persistent oscillation such as $\sin x$, because it has no eventual sign and neither tends to zero nor infinity. Complex exponential monomials

$$e^{i\phi(x)}$$

can be adjoined, but their magnitude may be 1, so phase order must be tracked separately from growth order. WKB and Fourier-type transseries therefore use a richer combination of exponential actions and oscillatory phases.

For z in a complex sector, e^{-Az} can itself switch between decay, growth, and oscillation as the argument rotates. Complex transseries are naturally sectorial.

35.5 Branching and logarithms

The complex logarithm is multivalued. A formal expression involving $\log z$ or z^α must be tied to a branch and a sector. Analytic continuation around the origin can add $2\pi i$ to a logarithm and change sector parameters. Formal algebra alone does not choose the branch.

Similarly, a Borel transform can live on a ramified Riemann surface rather than the punctured plane. Alien calculus records continuation among sheets and around singularities.

35.6 Not every function has a useful transseries

Some functions have dense oscillations, natural boundaries, or asymptotic behavior not captured by a selected monomial scale. A function can possess a Poincaré expansion yet fail to be resurgent. It can also require generalized monomials not present in the basic field.

The correct conclusion is not that transseries failed, but that every formal language has a domain. One should ask:

1. Which scales are generated by the problem?
2. Is the support well based?
3. What analytic continuation is available?
4. Which summation method matches the coefficient growth?

35.7 Formal existence versus analytic existence

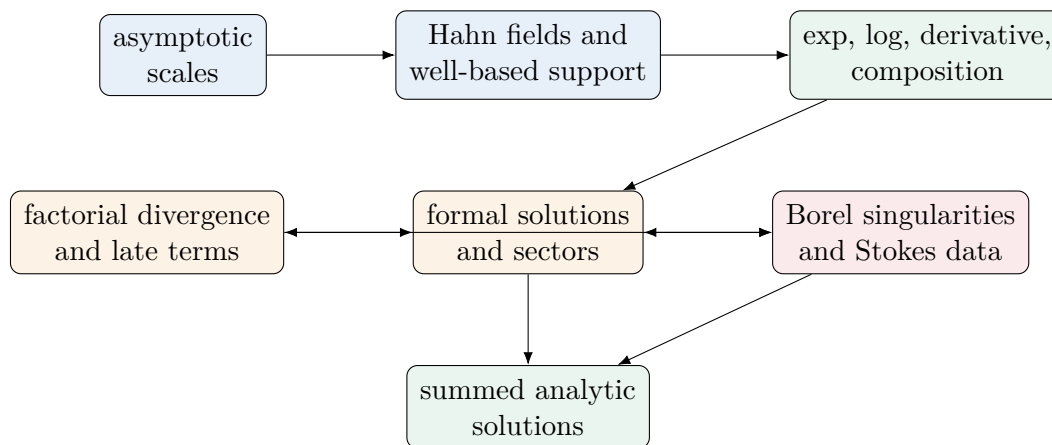
A formal transseries solution can often be constructed recursively even when no analytic solution has yet been proved. Turning it into a function requires additional theorems: summability, fixed-point estimates, sectorial analyticity, or integral representations. Conversely, an analytic solution may exist but have no expansion in a chosen transseries class.

Keep the logical levels separate:

formal construction $\neq$ summability $\neq$ global analytic continuation.

A rigorous argument states which level has been established.

35.8 A compact map of the subject



Chapter summary

“Transseries” names both a formal ordered field and a sectorial language for resurgent asymptotics. Basic logarithmic-exponential fields have elementwise finite height and depth; hyperseries extend them. Oscillation, branching, and nonresurgent singularity patterns require extra structure. Formal existence, summability, and global analytic existence are distinct claims.

Chapter 36

A brief historical and bibliographic guide

36.1 Long before the name

Asymptotic expansions are older than transseries. Euler freely manipulated divergent series in the eighteenth century. Poincaré clarified asymptotic expansions in the nineteenth century. Stokes discovered that asymptotic coefficients of special functions change their effective form across rays in the complex plane. Hardy’s *Orders of Infinity* organized logarithmic and exponential growth scales [17].

These developments supplied the analytic motivation: ordinary power series were too small a language for many limiting problems.

36.2 Generalized power series

Hahn’s 1907 construction placed formal sums over ordered abelian groups on a firm algebraic foundation [16]. Neumann’s lemma later supplied a powerful support theorem ensuring that products and geometric-series inverses remain well defined [19]. Modern transseries inherit this Hahn-field backbone.

The lesson of this algebraic line is that “infinite formal expression” is not enough. One must choose an ordered monomial group and a support condition that makes every coefficient a finite sum.

36.3 Dahn–Göring, Écaille, and independent origins

Logarithmic-exponential transseries were introduced in closely related forms by Dahn and Göring in model-theoretic work and by Jean Écaille in his theory of analyzable functions and resurgence [9, 11]. The two traditions emphasized different questions:

- ordered fields, exponentiation, and logical structure on one side;
- divergent asymptotics, alien calculus, and Stokes phenomena on the other.

The modern subject benefits from both.

36.4 Computer algebra and van der Hoeven's synthesis

Joris van der Hoeven developed transseries as a systematic computational language, including algorithms for arithmetic, composition, differentiation, integration, differential equations, and generalized summation [27, 28]. His monograph is a central source for the formal and algorithmic viewpoint. Edgar's survey gives a shorter, non-specialist construction with particular attention to grids and witnesses [13].

36.5 Model theory and H -fields

The papers of van den Dries, Macintyre, and Marker connected logarithmic-exponential series with o-minimal structures and Hardy fields [25, 26]. The later monograph of Aschenbrenner, van den Dries, and van der Hoeven developed the broad asymptotic differential algebra and model theory summarized in Chapter 30 [1].

36.6 Resurgence in analysis and physics

Costin developed rigorous generalized Borel summation and transseries for broad classes of nonlinear differential systems [8, 6, 7]. Sauzin and Dorigoni provide entry points to Écaille's resurgent framework [22, 10]. The extensive primer of Aniceto, Başar, and Schiappa explains large-order relations, nonlinear resonance, alien calculus, and physical applications [2].

36.7 Surreals and larger scales

Berarducci and Mantova constructed compatible transserial derivations on the surreal numbers and interpreted transseries as germs of surreal functions [4, 3]. Logarithmic hyperseries and transexponential fields extend the scale beyond ordinary finite-height logarithmic-exponential series [24, 21].

36.8 Suggested next reading

After this tutorial, choose according to purpose:

Formal calculations

Edgar's *Transseries for Beginners*, then van der Hoeven's *Transseries and Real Differential Algebra*.

Ordered differential algebra

Aschenbrenner–van den Dries–van der Hoeven, *Asymptotic Differential Algebra and Model Theory of Transseries*.

Borel summation and rigorous ODEs

Costin's *Asymptotics and Borel Summability*.

Alien calculus

Dorigoni's introduction, then Sauzin's notes and Écaille's original works.

Physics and detailed examples

Aniceto–Başar–Schiappa's primer.

Surreal connections

Berarducci–Mantova and Aschenbrenner’s essay on numbers, germs, and transseries.

Chapter summary

Transseries grew from several lines: classical asymptotics and Stokes phenomena, Hahn’s generalized series, model theory, Écaille’s resurgence, and computer algebra. The modern subject is not owned by one notation. Its unity lies in ordered asymptotic scales, legal infinite algebra, and the recovery of exponentially small analytic information.

Part VIII

Problems, solutions, and reference material

Chapter 37

Guided exercises

The exercises are grouped roughly by topic, not strictly by difficulty. Most require only algebra and calculus. A star marks a problem that introduces a slightly more advanced idea. Complete solutions begin in Chapter 38.

37.1 Asymptotic scales and first transseries

Exercise 37.1 (Ordering a mixed scale). Arrange the following monomials from largest to smallest as $x \rightarrow +\infty$:

$$e^{e^x}, \quad e^{x^2}, \quad x^x, \quad x^{-100}e^{3x}, \quad x^{100}, \quad (\log x)^7, \quad 1, \quad x^{-2}, \quad e^{-\sqrt{x}}, \quad e^{-x}.$$

Justify every neighboring comparison by taking a quotient or logarithm.

Exercise 37.2 (A rational expansion). Expand

$$\frac{x^2 + 1}{x^2 - x}$$

through the term of order x^{-4} , and verify the result by multiplying by the denominator.

Exercise 37.3 (Beyond every power). Prove that

$$e^{-\sqrt{x}} \prec x^{-N}$$

for every fixed $N > 0$. Then compare e^{-x} and $e^{-\sqrt{x}}$.

Exercise 37.4 (Flat ambiguity). Suppose

$$f(x) \sim_{\text{as}} \sum_{n \geq 0} a_n x^{-n} \quad (x \rightarrow +\infty).$$

Show directly from the definition that $f(x) + Ce^{-x}$ has the same asymptotic power series for every constant C . Why does this not imply that all the functions are equal?

Exercise 37.5 (Additive and multiplicative normal form). Let

$$T = 3e^{x^2+x} - 5x^{100}e^{x^2} + 7x^{-2}.$$

Find its dominant monomial, leading coefficient, additive decomposition into giant/constant/small parts, and multiplicative form $T = am(1 + U)$ with $U \prec 1$.

37.2 Supports and Hahn-series algebra

Exercise 37.6 (Which supports are legal?). For each formal sum below, decide whether its support is well based in the large- x dominance order used in this book:

$$\sum_{n \geq 0} x^{-n}, \quad \sum_{n \geq 0} x^n, \quad \sum_{n \geq 1} x^{-1/n}, \quad \sum_{n \geq 0} e^{-nx}.$$

Explain the obstruction in each illegal case.

Exercise 37.7 (Why a product coefficient is finite). Let

$$A = \sum_{n \geq 0} x^{-n}, \quad B = \sum_{n \geq 0} (n+1)x^{-n}.$$

Compute the coefficient of x^{-k} in AB . Why are only finitely many pairs of terms involved for each fixed k ? Sum the answer in closed form as an ordinary rational function and check it.

Exercise 37.8 (A two-scale inverse). Expand

$$\frac{1}{1 + x^{-1} + e^{-x}}$$

through x^{-3} in the purely algebraic sector and through $x^{-2}e^{-x}$ in the sector linear in e^{-x} . Ignore terms containing e^{-2x} . Explain why this “rectangular” truncation is not an initial segment of the full dominance order.

Exercise 37.9 (Logarithm of a perturbed monomial). Find the expansion through x^{-3} of

$$\log \left[x^3 \left(1 + \frac{2}{x} - \frac{1}{x^2} \right) \right].$$

Exercise 37.10 (Asymptotic integration). Seek a formal antiderivative of e^{x^2} in the form

$$I(x) = \frac{e^{x^2}}{2x} \sum_{n=0}^{\infty} c_n x^{-2n}.$$

Derive a recurrence for c_n and compute c_0, c_1, c_2, c_3 . Verify the result by differentiation through the displayed orders.

37.3 Composition, inverse, and recursive equations

Exercise 37.11 (A tiny inner perturbation). Expand $e^{x+e^{-x}}$ through the term proportional to e^{-2x} after collecting absolute transmonomials. Check the same result using Taylor’s formula for $T(x+U)$.

Exercise 37.12 (Inverse of $t + \log t$). The inverse of $T(t) = t + \log t$ is $S(x) = W(e^x)$. Put $L = \log x$ and verify by composition that

$$S(x) = x - L + \frac{L}{x} + \frac{L^2/2 - L}{x^2} + \frac{L^3/3 - 3L^2/2 + L}{x^3} + O\left(\frac{L^4}{x^4}\right).$$

Exercise 37.13 (An exponential fixed point). Solve

$$Y = x + e^{-Y}$$

through the fifth power of $q = e^{-x}$. First derive the coefficient recursion by substitution. Then recognize the closed form involving Lambert W .

Exercise 37.14 (A logarithmic fixed point). Find a transseries solution through x^{-3} of

$$U = \log(x + U).$$

Write the coefficients as polynomials in $L = \log x$ and verify the residual.

Exercise 37.15 (Riccati coefficient recursion). For

$$y' + y = \frac{1}{x} + y^2,$$

insert $y = \sum_{n \geq 1} a_n x^{-n}$. Derive the recurrence for a_n and compute $a_1, \dots, a_6$.

Exercise 37.16 (A boundary-layer parameter). For

$$\varepsilon y' + y = \frac{1}{1+x}, \quad y(0) = y_0,$$

write the exact solution with the integral beginning at 0. Show formally that, near the outer region, the coefficient of $e^{-x/\varepsilon}$ begins

$$C(\varepsilon) \sim_{\text{as}} y_0 - 1 - \varepsilon - 2!\varepsilon^2 - 3!\varepsilon^3 - \dots.$$

Interpret this result without pretending that the factorial series converges.

37.4 Borel summation and Stokes jumps

Exercise 37.17 (An alternating Euler series). Let

$$\widehat{F}(z) = \sum_{n=0}^{\infty} \frac{(-1)^n n!}{z^{n+1}}.$$

Compute its Borel transform. Show that its positive-direction Borel sum is unambiguous for $z > 0$, and verify directly that the summed function satisfies

$$F' - F = -\frac{1}{z}.$$

Exercise 37.18 (Convolution in action). Using Exercise 37.17, compute

$$\mathcal{B}(\widehat{F}^2)(\xi)$$

by convolution and simplify the integral to an elementary function.

Exercise 37.19 (The sign of the Euler jump). Starting from

$$\mathcal{B}\widehat{E}(\xi) = \frac{1}{1-\xi},$$

use a small contour around $\xi = 1$ to derive

$$\mathcal{S}_{0+}\widehat{E} - \mathcal{S}_{0-}\widehat{E} = 2\pi i e^{-z}$$

with the lateral convention of Chapter 24. Draw the orientation and account for both minus signs.

Exercise 37.20 (A pole predicts late coefficients). Suppose

$$\mathcal{B}\widehat{\Phi}(\xi) = h(\xi) + \frac{r}{\xi - A},$$

where h is analytic in a disk larger than $|A|$. Derive the leading large- n behavior of the coefficients in $\widehat{\Phi}(z) = \sum a_n z^{-n-1}$. What changes if two poles have the same modulus?

Exercise 37.21 (Least-term estimate). For $z = 100$, estimate the best truncation index and least-term magnitude of

$$\sum_{n \geq 0} \frac{n!}{z^{n+1}}.$$

Use Stirling's formula. Compare the answer with the exponential scale e^{-100} .

Exercise 37.22 (Higher Gevrey growth). Consider $a_n = (2n)!$. Determine its Gevrey order under the bound (22.6). Show that the ordinary Borel series $\sum a_n \xi^n / n!$ still has radius zero. What kind of generalized Borel transform is suggested?

Exercise 37.23 (Oscillation from conjugate singularities). Let

$$\widehat{\varphi}(\xi) = \frac{1}{1 - 2(\cos \theta)\xi + \xi^2}, \quad 0 < \theta < \pi.$$

Find its Taylor coefficients and hence the coefficients a_n of the formal inverse-power series whose Borel transform is $\widehat{\varphi}$. Explain the oscillation in terms of the two singularities $e^{\pm i\theta}$.

Exercise 37.24 (Median value of the Euler integral). For real $z > 0$, show that

$$\frac{1}{2} (\mathcal{S}_{0+}\widehat{E} + \mathcal{S}_{0-}\widehat{E})$$

equals the Cauchy principal value of

$$\int_0^\infty \frac{e^{-z\xi}}{1-\xi} d\xi.$$

Why is this average real? How does it relate to median resummation in this linear example?

37.5 Resurgence, saddles, and Stokes data

Exercise 37.25 (Bridge equation gives a translation). Assume

$$\dot{\Delta}_A \widehat{\Phi} = S_1 \partial_\sigma \widehat{\Phi}$$

and that no other singularity lies on the chosen Stokes ray. Derive the action of the Stokes automorphism on σ . Then derive its half-power used in median summation.

Exercise 37.26 (Airy prefactor). Insert

$$y(x) = e^{\lambda(2/3)x^{3/2}} x^\alpha, \quad \lambda = \pm 1,$$

into $y'' - xy = 0$. After the leading x terms cancel, determine α from the next largest power. What is the size of the residual after this choice?

Exercise 37.27 (Complex saddles of a quartic integral). Consider

$$I(g) = \int_{-\infty}^{\infty} \exp \left[-\frac{1}{g} \left(\frac{x^2}{2} + \frac{x^4}{4} \right) \right] dx, \quad g > 0.$$

Find all critical points of the action and their action values. Rescale $x = \sqrt{g}t$ and derive the formal perturbative coefficients about $x = 0$. Explain why they alternate and grow factorially, and relate the sign to the nearest complex saddles.

Exercise 37.28 (Adding actions). Suppose a nonlinear equation has sectors with actions A and B . Show algebraically and in the Borel plane why a sector with action $A + B$ should be expected. What special issue occurs when $B = 2A$ and the equation already has an independent sector of action $2A$?

Exercise 37.29 (A transasymptotic logistic solution). For the exact logistic solution

$$y(x) = \frac{1}{1 + Ae^{-x}},$$

set $\tau = Ae^{-x}$. Show that the sector expansion is a geometric series in τ , while the transasymptotic rearrangement sums it to one coefficient function $F_0(\tau)$. Explain which representation is preferable when $|\tau| \ll 1$ and when $\tau = O(1)$.

37.6 Structural and computational questions

Exercise 37.30 (H -field intuition). For each of the germs

$$x^3 + \log x, \quad 7 + x^{-1}, \quad e^{-x},$$

identify whether it is larger than every constant, bounded with a constant part, or infinitesimal. Check the two H -field axioms on these examples.

Exercise 37.31 (Logarithmic derivatives and scale). Compute the logarithmic derivatives of

$$x^a (\log x)^b, \quad e^{x^p}, \quad e^{e^x}.$$

Order the resulting logarithmic derivatives for fixed nonzero a, b and $p > 1$. What do they say about sensitivity of the original monomials to a small change in x ?

Exercise 37.32 (A surreal evaluation). Formally evaluate

$$T(x) = x^2 + 3x + \log x + 5 + e^{-x}$$

at $x = \omega$. Identify the leading Conway monomial and order every displayed term. Which term is infinitesimal? Why does this evaluation not by itself define function composition on all surreal numbers?

Exercise 37.33 (A difference-equation check). Verify by expansion through x^{-5} that

$$y(x) = \log x - \frac{1}{2x} - \frac{1}{12x^2} + \frac{1}{120x^4} + O(x^{-6})$$

satisfies

$$y(x+1) - y(x) = \frac{1}{x} + O(x^{-6}).$$

Why is an additive constant not determined by the difference equation?

Exercise 37.34 (A finite ratio grid). Let

$$\mu_1 = x^{-1}, \quad \mu_2 = e^{-x},$$

and consider

$$T = 1 + 2x^{-1} + 3x^{-2} + e^{-x}(1 - 4x^{-1} + 7x^{-3}) + 5e^{-2x}x^{-2}.$$

Represent every monomial by an exponent vector (k_1, k_2) with $\mu_1^{k_1}\mu_2^{k_2}$. Draw the occupied lattice points. Give a rectangular cut that retains all displayed terms and state why it is not the same as a single dominance cut.

Exercise 37.35 (Exact equality or equality to a cut?). A symbolic program expands

$$\log(e^x(1 + e^{-x})) - x - \log(1 + e^{-x})$$

through e^{-20x} and obtains zero. What can it conclude? Now simplify the expression using exact formal identities. Contrast the two arguments.

Exercise 37.36 (Classify the claim). Classify each statement as formal, asymptotic, summability, or global analytic. Explain why no statement automatically implies all the others.

- (a) A recursive transseries solves an ODE coefficient by coefficient.
- (b) Its Borel transform continues along a ray with exponential growth bounds.
- (c) The Borel sum solves the ODE in a sector.
- (d) That solution extends meromorphically to the whole plane.
- (e) The residual after N terms is $O(x^{-N-1})$.

Exercise 37.37 (Capstone: the Euler equation). For $y' + y = 1/z$, carry out the complete workflow:

- (a) derive the perturbative recurrence;
- (b) identify the homogeneous sector;
- (c) compute the Borel transform;
- (d) calculate the lateral jump;
- (e) derive the parameter transformation;
- (f) estimate optimal truncation;
- (g) verify that the Borel sums satisfy the ODE.

State clearly which steps are formal and which are analytic.

Chapter 38

Complete solutions

38.1 Solutions to asymptotic-scale exercises

Solution to Exercise 37.1

The order is

$$\begin{aligned} e^{e^x} &\succ e^{x^2} \succ x^x \succ x^{-100}e^{3x} \succ x^{100} \succ (\log x)^7 \\ &\succ 1 \succ x^{-2} \succ e^{-\sqrt{x}} \succ e^{-x}. \end{aligned}$$

For the first comparison, take logarithms of the quotient:

$$\log \frac{e^{e^x}}{e^{x^2}} = e^x - x^2 \rightarrow +\infty.$$

Next, $x^x = e^{x \log x}$ and $x^2 - x \log x = x(x - \log x) \rightarrow +\infty$. Also,

$$\log \frac{x^x}{x^{-100}e^{3x}} = x \log x - 3x + 100 \log x \rightarrow +\infty,$$

and

$$\log \frac{x^{-100}e^{3x}}{x^{100}} = 3x - 200 \log x \rightarrow +\infty.$$

Every positive power of x dominates every power of $\log x$, while $(\log x)^7 \rightarrow \infty$. The comparisons around 1 are immediate. Finally, $x^{-2}/e^{-\sqrt{x}} = x^{-2}e^{\sqrt{x}} \rightarrow \infty$, and

$$\frac{e^{-\sqrt{x}}}{e^{-x}} = e^{x-\sqrt{x}} \rightarrow \infty.$$

Solution to Exercise 37.2

Divide numerator and denominator by x^2 :

$$\frac{x^2 + 1}{x^2 - x} = \frac{1 + x^{-2}}{1 - x^{-1}}.$$

Using $(1 - u)^{-1} = 1 + u + u^2 + \dots$ with $u = x^{-1}$,

$$(1 + x^{-2})(1 + x^{-1} + x^{-2} + x^{-3} + x^{-4} + \dots) = 1 + x^{-1} + 2x^{-2} + 2x^{-3} + 2x^{-4} + O(x^{-5}).$$

Multiplying by $1 - x^{-1}$ gives

$$1 + x^{-2} + O(x^{-5}),$$

which is the normalized numerator to the required order.

Solution to Exercise 37.3

Set $t = \sqrt{x}$. Then

$$x^N e^{-\sqrt{x}} = t^{2N} e^{-t}.$$

Repeated use of l'Hôpital's rule, or the fact that the exponential series contains a term larger than any fixed power, gives $t^{2N} e^{-t} \rightarrow 0$. Hence $e^{-\sqrt{x}} \prec x^{-N}$. Moreover,

$$\frac{e^{-x}}{e^{-\sqrt{x}}} = e^{-x+\sqrt{x}} \rightarrow 0,$$

so $e^{-x} \prec e^{-\sqrt{x}}$.

Solution to Exercise 37.4

Fix N . Since $e^{-x} = o(x^{-N})$,

$$[f(x) + Ce^{-x}] - \sum_{n=0}^{N-1} a_n x^{-n} = \left[f(x) - \sum_{n=0}^{N-1} a_n x^{-n} \right] + Ce^{-x} = O(x^{-N}).$$

This holds for every N , so all the functions have the same Poincaré expansion. They are not equal because their pairwise differences are $(C_1 - C_2)e^{-x}$, nonzero when $C_1 \neq C_2$. The power scale is simply too coarse to see that difference.

Solution to Exercise 37.5

The first two terms share the large factor e^{x^2} , but their quotient is

$$\frac{x^{100} e^{x^2}}{e^{x^2+x}} = x^{100} e^{-x} \rightarrow 0.$$

Thus

$$\text{mag } T = e^{x^2+x}, \quad \text{lc}(T) = 3.$$

Both exponential terms are giants and $7x^{-2}$ is small, so

$$T_{\text{giant}} = 3e^{x^2+x} - 5x^{100}e^{x^2}, \quad T_{\text{constant}} = 0, \quad T_{\text{small}} = 7x^{-2}.$$

Factoring the leading term gives

$$T = 3e^{x^2+x} \left(1 - \frac{5}{3}x^{100}e^{-x} + \frac{7}{3}x^{-2}e^{-x^2-x} \right).$$

Both correction monomials tend to zero, so the parenthesized correction is $1 + U$ with $U \prec 1$.

38.2 Solutions to support and algebra exercises**Solution to Exercise 37.6**

The supports of

$$\sum_{n \geq 0} x^{-n} \quad \text{and} \quad \sum_{n \geq 0} e^{-nx}$$

are well based: each has a largest monomial 1, and moving through the index n gives a strictly decreasing chain.

The support $\{x^n : n \geq 0\}$ is illegal for a large- x transseries because it has no largest monomial and contains the infinite increasing chain $1 \prec x \prec x^2 \prec \dots$.

The support $\{x^{-1/n} : n \geq 1\}$ is also illegal. Its monomials increase toward 1:

$$x^{-1} \prec x^{-1/2} \prec x^{-1/3} \prec \dots,$$

and no member is largest. The fact that the exponents converge does not rescue the support; the order condition is combinatorial, not topological.

Solution to Exercise 37.7

To obtain x^{-k} , choose x^{-j} from A and $x^{-(k-j)}$ from B . The allowed values are $j = 0, \dots, k$, so the coefficient is the finite sum

$$\sum_{j=0}^k (k-j+1) = 1 + 2 + \dots + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Writing $r = x^{-1}$,

$$A = \frac{1}{1-r}, \quad B = \frac{1}{(1-r)^2}, \quad AB = \frac{1}{(1-r)^3}.$$

The binomial expansion

$$(1-r)^{-3} = \sum_{k \geq 0} \binom{k+2}{2} r^k$$

confirms the coefficient.

Solution to Exercise 37.8

Put $a = x^{-1}$ and $q = e^{-x}$. Expand first in q :

$$\frac{1}{1+a+q} = \frac{1}{1+a} - \frac{q}{(1+a)^2} + O(q^2).$$

Now

$$\begin{aligned} (1+a)^{-1} &= 1 - a + a^2 - a^3 + O(a^4), \\ (1+a)^{-2} &= 1 - 2a + 3a^2 - 4a^3 + \dots \end{aligned}$$

Therefore the requested two-scale truncation is

$$1 - x^{-1} + x^{-2} - x^{-3} - e^{-x} + 2x^{-1}e^{-x} - 3x^{-2}e^{-x} + \text{omitted terms.} \quad (38.1)$$

Every algebraic monomial x^{-N} dominates e^{-x} . Hence a genuine initial segment reaching the term e^{-x} would have to retain infinitely many algebraic powers. The finite selection in (38.1) is a useful rectangular cut in the (a, q) exponent lattice, not a single cut in total dominance order.

Solution to Exercise 37.9

Let $a = x^{-1}$ and $u = 2a - a^2$. Then

$$\log \left[x^3(1+u) \right] = 3 \log x + \log(1+u).$$

Through a^3 ,

$$\begin{aligned} u &= 2a - a^2, \\ u^2 &= 4a^2 - 4a^3 + O(a^4), \\ u^3 &= 8a^3 + O(a^4). \end{aligned}$$

Thus

$$\begin{aligned} \log(1+u) &= u - \frac{u^2}{2} + \frac{u^3}{3} + O(a^4) \\ &= 2a - 3a^2 + \frac{14}{3}a^3 + O(a^4). \end{aligned}$$

Therefore

$$\boxed{3 \log x + \frac{2}{x} - \frac{3}{x^2} + \frac{14}{3x^3} + O(x^{-4})}.$$

Solution to Exercise 37.10

Write

$$I = e^{x^2} F, \quad F = \frac{1}{2x} \sum_{n \geq 0} c_n x^{-2n}.$$

The equation $I' = e^{x^2}$ becomes

$$F' + 2xF = 1.$$

Equivalently, put $F = \sum_{n \geq 0} a_n x^{-2n-1}$, so $a_n = c_n/2$. Then

$$2a_0 = 1, \quad 2a_n = (2n-1)a_{n-1} \quad (n \geq 1).$$

Hence

$$c_0 = 1, \quad c_n = \frac{2n-1}{2} c_{n-1},$$

and

$$c_0 = 1, \quad c_1 = \frac{1}{2}, \quad c_2 = \frac{3}{4}, \quad c_3 = \frac{15}{8}.$$

Thus

$$I(x) \sim_{\text{as}} \frac{e^{x^2}}{2x} \left(1 + \frac{1}{2x^2} + \frac{3}{4x^4} + \frac{15}{8x^6} + \cdots \right).$$

Differentiating and using the recurrence cancels every negative even power in $F' + 2xF$ successively, leaving 1 to the computed order.

38.3 Solutions to composition and recursion exercises

Solution to Exercise 37.11

With $q = e^{-x}$,

$$e^{x+q} = e^x e^q = e^x \left(1 + q + \frac{q^2}{2} + \frac{q^3}{6} + \cdots \right).$$

After multiplying by e^x ,

$$e^{x+e^{-x}} = e^x + 1 + \frac{1}{2}e^{-x} + \frac{1}{6}e^{-2x} + O(e^{-3x}).$$

Taylor's formula gives the same result because every derivative of $T(x) = e^x$ equals e^x :

$$T(x+q) = \sum_{n \geq 0} \frac{T^{(n)}(x)}{n!} q^n = e^x \sum_{n \geq 0} \frac{q^n}{n!}.$$

Solution to Exercise 37.12

Write

$$S = x + s_0 + \frac{s_1}{x} + \frac{s_2}{x^2} + \frac{s_3}{x^3},$$

where

$$s_0 = -L, \quad s_1 = L, \quad s_2 = \frac{L^2}{2} - L, \quad s_3 = \frac{L^3}{3} - \frac{3}{2}L^2 + L.$$

Since

$$\log S = L + \log \left(1 + \frac{s_0}{x} + \frac{s_1}{x^2} + \frac{s_2}{x^3} + \cdots \right),$$

the logarithm through x^{-3} is

$$\log S = L + \frac{s_0}{x} + \frac{s_1 - s_0^2/2}{x^2} + \frac{s_2 - s_0s_1 + s_0^3/3}{x^3} + O(L^4x^{-4}).$$

Substitution gives

$$\log S = L - \frac{L}{x} + \frac{L - L^2/2}{x^2} + \frac{3L^2/2 - L - L^3/3}{x^3} + O(L^4x^{-4}).$$

Adding S cancels every displayed correction and yields $S + \log S = x + O(L^4x^{-4})$. Thus $T(S(x)) = x$ to the stated order.

Solution to Exercise 37.13

Set $Y = x + \delta$ and $q = e^{-x}$. The equation becomes

$$\delta = qe^{-\delta}. \tag{38.2}$$

Let $\delta = c_1q + c_2q^2 + \cdots$. Expanding $e^{-\delta} = 1 - \delta + \delta^2/2 - \cdots$ and comparing powers gives

$$c_1 = 1, \quad c_2 = -1, \quad c_3 = \frac{3}{2}, \quad c_4 = -\frac{8}{3}, \quad c_5 = \frac{125}{24}.$$

For the closed form, multiply (38.2) by e^δ :

$$\delta e^\delta = q,$$

so $\delta = W(q)$. Lagrange inversion gives

$$W(q) = \sum_{n \geq 1} \frac{(-n)^{n-1}}{n!} q^n.$$

Hence

$$Y = x + q - q^2 + \frac{3}{2}q^3 - \frac{8}{3}q^4 + \frac{125}{24}q^5 + O(q^6).$$

Solution to Exercise 37.14

Let $L = \log x$ and seek

$$U = L + \frac{a_1}{x} + \frac{a_2}{x^2} + \frac{a_3}{x^3} + \cdots,$$

where the a_j are polynomials in L . Since

$$\log(x + U) = L + \log\left(1 + \frac{U}{x}\right) = L + \frac{U}{x} - \frac{U^2}{2x^2} + \frac{U^3}{3x^3} + \cdots,$$

comparison gives

$$\begin{aligned} a_1 &= L, \\ a_2 &= a_1 - \frac{L^2}{2} = L - \frac{L^2}{2}, \\ a_3 &= a_2 - La_1 + \frac{L^3}{3} = L - \frac{3}{2}L^2 + \frac{1}{3}L^3. \end{aligned}$$

Thus

$$U = L + \frac{L}{x} + \frac{L - L^2/2}{x^2} + \frac{L - 3L^2/2 + L^3/3}{x^3} + O(L^4x^{-4}).$$

Substituting this expression into the logarithm reproduces it through x^{-3} , so the residual is $O(L^4x^{-4})$.

Solution to Exercise 37.15

The coefficient of x^{-1} gives $a_1 = 1$. For $m \geq 2$, the derivative contributes $-(m-1)a_{m-1}$ to the coefficient of x^{-m} , while y^2 contributes $\sum_{j=1}^{m-1} a_j a_{m-j}$. Hence

$$a_m = (m-1)a_{m-1} + \sum_{j=1}^{m-1} a_j a_{m-j}. \quad (38.3)$$

Successive values are

$$\begin{aligned} a_1 &= 1, \\ a_2 &= 1 + 1 = 2, \\ a_3 &= 2 \cdot 2 + (1 \cdot 2 + 2 \cdot 1) = 8, \end{aligned}$$

$$\begin{aligned}
a_4 &= 3 \cdot 8 + (1 \cdot 8 + 2 \cdot 2 + 8 \cdot 1) = 44, \\
a_5 &= 4 \cdot 44 + (44 + 16 + 16 + 44) = 296, \\
a_6 &= 5 \cdot 296 + (296 + 88 + 64 + 88 + 296) = 2312.
\end{aligned}$$

The rapidly growing numbers signal factorial divergence.

Solution to Exercise 37.16

Multiplying the ODE by $e^{x/\varepsilon}$ and integrating from 0 to x gives

$$y(x) = e^{-x/\varepsilon} \left[y_0 + \frac{1}{\varepsilon} \int_0^x \frac{e^{t/\varepsilon}}{1+t} dt \right]. \quad (38.4)$$

A formal outer particular solution is

$$\hat{y}_{\text{out}}(x, \varepsilon) = \sum_{n \geq 0} \frac{n! \varepsilon^n}{(1+x)^{n+1}}.$$

The complete formal solution is $\hat{y}_{\text{out}} + C(\varepsilon)e^{-x/\varepsilon}$. Imposing the boundary condition at $x = 0$ therefore gives, formally,

$$C(\varepsilon) = y_0 - \hat{y}_{\text{out}}(0, \varepsilon) \sim_{\text{as}} y_0 - 1 - \varepsilon - 2!\varepsilon^2 - 3!\varepsilon^3 - \dots.$$

This is an asymptotic description of the boundary-layer amplitude, not a convergent formula. It may be optimally truncated or Borel summed in an appropriate direction. Its role is to match the boundary data while the factor $e^{-x/\varepsilon}$ makes the correction disappear in the outer region.

38.4 Solutions to Borel and Stokes exercises

Solution to Exercise 37.17

Dividing each coefficient by $n!$ gives

$$\mathcal{B}\hat{F}(\xi) = \sum_{n \geq 0} (-1)^n \xi^n = \frac{1}{1+\xi}.$$

Its only finite singularity is at $\xi = -1$, so the positive Laplace ray is clear. For $z > 0$,

$$F(z) = \int_0^\infty \frac{e^{-z\xi}}{1+\xi} d\xi \quad (38.5)$$

converges absolutely. Differentiate under the integral sign:

$$\begin{aligned}
F'(z) &= - \int_0^\infty \frac{\xi e^{-z\xi}}{1+\xi} d\xi \\
&= - \int_0^\infty e^{-z\xi} d\xi + \int_0^\infty \frac{e^{-z\xi}}{1+\xi} d\xi \\
&= -\frac{1}{z} + F(z).
\end{aligned}$$

Thus $F' - F = -1/z$. Watson's lemma recovers the alternating factorial expansion.

Solution to Exercise 37.18

By the convolution rule,

$$\mathcal{B}(\widehat{F}^2)(\xi) = \int_0^\xi \frac{d\eta}{(1+\eta)(1+\xi-\eta)}.$$

Use

$$\frac{1}{(1+\eta)(1+\xi-\eta)} = \frac{1}{\xi+2} \left(\frac{1}{1+\eta} + \frac{1}{1+\xi-\eta} \right).$$

Both terms integrate to $\log(1+\xi)$, so

$$\boxed{\mathcal{B}(\widehat{F}^2)(\xi) = \frac{2\log(1+\xi)}{\xi+2}}.$$

The apparent singularity at $\xi = -2$ is entangled with the chosen branch of the logarithm; convolution has propagated the original singularity at -1 and created the summed location -2 .

Solution to Exercise 37.19

The upper contour runs from left to right above the pole; subtracting the lower integral means traversing the lower contour from right to left. Together they form a small clockwise loop around $\xi = 1$. Therefore

$$\mathcal{S}_{0+}\widehat{E} - \mathcal{S}_{0-}\widehat{E} = -2\pi i \operatorname{Res}_{\xi=1} \frac{e^{-z\xi}}{1-\xi}.$$

The residue is

$$\lim_{\xi \rightarrow 1} (\xi - 1) \frac{e^{-z\xi}}{1-\xi} = -e^{-z}.$$

The clockwise orientation supplies one minus sign and the denominator $1-\xi$ supplies the other:

$$-2\pi i(-e^{-z}) = 2\pi i e^{-z}.$$

Solution to Exercise 37.20

Expand the pole at the origin:

$$\frac{r}{\xi - A} = -r \sum_{n \geq 0} \frac{\xi^n}{A^{n+1}}.$$

Since $\widehat{\mathcal{B}\Phi} = \sum a_n \xi^n / n!$, the pole contributes

$$a_n = -r \frac{n!}{A^{n+1}}.$$

The coefficients of h are exponentially smaller relative to this term because h is analytic in a larger disk. Thus

$$a_n \sim -r \frac{n!}{A^{n+1}}.$$

If several singularities have the same modulus, their contributions must be added. For a conjugate pair, this sum often looks like $n!|A|^{-n} \cos(n\theta + \phi)$ and oscillates. A simple ratio test then need not converge.

Solution to Exercise 37.21

For $t_n = n!/100^{n+1}$,

$$\frac{t_{n+1}}{t_n} = \frac{n+1}{100}.$$

The terms decrease until $n+1$ reaches 100. The two terms t_{99} and t_{100} are equal and minimal, so one truncates around index 99 or 100. Stirling's formula gives

$$t_{100} = \frac{100!}{100^{101}} \sim \sqrt{\frac{2\pi}{100}} e^{-100} \approx 0.2507 e^{-100} \approx 9.3 \times 10^{-45}.$$

Thus the least term has exactly the predicted exponential scale e^{-100} , modulo an algebraic prefactor.

Solution to Exercise 37.22

The coefficients satisfy the Gevrey-2 bound with $C = A = 1$:

$$a_n = \Gamma(1 + 2n).$$

The ordinary Borel coefficients are $b_n = (2n)!/n!$. Their ratio is

$$\frac{b_{n+1}}{b_n} = \frac{(2n+2)(2n+1)}{n+1} = 2(2n+1) \rightarrow \infty.$$

Hence $\sum b_n \xi^n$ has radius zero. One should divide by $\Gamma(1+2n)$, or use the equivalent Borel transform of the corresponding higher order. Terminology alternates between “Gevrey-2 summation” and a reciprocal k -summability convention, so the gamma factor is safer than the bare name.

Solution to Exercise 37.23

The generating function for Chebyshev polynomials of the second kind is

$$\frac{1}{1 - 2t\xi + \xi^2} = \sum_{n \geq 0} U_n(t) \xi^n.$$

For $t = \cos \theta$,

$$U_n(\cos \theta) = \frac{\sin((n+1)\theta)}{\sin \theta}.$$

Therefore

$$a_n = n! \frac{\sin((n+1)\theta)}{\sin \theta}.$$

The denominator factors as

$$(1 - e^{i\theta}\xi)(1 - e^{-i\theta}\xi),$$

so the Borel singularities are at $\xi = e^{\pm i\theta}$. Their two factorial-over-power contributions have conjugate phases and combine into the sine. This is the simplest model of oscillatory large-order behavior.

Solution to Exercise 37.24

Remove a symmetric interval $(1 - \delta, 1 + \delta)$ around the pole. The upper and lower indentation integrals have opposite imaginary parts. Their average tends, as $\delta \downarrow 0$, to

$$\text{PV} \int_0^\infty \frac{e^{-z\xi}}{1 - \xi} d\xi.$$

For real $z > 0$, complex conjugation maps the upper contour to the lower contour, so the two lateral sums are conjugates and their average is real.

In this one-pole linear example, the symmetric average is the scalar shadow of median re-summation. In the full transseries, one more precisely applies half of the Stokes automorphism to the parameter and then takes a lateral sum. The two descriptions agree after the chosen homogeneous-sector normalization is taken into account.

38.5 Solutions to resurgence and saddle exercises**Solution to Exercise 37.25**

The Stokes automorphism is

$$\mathfrak{S} = \exp(\dot{\Delta}_A) = \exp(S_1 \partial_\sigma).$$

For any formal function of the parameter,

$$\exp(S_1 \partial_\sigma) F(\sigma) = \sum_{n \geq 0} \frac{S_1^n}{n!} F^{(n)}(\sigma) = F(\sigma + S_1).$$

Hence

$$\mathfrak{S} \hat{\Phi}(z, \sigma) = \hat{\Phi}(z, \sigma + S_1).$$

Its formal half-power is

$$\mathfrak{S}^{1/2} = \exp\left(\frac{S_1}{2} \partial_\sigma\right),$$

which translates σ by $S_1/2$. Signs reverse if the opposite convention for relating upper and lower lateral sums is adopted.

Solution to Exercise 37.26

Let

$$\phi(x) = \lambda \frac{2}{3} x^{3/2}.$$

Then

$$\frac{y'}{y} = \phi' + \frac{\alpha}{x} = \lambda \sqrt{x} + \frac{\alpha}{x},$$

and

$$\begin{aligned} \frac{y''}{y} &= \left(\lambda \sqrt{x} + \frac{\alpha}{x}\right)^2 + \frac{\lambda}{2\sqrt{x}} - \frac{\alpha}{x^2} \\ &= x + \lambda \left(2\alpha + \frac{1}{2}\right) x^{-1/2} + (\alpha^2 - \alpha) x^{-2}. \end{aligned}$$

The leading x cancels against the xy term in Airy's equation. The next term vanishes when

$$2\alpha + \frac{1}{2} = 0, \quad \alpha = -\frac{1}{4}.$$

The remaining relative residual is

$$(\alpha^2 - \alpha)x^{-2} = \frac{5}{16}x^{-2}.$$

The first fluctuation correction in the WKB series cancels this residual.

Solution to Exercise 37.27

The action is

$$S(x) = \frac{x^2}{2} + \frac{x^4}{4}, \quad S'(x) = x(1 + x^2).$$

Thus the saddles are

$$x_0 = 0, \quad x_{\pm} = \pm i.$$

Their actions are

$$S(0) = 0, \quad S(\pm i) = -\frac{1}{2} + \frac{1}{4} = -\frac{1}{4}.$$

Now put $x = \sqrt{g}t$:

$$\begin{aligned} I(g) &= \sqrt{g} \int_{-\infty}^{\infty} e^{-t^2/2} e^{-gt^4/4} dt \\ &\sim_{\text{as}} \sqrt{g} \sum_{n \geq 0} \frac{(-g/4)^n}{n!} \int_{-\infty}^{\infty} t^{4n} e^{-t^2/2} dt. \end{aligned}$$

The Gaussian moment is

$$\sqrt{2\pi}(4n-1)!!,$$

so

$$I(g) \sim_{\text{as}} \sqrt{2\pi g} \sum_{n \geq 0} c_n g^n, \quad c_n = (-1)^n \frac{(4n-1)!!}{4^n n!} = (-1)^n \frac{(4n)!}{2^{4n} (2n)! n!}. \quad (38.6)$$

Their ratio is

$$\frac{c_{n+1}}{c_n} = -\frac{(4n+1)(4n+3)}{4(n+1)} \sim -4n.$$

Thus the coefficients grow factorially and alternate. The law $a_{n+1}/a_n \sim n/A$ points to $A = -1/4$, exactly the action difference from the origin to either complex saddle. A negative Borel action produces alternating signs and a singularity away from the positive summation ray.

Solution to Exercise 37.28

In the original variable,

$$\left(e^{-Az} \widehat{\Phi}_A\right) \left(e^{-Bz} \widehat{\Phi}_B\right) = e^{-(A+B)z} (\widehat{\Phi}_A \widehat{\Phi}_B).$$

Therefore a nonlinear product forces a sector of action $A + B$. In the Borel plane, multiplication becomes convolution. Singular behavior near A convolved with singular behavior near B can pinch the convolution integral at $A + B$, creating a singularity there.

If $B = 2A$, the action $2A$ may arise both as an independent elementary sector and as the product of two A sectors. Their recursions mix. This is resonance: the same transmonomial receives contributions from different mechanisms. Additional powers of z , logarithms, or a larger parameter space may be needed to solve the resulting compatibility equations.

Solution to Exercise 37.29

With $\tau = Ae^{-x}$,

$$y = \frac{1}{1+\tau} = 1 - \tau + \tau^2 - \tau^3 + \cdots \quad (|\tau| < 1).$$

The sector expansion is therefore the geometric transseries $\sum_{k \geq 0} (-1)^k A^k e^{-kx}$. The transasymptotic rearrangement sums all those sectors into

$$F_0(\tau) = \frac{1}{1+\tau},$$

with no inverse- x corrections in this exact example. When $|\tau| \ll 1$, the sector expansion displays the hierarchy transparently. When $\tau = O(1)$, no finite number of sectors is uniformly accurate, while $F_0(\tau)$ remains the right compact description except near its pole $\tau = -1$.

38.6 Solutions to structural and computational exercises**Solution to Exercise 37.30**

The germ $x^3 + \log x$ exceeds every real constant for large x , and

$$(x^3 + \log x)' = 3x^2 + \frac{1}{x} > 0.$$

This illustrates the first H -field axiom.

The germ $7 + x^{-1}$ is bounded and decomposes as the constant 7 plus the infinitesimal x^{-1} . The germ e^{-x} is itself infinitesimal, with constant part 0. These illustrate $\mathcal{O} = C + \mathfrak{o}$. The examples do not prove the axioms for an entire field, but they show exactly what the axioms are abstracting.

Solution to Exercise 37.31

Direct calculation gives

$$\begin{aligned} (x^a (\log x)^b)^\dagger &= \frac{a}{x} + \frac{b}{x \log x}, \\ (e^{x^p})^\dagger &= px^{p-1}, \\ (e^{e^x})^\dagger &= e^x. \end{aligned}$$

For fixed nonzero a, b and $p > 1$,

$$e^x \succ px^{p-1} \succ \frac{a}{x} + \frac{b}{x \log x}.$$

The logarithmic derivative is the fractional rate of change: for small h ,

$$\frac{f(x+h)}{f(x)} \approx \exp\left(h \frac{f'}{f}\right).$$

Thus e^{e^x} is extraordinarily sensitive to a small shift in x , while a power-log monomial changes only by a relative amount of order h/x .

Solution to Exercise 37.32

The formal evaluation is

$$T(\omega) = \omega^2 + 3\omega + \log \omega + 5 + e^{-\omega}.$$

Its displayed terms have order

$$\omega^2 \succ \omega \succ \log \omega \succ 1 \succ e^{-\omega}.$$

The leading Conway monomial is ω^2 , and $e^{-\omega}$ is a positive infinitesimal smaller than every power ω^{-n} .

Evaluation at one infinite argument treats the transseries as a value in an ordered exponential field. Composition requires more: it must assign values coherently as the argument varies through positive infinite surreals and obey the chain rule. Arbitrary surreal numbers do not automatically carry such a compatible function-composition structure.

Solution to Exercise 37.33

Use

$$\begin{aligned} \log(x+1) - \log x &= x^{-1} - \frac{1}{2}x^{-2} + \frac{1}{3}x^{-3} - \frac{1}{4}x^{-4} + \frac{1}{5}x^{-5} - \frac{1}{6}x^{-6} + \cdots, \\ (x+1)^{-1} - x^{-1} &= -x^{-2} + x^{-3} - x^{-4} + x^{-5} - x^{-6} + \cdots, \\ (x+1)^{-2} - x^{-2} &= -2x^{-3} + 3x^{-4} - 4x^{-5} + 5x^{-6} + \cdots, \\ (x+1)^{-4} - x^{-4} &= -4x^{-5} + 10x^{-6} + \cdots. \end{aligned}$$

After multiplying by $-1/2$, $-1/12$, and $1/120$, respectively, the coefficients of $x^{-2}, \dots, x^{-6}$ all cancel. Therefore

$$y(x+1) - y(x) = \frac{1}{x} + O(x^{-7}),$$

which is stronger than requested. If y is a solution, so is $y + C$, because the forward difference annihilates constants. A boundary condition or normalization is needed to choose C .

Solution to Exercise 37.34

The occupied exponent vectors are

$$(0, 0), (1, 0), (2, 0), (0, 1), (1, 1), (3, 1), (2, 2).$$

Here (k_1, k_2) represents $x^{-k_1}e^{-k_2x}$. A rectangle such as

$$0 \leq k_1 \leq 3, \quad 0 \leq k_2 \leq 2$$

contains every displayed point, though it also contains unoccupied points. It is not a dominance initial segment. For every N , the omitted algebraic monomial x^{-N} with vector $(N, 0)$ dominates every monomial with $k_2 \geq 1$, including terms retained in the rectangle.

Solution to Exercise 37.35

The computation through e^{-20x} proves only that the expression is $O(e^{-21x})$ relative to that chosen grid, or more carefully that all coefficients above the cut vanish. It does not by itself prove exact zero.

Exact formal identities do prove zero:

$$\begin{aligned}\log(e^x(1 + e^{-x})) &= \log(e^x) + \log(1 + e^{-x}) \\ &= x + \log(1 + e^{-x}),\end{aligned}$$

where positivity and the formal logarithm rules are valid. Subtracting the last two terms gives exactly zero. The first argument is finite evidence; the second is an identity in the transseries field.

Solution to Exercise 37.36

- (a) This is a *formal* statement: coefficients cancel in the formal algebra.
- (b) This is a *summability hypothesis*: it supplies the continuation and growth needed for a Laplace integral.
- (c) This is a *sectorial analytic* conclusion obtained from summation.
- (d) This is a *global analytic* continuation statement, much stronger than sectorial existence.
- (e) This is an *asymptotic* estimate for finite truncations.

A formal solution can fail to be summable. A summable sector can have singularities preventing global continuation. A function can have an asymptotic series without that series being Borel summable. Each arrow requires its own hypotheses.

Solution to Exercise 37.37

(a) **Formal recurrence.** Put $\hat{y} = \sum_{n \geq 0} a_n z^{-n-1}$. Then

$$\hat{y}' + \hat{y} = \frac{1}{z}$$

gives $a_0 = 1$ and $a_n = na_{n-1}$, hence $a_n = n!$.

(b) **Homogeneous sector.** The homogeneous equation $y' + y = 0$ gives Ce^{-z} . Thus

$$\hat{y}(z; C) = \sum_{n \geq 0} \frac{n!}{z^{n+1}} + Ce^{-z}.$$

This step is formal algebra plus an exact elementary ODE calculation.

(c) **Borel transform.** Dividing by $n!$ gives

$$\mathcal{B}\hat{E}(\xi) = \sum_{n \geq 0} \xi^n = \frac{1}{1 - \xi}.$$

The equality begins near $\xi = 0$ and its rational continuation is analytic away from 1.

(d) **Lateral jump.** The residue calculation of Exercise 37.19 yields

$$\mathcal{S}_{0+}\hat{E} - \mathcal{S}_{0-}\hat{E} = 2\pi ie^{-z}.$$

This is analytic: it uses contour integrals and continuation.

(e) **Parameter transformation.** Representing one analytic solution on both sides requires

$$C_+ = C_- - 2\pi i.$$

This is the Stokes change of asymptotic coordinate.

(f) **Optimal truncation.** The ratio of consecutive terms is $(n+1)/|z|$, so truncate near $n \approx |z|$. The least term is of order $|z|^{-1/2}e^{-|z|}$ up to a constant, matching the homogeneous exponential scale.

(g) **ODE verification.** For a nonsingular ray,

$$E_\theta(z) = \int_0^{e^{i\theta}\infty} \frac{e^{-z\xi}}{1-\xi} d\xi.$$

Then

$$\begin{aligned} E'_\theta(z) + E_\theta(z) &= \int \frac{(1-\xi)e^{-z\xi}}{1-\xi} d\xi \\ &= \int_0^{e^{i\theta}\infty} e^{-z\xi} d\xi = \frac{1}{z}. \end{aligned}$$

Differentiation under the integral and convergence are analytic inputs. The full solution adds Ce^{-z} , which solves the homogeneous equation exactly.

This one example contains the formal sector construction, factorial divergence, Borel continuation, Stokes jump, parameter change, and numerical least-term estimate in their simplest possible forms.

Chapter 39

Formula sheet and problem-solving checklist

39.1 Dominance rules worth memorizing

For every fixed positive a, b, c ,

$$\begin{aligned}e^{x^a} &\succ x^b \succ (\log x)^c \succ 1, \\x^{-b} &\succ e^{-x^a}, \\e^{x^a} &\succ e^{x^b} \quad (a > b), \\e^{-x^a} &\prec e^{-x^b} \quad (a > b).\end{aligned}$$

More generally,

$$e^A \succ e^B \iff A - B \rightarrow +\infty$$

for real asymptotic exponents. Take logarithms before trying to compare complicated positive monomials.

39.2 Normal forms

For nonzero T , the additive leading form is

$$T = a\mathfrak{m} + \text{smaller terms}, \quad a = \text{lc}(T), \quad \mathfrak{m} = \text{mag}(T).$$

If $T > 0$, the multiplicative form is

$$T = a\mathfrak{m}(1 + U), \quad a > 0, \quad U \prec 1.$$

This one factorization unlocks all elementary functions:

$$\begin{aligned}T^{-1} &= a^{-1}\mathfrak{m}^{-1} \sum_{n \geq 0} (-U)^n, \\T^r &= a^r \mathfrak{m}^r \sum_{n \geq 0} \binom{r}{n} U^n, \\\log T &= \log a + \log \mathfrak{m} + \sum_{n \geq 1} \frac{(-1)^{n+1}}{n} U^n.\end{aligned}$$

For $T = L + c + S$ with L purely large, $c \in \mathbb{R}$, and $S \prec 1$,

$$e^T = e^{L+c} \sum_{n \geq 0} \frac{S^n}{n!}.$$

39.3 Differential formulas

$$\begin{aligned} (TU)' &= T'U + TU', \\ (T \circ U)' &= (T' \circ U)U', \\ (e^T)' &= T'e^T, \\ (\log T)' &= \frac{T'}{T}, \\ (T^r)' &= rT^{r-1}T'. \end{aligned}$$

The logarithmic derivative

$$T^\dagger = \frac{T'}{T}$$

is often easier to compare than T' itself.

For asymptotic integration of $e^{P(x)}$, set

$$I = e^{P(x)}F(x), \quad F' + P'F = Q$$

when $I' = Qe^P$. Begin with $F \sim Q/P'$ and iterate the derivative correction.

39.4 Composition and inversion

When $U \prec x$ is small enough for Taylor expansion,

$$T(x+U) = \sum_{n \geq 0} \frac{T^{(n)}(x)}{n!} U^n.$$

To invert $T = x + R$ with $R \prec x$, either solve

$$S = x - R \circ S$$

by contraction or use Newton iteration

$$S_{n+1} = S_n - \frac{T \circ S_n - x}{T' \circ S_n}.$$

Always verify by the residual $T \circ S - x$.

39.5 Factorial divergence and optimal truncation

If

$$a_n \sim K \frac{\Gamma(n+\beta)}{A^{n+\beta}},$$

then

$$\frac{a_n}{a_{n-1}} \sim \frac{n+\beta-1}{A}, \quad N_* \approx |Az|,$$

and the least term has an exponential factor $e^{-|Az|}$.

39.6 Borel–Laplace formulas

For

$$\widehat{\Phi}(z) = \sum_{n \geq 0} a_n z^{-n-1},$$

use

$$\mathcal{B}\widehat{\Phi}(\xi) = \sum_{n \geq 0} \frac{a_n}{n!} \xi^n,$$

$$\mathcal{S}_\theta \widehat{\Phi}(z) = \int_0^{e^{i\theta}\infty} e^{-z\xi} \mathcal{B}\widehat{\Phi}(\xi) d\xi,$$

$$\mathcal{B}(\widehat{\Phi}\widehat{\psi}) = (\mathcal{B}\widehat{\Phi}) * (\mathcal{B}\widehat{\psi}).$$

A singularity at $\xi = A$ predicts e^{-Az} . A simple pole $r/(\xi - A)$ predicts

$$a_n \sim -r \frac{n!}{A^{n+1}}.$$

39.7 A ten-question checklist

When confronting a new problem, ask:

1. What is the limiting variable and in which complex sector does it move?
2. What are the candidate dominant balances?
3. Which monomials form the smallest closed asymptotic scale?
4. Is the proposed support well based?
5. Which coefficients or sector amplitudes are free?
6. Does substitution produce a recurrence with factorial late growth?
7. What actions are suggested by ratios or saddle differences?
8. Where are the Borel singularities and singular directions?
9. How must transseries parameters transform across those directions?
10. Have formal residuals, large-order predictions, and numerical values all been checked independently?

Chapter 40

Glossary

Action

A quantity A appearing in an exponential scale e^{-Az} or $e^{-A/\varepsilon}$. In saddle-point problems it is often a difference of critical values of an exponent.

Alien derivative

A derivation Δ_ω that extracts resurgent singular information at the Borel location ω .

Anti-Stokes line

A convention-dependent term often used for a ray on which two exponentials have equal magnitude. This book prefers the explicit phrase *equal-magnitude line*.

Asymptotic couple

An ordered value group together with a map recording the valuation of logarithmic derivatives.

Asymptotic expansion

A formal series whose finite truncations approximate a function with successively smaller remainders in a specified limit.

Asymptotic scale

An ordered sequence or family $\phi_0 \succ \phi_1 \succ \cdots$ used as the coordinate axes of an expansion.

Beyond all orders

Smaller than every term in a specified algebraic scale, as e^{-x} is smaller than every x^{-N} .

Borel plane

The ξ -plane in which a factorially divergent series becomes a locally convergent Borel transform.

Borel sum

An analytic function obtained by Borel transformation, analytic continuation, and a directional Laplace integral.

Borel–Leroy transform

A Borel transform dividing coefficients by $\Gamma(n + \beta)$ or a related gamma factor adapted to an algebraic prefactor.

Borel–Padé method

A numerical method that approximates a finite Borel series by a rational Padé approximant before Laplace integration.

Bridge equation

An identity equating alien differentiation of a transseries with a vector field in its free parameters.

Contractive map

A formal map that makes differences asymptotically smaller. Iteration then fixes successive transmonomials and produces a unique fixed point in a complete formal setting.

Dominance

The comparison $f \prec g$, meaning $f/g \rightarrow 0$ analytically or that the leading monomial of f is smaller formally.

Dulac series

A generalized expansion involving powers and logarithms, common in local analytic dynamics and return maps.

Endless continuability

The ability to analytically continue a Borel germ along arbitrary finite paths after avoiding a locally finite singular set.

Exponential height

The number of recursive exponential levels needed to construct a transmonomial.

Flat function

A function smaller than every term of a chosen asymptotic scale. Flat functions cause nonuniqueness of ordinary asymptotic expansions.

Formal series

A symbolic sum manipulated coefficientwise without an initial convergence claim.

Froissart doublet

A nearby pole-zero pair in a Padé approximant, usually a numerical artifact.

Gevrey order

A measure of coefficient growth using bounds such as $|a_n| \leq CA^n \Gamma(1 + sn)$.

Giant part

The sum of terms in a transseries whose monomials dominate 1.

Grid

A finitely generated set of monomials, typically $\mathfrak{m}_0 \mu_1^{k_1} \cdots \mu_r^{k_r}$ with lower bounds on the integers k_j .

Hahn field

A field of generalized series with coefficients in a field and exponents or monomials in an ordered abelian group, subject to a well-ordered support condition.

Hardy field

A differential field of germs at infinity of real functions, ordered by eventual sign.

Height wins

The rule that a genuinely higher exponential level dominates all lower-height monomials, provided its top exponent is positive and purely large.

Homogeneous sector

A free solution of the linearized or homogeneous equation, often multiplied by a transseries parameter.

Hyperasymptotics

Repeated optimal truncation and re-expansion of exponentially small remainders using neighboring saddles or Borel singularities.

Hyperseries

Extensions of ordinary transseries allowing longer, often transfinite, logarithmic or exponential hierarchies.

 H -field

An ordered differential field axiomatizing basic eventual behavior of Hardy-field germs and transseries.

Instanton

A nontrivial finite-action saddle in a semiclassical problem, contributing a sector such as $e^{-A/g}$.

Laplace direction

The ray along which the Borel transform is integrated against $e^{-z\xi}$.

Lateral sum

A Borel–Laplace sum taken just above or below a singular direction.

Leading coefficient

The coefficient multiplying the largest monomial of a nonzero transseries.

Leading monomial

The largest monomial in the support of a nonzero transseries.

Least term

The smallest term in magnitude of a divergent asymptotic series at a fixed large argument.

Logarithmic depth

The number of iterated logarithmic substitutions needed to express an element in a chosen transseries hierarchy.

Logarithmic derivative

$T^\dagger = T'/T$, the relative rate of change of T .

Median summation

A symmetric resummation applying half a Stokes transformation so that conjugate lateral ambiguities cancel under suitable reality conditions.

Monomial group

An ordered multiplicative abelian group whose elements label asymptotic sizes.

Multisummability

Compatible summation of several components of a formal solution having distinct Gevrey levels.

Non-Archimedean

A setting with infinitesimals and infinitely large elements, where the largest term usually controls a sum.

Optimal truncation

Stopping a divergent asymptotic series near its least term instead of summing indefinitely.

Purely large

A transseries all of whose monomials dominate 1 and with no constant or small part.

Ratio set

A finite collection of small monomials used as generators for a grid.

Resonance

A coincidence among action combinations or characteristic exponents that forces sector mixing and may require logarithmic terms.

Resurgence

The property that Borel singularities of one formal sector contain controlled information about other sectors, yielding large-order relations and Stokes data.

Sector

A component of a transseries with a fixed exponential action and algebraic prefactor, usually carrying its own fluctuation series.

Singulant

A possibly variable action function governing a late-term ansatz and an exponential remainder, especially in exponential asymptotics.

Small part

The sum of terms whose monomials are dominated by 1.

Stokes automorphism

The formal automorphism relating lower and upper lateral resummations across a singular direction.

Stokes constant

A normalization-dependent coefficient measuring the coupling between resurgent sectors across a Borel singularity.

Stokes phenomenon

The change of an asymptotic representation, often the coefficient of an exponentially small sector, as a complex direction is crossed.

Support

The set of monomials having nonzero coefficients in a generalized series.

Summable family

A family of transseries for which the union of supports is well based and each monomial receives contributions from only finitely many family members.

Transasymptotic rearrangement

A reorganization that treats a combined variable such as $\tau = \sigma e^{-Az} z^\beta$ as order one and sums infinitely many sectors into coefficient functions of τ .

Transmonomial

A formal symbol representing one asymptotic magnitude, built recursively from powers, exponentials, and logarithms.

Transseries

A well-based generalized series of transmonomials; in resurgent analysis, also a sectorial expansion combining divergent power series with exponential scales and free parameters.

Transexponential

A scale beyond all finite iterates of the ordinary exponential, introduced to solve certain iteration equations.

Valuation

An algebraic measure of smallness. In a transseries field it is determined by the leading monomial, with larger valuation corresponding to smaller magnitude.

Well-based support

A support ordered so that every nonempty subset has a largest monomial, or equivalently has no infinite chain increasing in dominance under the conventions of this book.

Witness

In grid-based transseries calculations, finite ratio-set data certifying a particular dominance or support relation.

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